Gertrude's Children's Hospital
Laboratory Handbook

For Clinicians

[Version 1, 2016]

Preface

Clinical laboratory testing plays a crucial role in the detection, diagnosis and treatment of disease. Laboratory tests help determine the presence, extent, or absence of disease and monitor the effectiveness of treatment. An estimated 60-70 percent of all decisions regarding a patient's diagnosis and treatment, hospital admission and discharge are based on laboratory test results.

Clinical laboratory personnel ensure that the correct test is performed on the right person, at the right time, producing accurate test results that enable clinicians to make the right diagnostic and therapeutic decisions using the appropriate level of health care resources.

It is the role of laboratory professionals to advise clinicians about the most appropriate tests for given clinical conditions. Through this partnership, the overall cost of testing is controlled and the quality of patient care is improved.

This Laboratory Handbook has been written to help you make the best use of the Pathology Laboratory services. It outlines the services provided, hours of operation, communication channels, handling of samples and interpretation of some test results.

The handbook is designed to provide details of the test repertoire, specimen requirements, reference ranges and test frequency. The information provided is an overview only and where more detailed information or advice is required, clinicians are encouraged to consult the laboratory. Further, those who may have any suggestions or criticisms of either the laboratory service or this handbook should be assured that their feedback is welcome.

Dr.Beatrice Kabera
Chief Pathologist.
Gertrude's Children's hospital

Foreward

We are determined to provide a responsive service to our patients and clinicians and are working on improving our customer focus and in adding value to everything we do. We always welcome comments and suggestions and would be glad to discuss ways of improving our services to all users. I hope this handbook will of serve to improve your experience in using the laboratory and enhance the care you deliver to your patients.

Dr. Thomas Ngwiri
Head Clinician, Gertrude's Children's Hospital

Quality is a strategic goal that delivers the best outcome for both clinicians and patients. The laboratory has a primary commitment to quality which clearly requires a close partnership between the providers and users of this service. We are committed to achieving and maintaining the highest possible standards. I am sure that the information contained in this handbook will prove invaluable in helping us to attain this goal.

Dr. Robert Nyarango
CEO, Gertrude's Children's Hospital

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LABORATORY ADDRESS AND CONTACTS

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Website:	http://www.gerties.org

Laboratory telephone extensions

Laboratory Branch	Extensions
Muthaiga Laboratory Chemistry/Hematology	252
Muthaiga Laboratory Manager	253
Muthaiga Pathologists Office	254
Muthaiga Laboratory Reception	255
Muthaiga Microbiology /Parasitology	256
Muthaiga Phlebotomy	257
Lavington Laboratory	0721394306 (Ext. 832)
Donholm Laboratory	0723719612 (Ext. 707)
Embakasi Laboratory	0714333300 (Ext. 723)
Nairobi West Laboratory	0735639063 (Ext. 533)
Pangani Laboratory	0711463020 (Ext. 527)
Komarock Laboratory	0720939357 (Ext .805)
Kitengela Laboratory	0734800433 (Ext.2004)
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CHAPTER 1

1.1 PURPOSE

This handbook provides an overview on the organization of GCH laboratory services. This handbook also gives pre-analytical information and guidance to laboratory service users when requesting tests and includes:

- Details of services provided
- Laboratory contact details and opening hours
- Details of phlebotomy services
- Instructions for completing sample and request form information
- Arrangements for transporting samples to the laboratories
- Point of care testing
- Test table of analyses provided including details of specific sample requirements

The Laboratory is responsible for performing all laboratory procedures requested by clinicians or others for the management of patients, and deals with testing for all patients (children and adults), although emphasis is put on pediatric laboratory testing, since Gertrude's Children's Hospital is paediatric hospital.

The laboratory provides an efficient, high quality, cost effective, clinical-based diagnostic service. We participate in both internal quality control and external quality assessment schemes in all areas of laboratory testing, with good performance. Our range of services covers phlebotomy, clinical chemistry, histopathology, hematology, microbiology and immunology. In addition, we run a hospital based blood bank at the Muthaiga Laboratory. The laboratory also administers the mortuary services for the hospital, and offers autopsy services.

1.3 ORGANIZATION OF THE LABORATORY SERVICES

1.3.1 VISION

To be the preferred provider of paediatric pathology services in the region catering for the diagnostic needs of our patients and other clients.

1.3.2 MISSION

To deliver quality results in an efficient and cost effective manner to patients / clients, in an environment of continuous change in all aspects of pathology.

1.3.3 ORGANIZATION OF THE LABORATORY

The laboratory is organized into several different sections:

- Reception
- Phlebotomy
- Microbiology and serology
- Hematology
- Biochemistry
- Anatomic Pathology
- Mortuary Services
- Pathologist Advisory/Consultancy Services

1.4 ADMINISTRATIVE STRUCTURE

1.4.1 TECHNICAL STAFF

Pathologists	Chief Pathologist, subspecialty pathology services in haematopathology, microbiology, histopathology, cytology and clinical chemistry. These must be holders of a Medical degree and Master of Medicine (MMed) in Pathology and Subspecialty.
Lab Manager	Head of the technical staff; holder of a Masters degree in a laboratory related field; administrative and managerial training.
Laboratory Quality manager	Oversee QMS Implementation within the laboratory Higher National Diploma (HND) /BSc medical laboratory sciences.
Chief Technologists	Heads of different sections within the laboratory; Higher National Diploma (HND) in subspecialties including microbiology, serology, clinical chemistry, Haematology and management
Technologists	HND entry level/BSc medical laboratory sciences
Phlebotomists	Trained in phlebotomy
Lab Attendants /Mortuary	Trained in Mortuary techniques

1.4.2 NON TECHNICAL STAFF

Assistants / Secretaries	Diploma/Certificate in basic secretarial skills; Specialization in specific area of assignment
Cleaners	O-Level Certificate/Housekeeping skills

1.5. WORKSTATION

1. Haematology, Coagulation& FlowCytometry (CD3/CD4/CD8)
2. Chemistry
3. Serology & Urinalysis
4. Microbiology
5. Blood Bank
6. Phlebotomy
7. Store room
8. Cross-check

1.6 QUALITY ASSURANCE

1.6.1 ACCREDITATION

Accreditation is an important milestone in the path of continuous quality improvement. Successful participation in accreditation programs is a long-term operational goal for medical laboratories. Universal access to testing remains an essential focus for laboratory services but it is similarly vital that existing testing services are of consistently high quality and that clinicians can confidently rely upon laboratory results for patient-care decisions.

The establishment of the Joint Commission International Accreditation (JCIA) hospital Accreditation Program provides an affordable and potentially ground-breaking opportunity to improve laboratory quality practices in Gertrude's Children's Hospital.

JCI Accreditation Program and ISO 15189 for medical laboratories in GCH laboratory will provide a feasible, affordable, and scalable procedure to improve the quality of testing services across the laboratory systems.

Participation in an ongoing accreditation program will ensure regular and rigorous assessment of laboratories, drive the consistent maintenance of high standards, improve the reliability of testing and bolster the confidence of clinicians to make use of laboratory data in diagnosis and treatment of patients. Accreditation is a critical milestone on the quality roadmap and steps should be taken without delay to put policies, planning and processes in place that focus laboratory strengthening efforts on this target.

1.6.1.1 EXTERNAL QUALITY ASSESSMENT AND INTERNAL QUALITY CONTROL

We have an established quality management system. All our laboratories participate in external quality assurance schemes to monitor the accuracy and precision of its analyses. Internal quality control is used to check the validity of results on a day-to-day basis.

1.6.1.2 QUALITY CONCERNS AND COMPLAINTS

If you have any issues about the quality of service you receive please contact the Laboratory Quality Manager, the Laboratory Manager and the Chief Pathologist.

1.6.2 TECHNICAL REQUIREMENTS

1.6.2.1 PERSONNEL

Our laboratory's management ensures personnel performing specific tasks are qualified based on appropriate education, training, experience, and/or demonstrated skills, as required. All the Technologists are registered by the Kenya Medical Laboratory and Technologists Board (KMLTTB).

Adequate supervision is provided for staffs that are undergoing training. Training needs are identified through induction criteria when a new employee is hired. Any routine training and re-training necessary for a person to perform a job effectively is specified in job descriptions and maintenance procedures as appropriate.
An annual training plan is established and maintained by the training committee; a detail of any training carried out is recorded on personnel training records and, where issued, certificates provided to the individual.

1.6.3 MANAGEMENT REQUIREMENTS
LEGAL IDENTITY

The laboratory is authorized by the Ministry of Health under the Kenya Medical Laboratory and Technologists Board (KMLTTB) and Kenya Medical Practitioners and Dentists Board (KMPDB).

CHAPTER 2 – GENERAL PROCEDURES

2.1 LABORATORY GENERAL

This section provides general information and procedures for use of the laboratory. The table outlines the processes from pre-analytical to post-analytical phase. Detailed information that is specific to each laboratory section is provided in the following chapters.

Step	What happens?	Who's responsible?	Procedures needed?	Pitfalls
PRE-ANALYTICAL PHASE				
Order placed	Clinician determines need	Clinician	Ordering protocols	• Unauthorized person ordering • Inappropriate order
Patient presents to laboratory	Laboratorian interacts with patient	Patient / Laboratorian	Customer Service	• Lack of timely service • Interaction not client-friendly
Requisition completed & reviewed by laboratory staff	Requisition reviewed for proper information	Clinician, Clerk, or Laboratorian	Criteria for specimen acceptability	• Incomplete patient data • Incomplete clinical history • Clerical errors
Specimen type determined for collection	Note specific test requested and determine what type of sample is needed	Laboratorian	• Specimen requirements for (venous) blood collection • SOP for each analyte	• Not checking or following specimen requirements • Inadequate communication to patients regarding specimen self-collection
Specimen collected	• Blood drawn from patient • Sputum, urine, stool or other specimen is collected	• Blood – Clinician or Laboratorian • Non-blood specimens – Clinician or patient	• Phlebotomy key competencies • Phlebotomy training checklist	• Blood – wrong tube, incorrect amount of blood, injury • Non-blood specimen collection – incorrect specimen or incorrect collection procedure; improper labeling
Specimen logged	Appropriate information recorded in specimen log	Laboratorian	Specimen management	• Clerical errors • Inadequate information
Specimen accepted or rejected	Specimen accepted or rejected based on meeting acceptance criteria	Laboratorian	• Specimen management • Criteria for specimen acceptability	• Unsatisfactory specimen • Specimens with hazardous handling conditions • Inadequately labeled specimen

Specimen assigned according to test request/s	• Requests reviewed for testing priority – STAT versus routine. • If multiple tests to be done, sequential workstations versus aliquoting • Centrifugation required • Send out versus in-house testing	Laboratorian	• Guidelines for STAT testing • Guidelines for multiple tests from one sample • Specific SOPs for each analyte • SOP for send • outs (specimens referred to other facilities for testing)	• Processing not performed in a timely fashion as ordered • Missing some tests on a requisition with multiple tests requested • Centrifuge not performed in a timely manner • Send out tests not referred in a timely manner or transported inappropriately
ANALYTICAL PHASE				
Routine quality checks completed	Prior to testing, determine if proper routine QC, reagent validation, equipment maintenance and calibration completed	Laboratorian	• SOP for each analyte • Guidelines for quality checks for all Log/Charts for each analyzer or test	• QC not done or out of control • Inadequate troubleshooting or follow up of QC • Improper calibration • Inadequate equipment maintenance
Specimen analyzed	Run analysis on specimen	Laboratorian	Specific SOPs for each analyte	• Not following SOP • Taking shortcuts
POST-ANALYTICAL PHASE				
Test results analyzed	• Review test results or accuracy, legibility & validity • Cross-checking • Assure proper quality monitoring	Laboratorian, Supervisor	Specific SOPs for each analyte	• Release of test results without validation or interpretation • Inadequate cross-checking
Test results recorded	• Transfer test results into logbook • Record results accurately	Laboratorian, Clerk	• Test Reporting SOP • Specimen Management	• Clerical errors • Analyte printout results listed in different order than logbook reporting columns
Test results communicated / reported	• Notify clinician of results via written report • Verbal reporting if necessary • Critical Values reporting Ensure that referral specimens are properly tracked	Laboratorian, Nurse	• Specimen management • Client satisfaction guidelines	• Results not communicated in a timely fashion • Results lost • Critical values not reported • Confidentiality breached • Failure to track referral specimens or failure to follow-up on overdue specimens
Documents and records maintained, filed & stored	• File & store results in a retrievable fashion • Transfer files to long term storage • Dispose of files at an appropriate time	Laboratorian	SOP for document & record management (Including Document & Record Retention)	• Unable to retrieve information when needed • Lack of adherence to document retention schedule • Water or moisture damage

2.2 LABORATORY HOURS

2.2.1 PHLEBOTOMY SERVICES

These are available 24 hours at the laboratory for both children and adults. The phlebotomists are specially trained in paediatric phlebotomy. The Main laboratory in Muthaiga is open 24 hours.

2.2.2 PROCESSING OF SPECIMENS

The laboratory receives specimens from inpatients, satellite clinics, doctors' clinics and other laboratories for processing at any time and on any day of the week, including weekends and public holidays.

2.3 REQUESTING LABORATORY TESTS

2.3.1 ELECTRONIC REQUESTS:

This is done online using the Laboratory Information System (LIS) - Kranium. All out-patient clinics, A&E, and inpatient staff will use Kranium to enter laboratory test requests except in the event of failure of the system. Refer to section below for information on how to request tests when Kranium is not working.

2.3.2 HAND-WRITTEN REQUESTS

These will be accepted in the laboratory for external patients referred from outside the hospital, who require only laboratory services. These requests must be legible and contain the following information:

- Patient's name (three names)
- Date of birth
- Contact telephone number, Fax number or P. O. Box Number
- Full name of referring clinician (e-mail address must be provided if results are to be transmitted electronically)
- Requested tests
- Relevant clinical information/history

Clinical details and the patient's age are particularly important in paediatric requesting so that laboratory staff may:

1. Understand the reason for the request
2. Interpret the results
3. Consider the need for further investigations
4. Advise and assist the clinical staff concerning the results obtained

This information will be uploaded on to the LIS (Kranium) and results will be transmitted electronically* or on a HARDCOPY TYPED template.

2.3.3 FAILURE OF THE LABORATORY INFORMATION SYSTEM (KRANIUM)

If this occurs, the following procedures will be followed by the laboratory and all services that submit specimens:

- Legibly written paper forms will be submitted with appropriate and properly labeled specimens
- Clinical Pathology (chemistry, hematology, microbiology etc.) will test all specimens and telephonically report all critical values, as well as scan or fax a copy of all STAT and ASAP results to the requesting clinician
- Routine results will be entered into Kranium once it is working. If the LIS is inoperative for more than six hours, routine test results will be dispatched to the clinician via emails or hardcopy delivery

2.3.4 TESTING PRIORITIES:

There are three categories of laboratory testing priorities: **Routine, ASAP, and STAT.** All requests will be assumed to be routine unless specifically ordered as STAT or ASAP. All laboratory testing is handled in as expeditious manner as possible.

Routine requests: Most routine tests have testing started as soon as the specimen is received. It is highly likely that the testing will be completed within two to four hours after receipt of the specimen; the clinical analyzers utilized in chemistry and hematology allow rapid throughput of patient specimens. All major analyzers are interfaced directly with Kranium to allow rapid and precise transfer of data. Some specimens are "batch tested" to reduce costs and improve efficiency; some common examples are Hormones, HIV/HBV/HCV by the long ELISA method, Haemoglobin electrophoresis, CD4 testing, and glycated hemoglobin.

ASAP (As Soon As Possible): This category is to be utilized for lab requests that are not emergencies, but require results to be received as quickly as possible. Testing will be performed as soon as possible. The maximum turn-around time for an ASAP procedure is at most within two (2) hours. The majority will be ready within the stipulated TAT. Results will be certified and authorized in Kranium and be available to the clinician within two (2) hours of specimen receipt. Any critical values will be relayed directly to the clinician over the telephone with read back verification. Multiple ASAP requests will be performed in the order received.

STAT: : STAT requests are for true emergencies that involve loss of life, limb, or eyesight. The maximum turn-around time for a STAT procedure is one (1) hour or less from specimen receipt. Results will be certified and authorized in Kranium and available to the requesting clinician. Any critical values will be relayed directly to the provider over the telephone, with read back verification.

When multiple STAT requests are received, testing is performed in the order specimens are received (unless a call for immediate priority is received from the unit nurse or clinician). THIS TESTING PRIORITY IS RESTRICTED TO TRUE EMERGENCIES AND ABUSE OF THIS PRIORITY IS STRICTLY DISCOURAGED.

Emergency Requests: STAT and ASAP are the two classifications of requests that are considered emergency or urgent. Ix A provides a list of STAT and ASAP procedures offered by the laboratory. Most these requests are expected to come from the wards and clinics.

2.3.5 SPECIAL TESTS:

The following tests must be scheduled through the laboratory prior to collection:

- Glucose Tolerance Tests
- Therapeutic phlebotomies
- Cytogenetic testing
- Flow cytometry
- Stimulation assays
- Bone Marrow Aspiration(BMA)
- Bone Marrow Trephine Biopsy
- Fine Needle Aspiration (FNA)

2.4 SPECIMEN COLLECTION:

Patient Preparation:

- Most tests do not require special preparation e.g. FBC,LFT'S, RFT,S e.t.c. these can be taken any time.
- Very few laboratory tests require patients to be prepared in a certain way

- As Specimens for some tests must be collected with the patient fasting, or with knowledge of when food was last taken.

BLOOD GLUCOSE

- Fasting blood sugar(FBS),
- For fasting blood sugar test, patient should not eat or drink anything other than water for at least 8 hrs before the test.
- If patient is on medication they may wait until they have had their blood collected before taking their morning dose.

Random blood sugar(RBS)

No special preparation is needed before taking the test.

Glucose Tolerance Test(Non Pregnant)

Patient should fast for 8 hours

Prenatal glucose tolerance test.

50 gram 1 hr glucose tolerance test(gestational diabetes screen)

No special preparation is required.

100 gram 3 hour glucose tolerance test(gestational diabetes)

100 GRAM 3 HOUR GLUCOSE TOLERANCE TEST (GESTATIONAL DIABETES)

This test should be performed in the morning after overnight fast of at least 8hrs and after at least 3 days of unrestricted diet.

Urine

Urine culture:

preferably morning urine but sample gotten from any time of the day can steel be used and the sample should reach the laboratory within one hour since the bacteria that might be in the urine could multiply and the elements that form part of it my be affected. High doses of vitamin c should be avoided as this could affect urinalysis results.

Lipid profile

Performed in the morning after overnight fast at least 8-12hours

Culture Samples:

For Samples for culture should be taken before the patient is subjected to antibiotics.

Occult Blood Test; In case of occult blood test and the specimen is stool, patient should be in a red meat diet free for three days. Also avoid foods rich in ions, aspirins, vitamin c, anti inflamatories, and vegetables such as broccoli, cucumber and carrots should be avoided.

Digoxin level;

Blood should be drawn 6-8hrs after the last dose of digoxin was taken.

Autoimmune antibodies testing

For centromere antibody, histone antibody, LA antibody (SSA), RO (SSA), scl-70antibody. For all these tests a prior ANA test findings are essential. Some tests must be collected in the basal state or with due regard to diurnal variations.

1. Hormonal suppression or stimulation tests require accurately timed collections.

2. Blood for drug monitoring assays is usually collected as a 'trough' (pre-dose) sample, but in certain cases (eg, flucytosine) a 'peak' sample is required.

CAUTION*Specimen containers **should not** be labeled in advance of receiving the specimen.

2.4.1 INPATIENTS

Specimens will be routinely collected by ward staff. Any STAT or ASAP requests must be promptly delivered to the laboratory. Phlebotomists called from the laboratory will only draw blood. If the phlebotomist is unable to collect the specimen because of the condition of the patient, he or she will notify the nursing supervisor and arrangements will then be made for the attending physician or his/her designee to obtain the blood. Collection of specimens other than blood, such as urine, feces, sputum, is the responsibility of the ward staff.

It is very important that any specimen collected by ward personnel be delivered to the lab as soon as possible. Time should not exceed 30 minutes as delays may cause erroneous results and necessitate repeat testing. If delays are expected, contact the lab for proper storage requirements. Specimens must never be placed in a refrigerator used for drugs or food. For specific instructions on sample collection requirements consult the appropriate section of this guide or Kranium, or call the laboratory.

2.4.2 OUTPATIENT

Persons requiring collection of laboratory specimens will present to the Outpatient Phlebotomy Room. Kranium will be checked for pending orders, or the patient may present a laboratory request slip from their doctor. Patients who have neither a valid order on Kranium nor a written order will be referred to their provider after the laboratory has made every attempt to assist the patient. Blood specimens will be collected in a manner that ensures proper testing and reduces the possibility of cross contamination and erroneous results. Depending on the method of blood collection, blood tubes are filled in a specific order.

2.4.3 ORDER OF DRAW:

Specimens collected using evacuated tubes with a tube holder, such as the Vacutainer system, will be filled in the following order:

1. Blood Cultures – SPS
2. Citrate Tube – coagulation studies
3. BD Vacutainer SST Gel Separator Tube - chemistry
4. Serum Tube (glass or plastic)
5. Heparin Tube – most chemistries
6. BD Vacutainer PST Gel Separator Tube with Heparin
7. EDTA Tube – CBC, CD4
8. Fluoride (glucose) Tube – glucose levels

2.4.4 ORDER OF TRANSFER:

Specimens collected in a syringe will be transferred into blood tubes in the following order.

1. Blood Cultures
2. Light blue (sodium citrate) – coagulation studies
3. Green (sodium or lithium heparin) – most chemistries
4. Lavender (EDTA) – CBC, CD4
5. Gray (sodium fluoride) - glucose

6. Marbled red/SST/red & yellow - chemistries
7. Plain red top – chemistries

All containers that hold the primary specimen (i.e. blood tubes, urine cups, culturettes) will be labeled with the patient's full name, hospital number, age or bar code. Histology, microbiology, and blood bank specimens for pre-transfusion testing have additional requirements that are described in the appropriate section of this booklet. UNLABELED, MISLABELED, INCOMPLETELY LABELLED OR ILLEGIBLE SPECIMENS WILL NOT BE ACCEPTED.

All standard precautions will be followed and Personal Protective Equipment (PPE) will be worn during collection of specimens.

As part of quality practice, efforts have been made to minimize the volume of blood collected from patients and special paediatric collection devices and containers, and combination of all possible tests under one accession number is done. For example, a request for CBC, Sickle Cell Screen, and an ESR will be combined under one accession number so only one specimen is collected to perform these tests. However, certain reference laboratories require one specimen per test (e.g., HIV PCR specimens). In these cases, it is impossible to minimize the amount of blood collected. If there are concerns about blood loss from a patient during specimen collection, contact the laboratory for guidance.

Specimens collected outside the hospital will be submitted in a container that does **NOT** contain any spillage, breakage, or other contamination and enclosed in a plastic biohazard bag. Larger biohazard bags or plastic containers with a sealed lid and a biohazard label are suitable for transporting larger items. The laboratory will dispose of any specimens that arrive broken, leaking or otherwise contaminated via Regulated Medical Waste (RMW). The requesting clinic will be notified to recollect the specimen.

Special specimen collection

- Patients with special sample collection requirements will report to the laboratory for instructions and/or containers. This includes 24-hour urine, semen, and fecal samples.
- Ward or clinic personnel or the requesting clinician will collect all microbiology specimens. These should be submitted to the laboratory immediately after collection to prevent loss of pathogenic organisms or overgrowth by contaminating organisms.
- Bone marrow aspiration/collection for patients will be performed by the hospital paediatricians or when specifically requested, by the clinical pathologist/haematologist. In the latter instances this must be scheduled at least 24hours in advance with the Laboratory.
- Cytology: Fluids such as pleural, peritoneal, CSF, urine, and other fluids for cytology must be submitted ASAP. See the Cytology portion of this booklet for specific information.

2.5 SPECIMEN REJECTION:

2.5.1 UNACCEPTABLE SPECIMENS:

Our quality programme mandates that there is an established written criterion for specimen rejection. Criteria for acceptance and rejection of specimens in the laboratory.GCH/LQM/011

Specimens submitted under the following conditions will be rejected:

- Blank request forms or inadequately identified patients on request forms. ALL request forms MUST have the patient's first and last names, date/time of collection (timed collections), test(s) to be performed and initials of persons collecting specimen.
- Inadequately identified/unidentified/mislabeled specimens - ALL specimens MUST have the patient's first and last names, date/time of collection (timed collections) and initials of persons collecting specimen.
- Clotted blood or incompletely filled tube for Complete Blood Count (CBC).
- Improperly filled anticoagulated tubes (coagulation).

- Contaminated or leaking specimens or containers.
- Incomplete timed specimens, such as 24-hour urines only partially collected.

Frequently, the laboratory receives specimens without a request form or an electronic order in Kranium. The laboratory will make every reasonable effort to maintain the testing integrity of the specimen (e.g. centrifuge, refrigerate, etc.) and to contact the submitting provider or clinic. Irretrievable specimens, such as pus from wounds or anaerobic sources, CSF, and blood cultures, will be processed to the fullest extent possible (e.g. cell count, culture processing) while trying to get the relevant missing details. Retrievable specimens are defined as routine blood specimens, throat cultures, and urine (random and 24-hour) and will be discarded. Documentation of all attempts to resolve these issues will be made into Kranium or in a ledger maintained for this purpose.

The following protocol will be utilized for notification when samples are considered unacceptable:

i. Irretrievable specimens (e.g. tissue, CSF, etc.) will be processed, if possible, and the appropriate clinic notified of specimen condition and the lack of orders.
ii. Samples with improper request forms, contaminated containers, or incomplete information will not be accepted. Ward or clinic personnel will be notified to pick up these specimens and resubmit them properly. Where the specimen is brought by a patient, the patient will be informed that the sample will be put in appropriate storage for 48 hours as efforts to contact the referring clinician for proper information are made. Where this is not successful the sample will be discarded and patient informed.
iii. For lipaemic, clotted, or haemolysed blood samples the laboratory will notify the clinician or ward/clinic. If the patient is an inpatient, it is the ward's responsibility to recollect. If the patient had blood collected as an outpatient, the laboratory will contact the patient for recollection.
iv. Sub-optimal microbiology cultures that are irreplaceable will be held for 48 hours before discarding. The clinician will be contacted as soon as possible after receipt. If the clinician requests, these specimens will be cultured and processed.
v. Specimen received without order will be held for the viable period of testing. For urine specimens and anticoagulated tubes this is 24 hours; serum specimens will be held for 7days.
vi. Specimens received without indication of who is requesting the test or where the patient is located (for inpatients and clinics) will be processed but it will be the responsibility of the clinician or patient to contact the lab for the results.

Specific Sample Rejection Criteria for Microbiology:

i. All unlabeled specimens
ii. Any specimen collected in an improper container
iii. Specimens which have been contaminated
iv. Sputum specimens that are determined to be primarily of salivary origin
v. Stool specimens not submitted in proper collection containers for test requested
vi. Specimen samples for culture that do not identify the source of inoculums
vii. Anaerobic cultures cannot be performed on the following specimens – gastric washings, sputum, throat, nasal, urine (except suprapubic), faeces, ileostomy and colostomy swabs, superficial skin, environment cultures, and vaginal/cervical swabs.
viii. Specimens such as urine or sputum in unsterile containers

Specific Sample Rejection Criteria for Cytology

i. Any unlabeled/mislabeled specimen.

2.6 STAT/ASAP/EMERGENCY PROCEDURE LIST:

Appendix 1 lists tests that will be handled as emergency requests. Tests essential for patient care which are required on an emergency basis but which are not listed below may be obtained if the clinician contacts a pathologist or the laboratory manager. Tests ordered STAT, which are not performed as emergency procedures will be handled as either routine or ASAP.

2. 7 ORDER COMMENTS:

Kranium allows brief comments to be entered in the "ORDER COMMENTS:" section of CHCS during order entry. Providers are encouraged to use this option to provide the laboratory with any important information (e.g. known cancer patient). If the order comment area does not provide sufficient room for necessary information the provider should call the laboratory.

2.8 REPORTING OF RESULTS:

All results of laboratory testing are reported via LIS (Kranium) if the requesting provider, clinic, ward or department is linked to LIS (Kranium). In cases where the requesting person/clinic is not linked to the LIS (Kranium), hard copy of the typed and printed results will be made.

Telephone reports will be made in the cases below:

i. **Alert Values and Reporting:** Alert or Critical Values are laboratory test values that vary significantly from the normal range in such a way to be life threatening. All results that fall outside normal ranges will be verified by checking the accuracy of quality control specimens and may undergo repeat testing. The clinician, clinic, and/or ward will be notified with-in one hour and contact documented in Kranium, Appendix 2 lists critical/alert values that were established in consultation with the clinical staff.

2.9. REFERENCE LABORATORIES

Where particular tests are not available in the laboratory, the specimen will be sent to a selected referral laboratory. The referral laboratories selected are accredited (ISO/CAP). In these cases, the requesting provider will be advised of the expected turnaround time. A list of referral laboratories used is attached (Appendix 3). The laboratory reviews this list annually in consultation with the medical staff to ensure any concerns or needs are met.

2.9.1 POINT-OF-CARE TESTING

Point-of-care testing (POCT) is defined by the College of American Pathologists (CAP) as "those analytical patient-testing activities provided within the institution, but performed outside the physical facilities of the clinical laboratories."

Standardization is the consistent use of the same instrument/reagent/test method for any particular analyte throughout the designated health care system. GCH laboratory has POCT for blood glucose using standardized glucometers.

Prior authorization by the Department of Pathology must be obtained before point-of-care-testing of any type is performed at any site in GCH. No point-of-care testing may be done without this authorization.For advice about existing POCT services or introduction of new devices please contact the laboratory quality manager or chief pathologist.

2.9.2 BLOOD COLLECTION, ADDITIVES AND ANTICOAGULANTS

i. Blood collection should be from peripheral veins of the upper limb at the cubital fossa in older children. Collection from the veins on the dorsum of the hand may be performed in younger children or when collection at the cubital fossa is contraindicated or not possible. When other sites are not available, blood may be collected from the veins at the inner aspect of the wrist.

ii. Collection of blood through lines that have been previously flushed with heparin should be avoided, if possible. If it is not possible to avoid using such lines and blood is drawn from an indwelling catheter, the line should be flushed with 5.0 mL saline and the first 3 -5 mL of blood (or six dead space volumes of the catheter) discarded. If the blood is obtained from an IV line approximately 6 times the line volume should be drawn off and discarded before the coagulation tube is filled. The laboratory should always be informed if a specimen is collected from an indwelling line.

2.9.2.1 TUBE ADDITIVES AND ANTICOAGULANTS.

Most specimen tubes contain additives. If the additive is an anticoagulant, the blood will not clot and the specimen will be whole blood that can be centrifuged to obtain plasma. All other additives and additive-free tubes (e.g. red top) produce serum specimens. An additive works optimally when the tube is filled to its stated volume and gently inverted immediately after collection to mix the additive with the blood. Specimen quality can be compromised if a tube is partially filled. Shaking or vigorous mixing can lead to hemolysis of red cells and compromise test results. Always ensure that the specimen is placed in the correct container and clearly labeled. Additive reliability is guaranteed until an expiration date on the label of the tube, if the tube is handled and stored properly. Expiration dates should be checked and expired tubes discarded.
The most common additives are categorized as follows:

Anticoagulants;

Prevent blood from clotting and include ethylenediaminetetraacetic acid (EDTA), citrates, heparin, and oxalates. Each is designed for use in certain types of testing, and it is important to use the correct one.

i) EDTA (Ethylene diaminetetraacetic acid):

EDTA is the anticoagulant of choice for most routine hematology studies - Complete/Total/Full blood counts (CBC/TBC/FBC). The lavender/purple topped vacuum blood tube contains EDTA. The tube must be correctly filled with blood otherwise the accuracy of testing is affected. Samples that are less than 50% of the correct volume will be rejected (due to inappropriate anticoagulant to blood ratio). Samples with clots will also be rejected.

ii) Sodium citrate

Sodium citrate is the anticoagulant of choice for coagulation studies. The blue topped vacuum blood tube contains sodium citrate. The tube must be properly filled and samples having less that 90% of the expected fill, or over-filled will be rejected as the results are likely to be erroneous due to inappropriate anticoagulant to blood ratio. Samples with clots will also be rejected.

iii) Heparin

Contains the anticoagulant Heparin combined with sodium, lithium, or ammonium ion. Heparin is an anticoagulant commonly used in chemistry and special chemistry testing (Sodium or Lithium heparin used for cytogenetic testing). Heparin prevents clotting by inhibiting thrombin and thromboplastin.

Antiglycolytic agents

Prevent glycolysis, which can decrease glucose concentration by up to 10 mg/dL per hour. The most common antiglycolytic agent, **sodium fluoride,** preserves glucose for up to 3 days and inhibits bacterial growth. It is often combined with potassium oxalate anticoagulant to provide plasma specimens. In addition to glucose, sodium fluoride is used to collect ethanol specimens to prevent an increase in alcohol due to fermentation by bacteria.

Clot activators

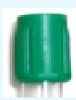
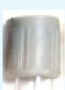
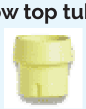
Are coagulation factors such as thrombin and substances such as glass (silica) particles and inert clays like diatomite (Celite) that enhance clotting by providing more surface for platelet activation. The clot activators in gel separator tubes and plastic red top tubes are typically silica.

Thixotropic gel

Separators are inert substances contained in or near the bottom of certain tubes. During centrifugation, the gel lodges between the cells and the fluid, forming a physical barrier that prevents the cells from metabolizing substances in the serum or plasma.

Trace Element–Free Tubes

Trace element–free tubes are as contamination free as possible. They are used to collect specimens for trace elements, toxicology, nutrients, and other tests that detect analytes found in the blood in such tiny amounts that trace elements leached into the blood from tube or stopper materials could falsely elevate test results. The tubes contain EDTA, heparin, or no additive.

Tube cap color	Additive	Function of Additive	Common laboratory tests	Mix by inverting
Light-blue 	3.2% Sodium citrate	Prevents blood from clotting by binding calcium	Coagulation	3 to 4 times
Red or gold (mottled or "tiger" top used with some tubes) 	Serum tube with or without clot activator or gel	Clot activator promotes blood clotting with glass or silica particles. Gel separates serum from cells	Chemistry, Immunology, Serology	5 times
Green 	Sodium or lithium heparin with or without gel	Prevents clotting by inhibiting thrombin and thromboplastin	Stat and routine chemistry, Cytogenetics, Karyotype	8 to 10 times
Lavender or pink 	Potassium EDTA	Prevents clotting by binding calcium	Hematology and blood bank	8 to 10 times
Gray 	Sodium fluoride, and sodium or potassium oxalate	Fluoride inhibits glycolysis; Oxalate prevents clotting by precipitating calcium	Glucose (especially when testing will be delayed), blood alcohol, lactic acid	8 to 10 times
Royal Blue top; 	Serum, no additive, or sodium heparin		For trace element, toxicology and nutritional chemistry determinations	8 – 10 gentle inversions
Yellow top tube 	ACD solution A or ACD solution B			8 to 10 times
BacT/ALERT® Culture Media 	30 mL of supplemented complex media containing adsorbent polymeric beads	Provides an optimal environment for the recovery of a wide range of organisms, including bacteria, fungi & mycobacteria	For detection of microbial growth from blood specimens (blood culture)	8 – 10 gentle inversions

2.9.3 SMALL VOLUMES

Paediatric tubes are available for both general hematology and coagulation testing. Heel/finger stick collection devices are also available from the laboratory. The tube must be thoroughly mixed immediately after collection as these samples tend to clot very quickly. Excessive squeezing should be avoided when collecting samples from finger/heel stick to avoid haemolysis and dilutional errors due to tissue fluids.

2.9.4 TESTING PHASES

The laboratory testing cycle can be outlined in phases as follows;

- The pre-analytic testing phase occurs first in the laboratory process. This phase may include specimen handling issues that occur even prior to the time the specimen is received in the laboratory. Important errors can occur during the pre-analytic phase with specimen handling and identification. Therefore, the pre-analytical phase must have rigorous control measures to avoid unwittingly allowing problems or errors to travel further "downstream."
- The second phase is the analytic phases. This phase includes what is usually considered the "actual" laboratory testing or the diagnostic procedures, processes, and products that ultimately provide results.
- The post-analytic phase is the final phase of the laboratory process. This phase culminates in the production of a final value, result, or in the case of histology, a diagnostic pathology report.

PRE-ANALYTICAL PHASE

Factors affecting blood collection

The following should be taken into account during sample collection:

- After food intake glucose, cholesterol, triglycerides, iron, inorganic phosphate and amino acids are present in elevated concentrations in blood.
- If the patient is moved from a recumbent to an upright position, the concentration of corpuscular and macromolecular substances such as leucocytes, erythrocytes, haemoglobin, haematocrit, total protein, enzymes, lipoproteins and protein-bound ions (e.g. calcium, iron) increases by up to 10 %.
- Some drugs may affect the test performed.
- Substantial diurnal variations can be observed in the case of some analytes, e. g. hormones (epinephrine, aldosterone, corticotropin, cortisol, norepinephrine, prolactin, somatotropin, and testosterone), electrolyte excretion in urine, serum hemoglobin and iron. Therefore, it is recommended to collect samples between 7 and 9 a.m.

2.9.5 SPURIOUS RESULTS DUE TO INAPPROPRIATE SPECIMEN COLLECTION AND HANDLING

Problem	Common Causes	Consequences
Delay in separation of serum	• Overnight storage • Delay in transit	Increased K^+, PO_4, ALT, LDH Decreased HCO_3, (Na^+ occasionally)
Storage	storing at 4oC	• Increased K^+ • Decreased HCO_3
Haemolysis	• Expelling blood through needle into tube • Over vigorous mixing of specimen • Storing specimen in freezer (-20˚C) • Excessive delay in transit • Leaving specimen in hot environment	• For haemolysed samples the following tests will NOT be reported: Folate, potassium, direct bilirubin, CK, IgM, LDH & zinc • For grossly haemolysed samples the following tests will not be reported: Folate, potassium, phosphate, magnesium, lithium, ALP, direct bilirubin, CK, LDH, IgM, triglycerides, urea & zinc • For very grossly haemolysed samples no results will be reported
Inappropriate sampling site	specimen taken from drip arm	• Increased drip analyte, e.g. glucose, K^+, Mg^{2+} • Dilutional effect
Incorrect container or anticoagulant	No enzyme inhibitor EDTA tube (red or yellow) or transferring blood from one tube to another	• Low glucose • Increased K^+ • Decreased Ca^{2+}, ALP, Mg^{2+}
Lipaemia	specimen taken after a fatty meal	Decreased Na^+

***One strategy that can be used to great advantage is to order more tests at one time on one sample rather than ordering one or two tests on many samples.**

CHAPTER 3-HOSPITAL BLOOD BANK

3.1 GENERAL INFORMATION:

The Hospital's Blood Bank (BB) is housed within the main laboratory at Muthaiga. The BB plays a vital role in ensuring that correct blood components are available for patients and follows policies and guidelines established by the National Blood Transfusion Services (NBTS) of Kenya.

3.2 FUNCTIONS OF THE BLOOD BANK

The hospital blood bank is responsible for:

- Timely response to requests for blood components
- Checking pre-transfusion samples and requests
- Ensuring that the blood products have been tested for HIV 1 &2, Hepatitis B and C, T. pallidumand Malaria
- Assessment of immunological compatibility between donor and recipient
- Providing blood components that are requested in the amount asked for
- Providing advice on the suitable blood component for each clinical condition
- Safe handling of blood components until collection from the Laboratory
- Inventory and stock management
- Interactions with the clinicians
- Laboratory investigation of transfusion reactions

The hospital BB obtains blood from the National Blood Transfusion Services whenever possible. It also obtains blood from voluntary, non- remunerated donors especially where platelets are required as these are prepared from fresh non-refrigerated blood.

The BB supplies the following blood products: Whole blood (fresh/stored), packed red cells (PRBC), fresh frozen plasma (FFP), cryoprecipitate (CRYO) and platelet concentrates (PC) obtained from single units of donated blood. **Clinicians requesting blood and blood products are asked to request for the specific product required and the amounts in line with the GCH andNBTS guidelines.**

3.3 BLOOD/BLOOD PRODUCT REQUESTS:

Requests for blood and blood products should be submitted on a properly filled pink blood request sheet, in duplicate providing the following details:

- Name of patient
- UHID No
- Date of Birth
- Ward
- Dr's Name
- Clinical diagnosis
- Blood Group (if known)
- Hemoglobin (if known)
- Details of previous transfusion reaction if present
- Type of blood product to be supplied and the amount
- Degree of urgency with which the blood is needed.

The collection of the specimen for grouping and crossmatching should be by the attending doctor, nurse or phlebotomist. The staff must certify on the requisition form that the specimen obtained is from the correctly identified patient. The following terms are used to denote the urgency of the request. Clinicians are requested to be as accurate as possible in their use of the terms.

i. **Routine** Type/Cross match requests should be received in the laboratory at least 12 hours prior to the scheduled surgery. Samples drawn from the patients up to 7 days prior to surgery may be used only if the patient does NOT have a history of transfusion within the preceding 90 days. Information on prior transfusions MUST be filled in the pink sheet with an indication of dates if within the previous

90 days. If the patient has been transfused within the preceding 90 days the prior surgery, blood MUST be drawn within 72 hours of the scheduled surgery.

ii. **ASAP (As Soon As Possible) blood demands:** Standard cross match procedures is utilized. Priority will be given over routine requests and blood is normally available within 60 minutes if the patient has no irregular antibodies and blood of the suitable group is available in the BB.

iii. **Urgent Requests:** The full cross match procedure will be started immediately upon receipt of the specimen. Normally, the cross match will be completed in approximately 30 minutes providing the patient has no irregular antibodies.

3.4 EMERGENCY RELEASE

The requesting physician or designee will sign an emergency release form to obtain release of uncross matched blood. A full cross match procedure will be started immediately upon receipt of the pre-transfusion blood sample. (**NOTE:** A sample of blood MUST be obtained from the patient **BEFORE** any transfusions are started.)

3.5 TESTING FOR TRANSFUSION TRANSMITTED INFECTIONS

All blood and blood products that are for transfusion are tested for HIV 1 and 2, Hepatitis B and C, Treponema pallidum and Malaria parasites using recommended testing methods and reagents.

3.6 COMPATIBILITY TESTING.

All donor blood and patient blood is ABO and Rh D grouped. Cross-matching is performed for each unit of blood to be issued. For cross-matching, red cells from compatible ABO and Rh donor are tested against the recipient's plasma to determine compatibility. Test results are valid for 72 hours after time of specimen collection. After 72 hours, a new specimen must be collected and submitted for testing.For blood products not requiring cross-matching (FFP, PC, CRYO) ABO compatible products will be issued. In case of Rh D negative females, the products issued will be from Rh D negative ABO compatible donated blood.

3.7 ISSUING BLOOD.

Blood/blood products will only be issued to an adequately trained individual. Counter-checking of the report of patient's blood group, pack number, crossmatch results and disease screening results will be done by the person collecting the unit and the technologist on duty.
In general, only one unit of blood will be issued at a time for any given patient in the ward. Exceptions to this policy will be dealt with on a case-by-case basis (e.g. emergency release or for patients in the operating room).

Cross-matched blood will be reserved in the Blood Bank for only 72 hours after the time of specimen collection unless specific instructions to the contrary are given to the laboratory.Once FFP is ordered, the ward will be notified to collect it from the laboratory once it has been thawed. It must be collected **IMMEDIATELY**. Any thawed units that are not transfused immediately after thawing must be returned to the laboratory for storage in the fridge (not re-frozen) for up to 24 hours. If not used within this time the unit will be discarded.

Any unused units of blood must be returned to the laboratory within 30 minutes of issue (except for units issued to the operating room (OR) and kept in the OR fridge). Units returned after this timeframe must be destroyed.

3.8 THERAPEUTIC PHLEBOTOMY

This will be performed on appointment only. Therapeutic phlebotomy must be accompanied by a doctor's request for the same, providing the following details:

- Patient name, age, sex
- Diagnosis/reason for phlebotomy

- Amount of blood to be withdrawn
- PCV/HCT cut-off value
- Requesting physician's signature and legibly printed name
- Clinic from which the consult originated

Therapeutic phlebotomy that requires concurrent administration of intravenous fluids or for patients who have circulatory or respiratory complications will NOT be undertaken in the laboratory.

3.9 TRANSFUSION PROCESS

- Two nurses will counter check the patient identity, full name and birth date with patient's wrist band.
- Start transfusion within 30 minutes of issue and transfuse blood in compliance with nursing standard operating procedures (SOPs). The requisite monitoring of the patient should be performed according to nursing SOPs.
- Blood must be completely transfused into a patient within 4 hours of issue (a National Blood Transfusion Services (NBTS) policy). If a patient has a medical condition that necessitates slower transfusion e.g. avoiding cardiac overload in a patient with cardiac failure, then the blood bank should be informed beforehand so that the unit can be aliquoted appropriately.
- After transfusion is complete, the empty bag must be returned to the laboratory. The original "Medical Record" copy will be posted in the patient chart.

3.10 RETURN OF UNUSED BLOOD PRODUCTS FROM WARDS

Blood that has been returned to the blood bank shall not be made available for reissue unless the following conditions have been met:

- The unit of blood has not been out of the Blood Bank for more than 30 minutes.
- The temperature of the blood has not exceeded 8C.
- The unit of blood has not been penetrated or entered in any way.
- The integral pilot segments are still attached to the unit.

Blood that has been returned to the Blood Bank from the OR must have been stored in the OR fridge the whole time and should not have attained a temperature greater than 8˚C. The blood must not have been out of the OR fridge for more than 15 min.

3.11 INVESTIGATION OF A SUSPECTED TRANSFUSION REACTION

In the event of a transfusion reaction, or a suspected reaction, the following steps must be performed.

1. Discontinue the transfusion immediately.
2. Maintain the patient's IV-line access with normal saline.
3. Notify the attending doctor.
4. Follow the nursing protocol for transfusion reactions.
5. Check the clerical information to ensure the patient was receiving the correct blood.
6. Inform the laboratory about the possible transfusion reaction.
7. Collect the following samples immediately and send to the laboratory.
8. Use the opposite arm to withdraw the following blood samples:
    a. Blood sample in EDTA-purple top tube (2 mL)
    b. Blood sample in plain tube; check the colour of plasma for haemolysis (5 mL)
    c. A sample of the first voided urine
9. Bring the samples to the laboratory, along with the remaining blood in the blood bag, together with the attached transfusion set and all the empty bags of transfused units
10. Ensure that the Laboratory requisition form for investigation of a transfusion reaction is filled in appropriately

Ensure that correct labeling of the samples is done.

An additional sample for culture may be required especially if bacterial contamination of the unit is suspected.

The results of the tests will be communicated to the ward in a full report of the investigations. The Hospital Transfusion committee will be furnished with a report.

CHAPTER 4: HAEMATOLOGY

The following section discusses general aspects of haematology testing, selection and interpretation of some of the tests.

4.3 FREQUENTLY REQUESTED (ROUTINE) HAEMATOLOGY TESTS

4.3.1 BLOOD COUNTS

This is one of the most requested tests. It is ordered most often in the investigation of non haematological disorders especially when secondary effects on blood cells are suspected. It is invaluable in the investigation of a primary haematological disorder.

Complete blood counts provide information on the cellular elements of the blood: white cells, red cells and platelets and, in addition, the haemoglobin, mean corpuscular volume (MCV), mean corpuscular haemoglobin (MCH), mean corpuscular haemoglobin concentration (MCHC) and other red cell and platelet parameters.

4.3.2 THE PERIPHERAL BLOOD FILM

This is a smear made from the peripheral blood and stained to allow microscopic examination of the blood cells. It is useful for examining the morphology of the blood cells and helps detect abnormal cells in the blood and parasites (e.g. malaria parasites). Thus, the peripheral blood film is an important indicator of haematological and other diseases and complements the findings in the blood counts. It should be performed in many cases where the blood counts are abnormal. It may, however, be relatively difficult to interpret and can be of limited specificity depending on the abnormality that is found.

4.3.3 RETICULOCYTE COUNT

Reticulocytes are immature red blood cells, typically composing about 1-2% of the red cells in the human body. They provide a good indication of the erythropoietic activity of the bone marrow. Their measurement is useful in the evaluation of anaemia and can give an indication of whether the anaemia is likely due to increased blood loss, destruction or sequestration in the body or underproduction in the bone marrow. The reticulocyte production index(RPI) is an important step in understanding whether the reticulocyte count is appropriate or inappropriate to the level of anaemia present. The reticulocyte count can be used to monitor the treatment of anaemia.

4.3.4 Interpretation of Reticulocyte Counts

Reticulocyte Count	Associated Conditions
High	Haemolytic conditions, recent blood loss, anaemia responding to treatment, hypersplenism
Low	Haematinic deficiency, anaemia of chronic disease, aplastic anaemia, red cell aplasia, chemotherapy, malignancy involving the bone marrow, erythropoietin production problems

4.3.5 ERYTHROCYTE SEDIMENTATION RATE (ESR)

This is the measurement of the rate at which red blood cells sediment in a period of 1 hour. It is a non-specific measure of inflammation, and of limited use as a screening test in asymptomatic patients. Currently other tests such as the C-reactive protein (CRP) are considered superior to this test. It tends to be higher in female adults as compared to males. The ESR also tends to rise with age and is also higher in pregnancy. Some drugs may cause a rise in the ESR (e.g. contraceptive pills, corticosteroids)

4.3.6 CAUSES OF RAISED AND LOWERED ESR

ESR	Associated conditions
High	Acute & chronic infection, autoimmune diseases (rheumatoid arthritis, rheumatic fever, SLE), cancers (lymphoproliferative diseases, multiple myeloma, other cancers), temporal arteritis
Low	Polycythaemia, sickle cell anaemia, hereditary spherocytosis, congestive heart failure

4.4 LABORATORY APPROACH FOR THE INVESTIGATION OF ANAEMIA

Anaemia can be defined as a condition in which the amount of haemoglobin is below the lower reference limit with respect to the age, sex and geographic location of an individual.

Investigations for anaemia should take into consideration the clinical features of the patient.

4.4.1 SCHEMATIC APPROACH TO LABORATORY INVESTIGATION OF ANAEMIA

4.4.2 INVESTIGATION OF HYPOPROLIFERATIVE ANAEMIA IN A CHILD

CBC & RETICULOCYTE

- Low reticulocyte count
  - Hypoproliferative anaemia (See 4.4.2)
- Elevated Reticulocyte count
  - **Consider:**
    Blood loss (see 4.5)
    Haemolysis, (see 4.4.4)
    Hypersplenism
    Response to Haematinic therapy

SPECIFIC MORPHOLOGIC CHANGES ON BLOOD FILM

Microcytes+/- hypochromia	Macrocytosis+/- hypersegmentation	Leukocytosis+/- thrombocytosis+/- basophilia polycythemia+/-	Thrombocytopenia, Leukopenia or Pancytopenia	Blast cells with leucocytosis/ pancytopenia
Iron deficiency anemia Anemia of chronic disease Thalassemia trait	Liver disease Vitamin B12 or Folic acid deficiency Medication effect	Reactive leukocytosis CML (Myeloproliferative disorder - uncommon)	Bone marrow suppression due to viral infection, HIV, drugs or irradiation Aplastic anemia/ Fanconi's anaemia/ Diamond Blackfan anaemia/ Shwachman Diamond syndrome	Acute leukemia Malignant lymphoma
Serum Iron / Serum Ferritin/ other iron studies Hb Analysis (Bone marrow iron assessment)	Liver function tests Vitamin B12 & folic acid assays Bone marrow studies	Bone marrow and trephine biopsy Cytogenetic analysis (PHI JAK2, V627G mutation, del (5q))	Bone marrow suppression due to viral infection, HIV, Drugs or Irradiation Aplastic anaemia/Fanconi's anaemia/Diamond-Blackfan anaemia/Shwachman Diamond syndrome	Bone marrow Immunophenotyping (flow cytometry or immunohisto-chemistry cytogenetic analysis

KEY

- Features present
- Presumptive diagnosis
- Possible Diagnostic / confirmatory tests

4.4.3 TESTS FOR HAEMOGLOBIN DEFECTS

Haemoglobin defects are broadly classified into two:

i. Abnormal structural variants –e.g. Hb S, Hb C, Hb D, Hb G, Hb E
ii. Thalassaemias (Alfa, beta, delta beta) due to an imbalance in the globin chain synthesis

Investigations for structural haemoglobinopathies include:

- Blood count, reticulocyte count and film examination – may provide suggestive features of haemoglobinopathye.g. sickle cells, target cells
- Sickling test – be positive in Hb AS (trait) and Hb SS; false positives in infants may occur
- Haemoglobin electrophoresis – will provide qualitative information on types of haemoglobin and structural variants; it can also be used to determine whether there is a deficiency of any normal form of hemoglobin, as in Thalassemia's
- Is-electric focusing (IEF) – a sensitive method to detect hemoglobin and structural variants; it can also be used to determine whether there is a deficiency of any normal form of hemoglobin, as in Thalassemia's
- Automated high performance liquid chromatography (HPLC) – sensitive qualitative and quantitative method for identification of haemoglobinopathies
- Hb solubility – detects the presence of HbS
- Tests for detection of unstable haemoglobins
- Tests for detection of Hb M
- Tests for detection of altered affinity Hb M

Investigations for Thalassaemias include:

- Blood count (including red cell indices and reticulocyte count) and film examination
- Hb A2 and Hb F quantification
- Haemoglobin electrophoresis
- Globin chain electrophoresis
- IEF
- HPLC
- DNA analysis

4.4.4 TESTS FOR HAEMOLYTIC DISORDERS (RED CELL MEMBRANE AND ENZYME DEFECTS)

Haemolysis is the abnormal breakdown of red blood cells (RBCs), either within the blood vessels (intravascular haemolysis) or elsewhere (usually the spleen, sometimes the liver or in the bone marrow) (extra vascular).Haemolytic disorders may be hereditary or acquired. The table below shows the hemolytic disorders.

The hemolytic disorders.

Figure 3: Suggested scheme for Investigation of a hemolytic disorder.

HEREDITARY HEMOLYTIC CONDITIONS	ACQUIRED HEMOLYTIC DISORDERS
Membrane defects i. Hereditary spherocytosis ii. Hereditary elliptocytosis iii. Hereditary ovalocytosis iv. Hereditary stomatocytosis **Enzyme defects** i. G6PD deficiency ii. Pyruvate kinase deficiency iii. Haemoglobinopathies	i. Parasitic infections (malaria) ii. Immune haemolytic anaemia (IHA); HTR, AIHA, PCH iii. Microangiopathic/mechanical HA, HUS, TTP, DIC iv. Paroxysmal nocturnal haemoglobinuria v. Toxins vi. Sepsis vii. Drug induced HA

KEY

G6PD	Glucose 6 Phosphate Dehydrogenase
AIHA	Autoimmune Haemolytic Anaemia
DIC	Disseminated Intravascular Coagulopathy
HA	Haemolytic Anaemia
HTR	Haemolytic Transfusion Reaction
HUS	Haemolytic Uraemic Syndrome
PCH	Paroxysmal Cold Haemoglobinuria
TTP	Thrombotic thrombocytopenic purpura

Suggested Schematic Representation for Investigation of a Haemolytic Disorder

The figure below outlines the suggested scheme for the investigation of a haemolytic disorder:

Evidence for Hemolysis:

- Blood count (including red cell indicies) and blood film examination
- Reticulocyte count
- Biochemical tests – bilirubin (unconjugated); haptoglobulin/haemopexin;
- Urine – Hb, hemosiderin, urobilinogen

Spherocytes or Suspected HS

DAT +ve – followed by other tests for IHA

OFT/glycerol lysis test – followed by other tests

Specific features

Investigation for: SCD, HUS, DIC, TTP etc

Other red cell morphology or

- G6PD deficiency screening
- Red cell enzyme analysis
- PNH testing
- Tests for unstable Hb etc

KEY

DAT	Direct Antiglobulin Test (Coombs test)
DIC	Disseminated Intravascular Coagulopathy
HB	Haemoglobin
HUS	Haemolytic Ureamic Syndrome
OFT	Osmotic Fragility Test
PNH	Paroxysmal Nocturnal Haemoglobinuria
TTP	Thrombotic Thrombocytopenic Purpura

4.5 INVESTIGATION OF A BLEEDING DISORDER

Bleeding disorders occur when the ability of the blood to clot is impaired. They may be hereditary or acquired the latter being more common. The table below shows the more common bleeding disorders.

Table 4 The bleeding disorders

Hereditary Bleeding Disorders	Acquired Bleeding Disorders
• Haemophilia A, B (FVIII, FIX) • von Willebrand's disease • Other factors – fibrinogen, VII, II, X etc. • Hereditary thrombocytopaenias and platelet function disorders • Hereditary vascular disorders	• DIC • Liver disease • Vitamin K deficiency • Anticoagulant use • Aspirin use • Massive transfusion • Thrombocytopaenia • Acquired vascular disorders

4.5.1 INVESTIGATION OF A BLEEDING DISORDER

Investigation of a bleeding disorder should start with a comprehensive history and physical examination. This is followed by laboratory investigations in a stepwise process, starting with simpler first-line tests (screening tests). Abnormalities detected in the first-line tests are then followed by more specific tests as is appropriate. In certain instances, the first-line tests may not show abnormalities despite significant family history of bleeding. Specific tests as appropriate would still be performed. The figure below shows a suggested scheme for the investigation of a bleeding disorder.

4.5.2 SCHEMATIC REPRESENTATION OF INVESTIGATION OF A HAEMOSTATIC DISORDER

Coagulation Screen (first-line test)
- Platelet count, FBC and film
- Prothrombin time (PT)
- Activated partical thromboplastin time (APTT)
- Thrombin time (TT)
- Bleeding time

ABNORMAL FIRST- LINE TEST | CONSIDER CLINICAL FEATURES

Secondary tests
- Factor assay – if suspect hereditary factor deficiency
- Liver function tests
- Platelet function tests
- Tests for the fibrinolytic pathway
- Renal function tests

4.5.3 INTERPRETATION OF COAGULATION TESTS

The table below shows the interpretation of coagulation screening tests (first-line tests).

TESTS					CONDITION
PT	APTT	TT	BT	PLT	
N	N	N	N	N	Normal haemostasis; mild/marked coagulation factor deficiency; fibrinolytic disorder
N	N	N	↑	LOW	Thrombocytopaenia
N	N	N	A	N	Platelet function disorder; vascular abnormality
↑	N	N	N	N	FVI deficiency; oral anticoagulation; Lupus anticoagulant; mild FII, V or X deficiency
N	↑	N	N	N	F VIII, IX, XI, XII; circulating anticoagulant (e.g. lupus); mild F II, V or X deficiency
N		N	A	N	von Willebrand's disease
↑	↑	N	N	N	Vit K deficiency; oral anticoagulants; F V, X or II deficiency; multiple factor deficiency e.g. liver failure; combine FVIII & V deficiency
↑	↑	↑	N	N	Heparin (large amount); liver disease; fibrinogen deficiency/ abnormality; hyperfibrinolysis
↑	↑	↑	A	LOW	Massive transfusion; liver disease
↑	↑	N or ↑	↑	LOW	Disseminated intravascular coagulation; acute liver disease

KEY

N Normal

A Abnormal

PLT Platelet count

↑ Increased

FIB Fibrinogen

BT Bleeding Time

F Factor

4.6 THROMBOTIC DISORDERS

Thrombotic disorders occur when the blood has a pathological tendency to clot. Thrombosis is caused by abnormalities in one or more of the following (Virchow's triad):

- The composition of the blood (hypercoagulability or thrombophilia)
- Quality of the vessel wall (endothelial cell injury)
- Nature of the blood flow (stasis, turbulence)

HEREDITARY THROMBOTIC DISORDERS (THROMBOPHILIAS)

These are largely due to:

- Deficiency of protein C, protein S, protein Z, or antithrombin
- Factor V Leiden mutation, which causes resistance to activated protein C (APC)
- Prothrombin 20210 gene mutation

ACQUIRED THROMBOTIC DISORDERS

These are mostly caused by:

- Atherosclerosis
- Cancer (promyelocytic leukemia; lung, breast, prostate, pancreas, stomach, and colon tumors)
- Heparin-induced thrombocytopenia
- Hyperhomocysteinaemia
- Severe infections (e.g. sepsis)
- Oral contraceptives that contain estrogen
- Stasis
- Tissue injury

4.6.1 INVESTIGATION OF INHERITED THROMBOTIC STATES

Clinical features and frequency of particular states amongst the population should guide the investigation. This should be a stepwise process, starting with simpler first-line tests:

- Protein C assay
- Protein S Assay
- Antithrombin
- Activated Protein C resistance (Factor V Leiden)
- Heparin Co-factor II assay

4.7 BONE MARROW EXAMINATION

This is the pathologic analysis of samples of bone marrow obtained by bone marrow biopsy (often called a trephine biopsy) and bone marrow aspiration. Bone marrow aspiration may be performed by a clinician who is conversant with the procedure. Bone marrow biopsy should be performed by a haematologist.

Indications for bone marrow examination:

- Cytopaenias: anaemia, neutropenia, thrombocytopenia or pancytopenia with no apparent cause
- Suspected myeloproliferative disorders
- Lymphoma
- Leukaemias
- Multiple Myeloma
- Staging of some paediatrics childhood tumours
- Investigation for splenomegaly

- Investigation for some infective diseases(cultures of the marrow should be also performed); e.g. disseminated fungal infection, TB, typhoid
- Storage disorders e.g. Gaucher's disease, Neimann Pick disease
- Follow-up for treatment of a haematological disorder.

Site for procedure:

- Children less than 18 months old - the tibial tuberosity are the ideal site.
- Children above 18 months - the ideal area are the posterior superior spine of the iliac crest. Other sites on the iliac crest may also be used but may be problematic in the younger age groups.
- Adults (16 years and over) - the preferred site is the posterior iliac crest, other areas on the iliac crest including the anterior superior iliac spine. Less suitable is the sternum.

Requesting bone marrow aspiration or trephine biopsy:

The laboratory should be contacted if bone marrow trephine biopsy is required or aspiration by a hematologist.

CHAPTER 5 - CLINICAL CHEMISTRY

5.1 INTRODUCTION

The Department aims to provide clinical users with a comprehensive service for biochemical tests which meets their needs for diagnosis and monitoring of conditions with a biochemical component. This involves provision of appropriate analytical tests of a high technical quality and with a satisfactory turnaround time, and provision of an interpretative service for these test results when required. Advice may be obtained on the selection of tests, planning of investigations, interpretation of results and therapeutic intervention, as appropriate.

Almost all biochemical results are numerical, therefore, users should be aware that many factors may influence these numbers in addition to the illness for which the patient is being investigated: such factors include:

- Age
- Sex
- Sampling times
- Diet
- Medication
- Biological (within-patient) variability
- Delay in analysis
- Analytical variability
- Interferences from drugs and diet

***We aim to provide advice on these matters, and to help us provide interpretation of results, relevant clinical details should be included on the request form.**

New tests may be introduced and others replaced, we aim to discuss potential developments with users. Urgent biochemistry requests will be fast tracked if they are clearly labeled as **URGENT.** Full details of the type of sample container required, special notes about collection, volumes, reference ranges and target turnaround times are listed on the APPENDIX 1.

5.2 OUR SERVICES

- Sample analysis (clinical and research)
- Consultation on data interpretation
- Assistance in new tests and preferred tests
- Send-out services for tests not offered in-house

5.3 REFERRED SPECIMENS

The special chemistry tests (endocrinology assays, Karyotyping, Inborn errors of metabolism screening tests) are outsourced from accredited laboratories within and outside the country. **Please provide clinical information with all outsourced tests request!**

5.4 DYNAMIC TESTS

The clinical pathologist should be contacted before all Dynamic tests are undertaken, or phone the laboratory directly and inform the staff.

The interpretation of these results requires a detailed patient information and diagnosis!

5.5 INTERPRETATIVE CLINICAL CHEMISTRY

Table showing some interpretations of common tests.

ANALYTE	MEDICAL RATIONALE	
	Increased in	**Decreased in**
• Alanine Aminotransferase (ALT), (SGPT)	• Viral hepatitis • Toxic hepatitis • Shock liver • Infectious mononucleosis • Alcoholic hepatitis • Polymyositis	• End stage liver disease • Renal haemodialysis and/or renal insufficiency
• Albumin	• Dehydration	• Renal disease i.e. nephrotic syndrome • Liver insufficiency with decreased albumin synthesis • Severe malnutrition • Acute inflammation • Chronic inflammation • Malignancy • Pregnancy • Burns
• Alkaline Phosphotase (ALP)	• Intra hepatic cholestasis e.g. liver infiltrated by carcinoma, leukaemia, tuberculosis, sarcoidosis, amyloidosis, fibrosis (cirrhosis) • Extrahepatic cholestasis, e.g. bile duct stone, neoplasms, atresia • Osteoblastic disease e.g. malignancy,secondary hyperparathyroidism, Paget's disease • Tumor • Pregnancy	Hypophosphatasemia is relatively rare inherited disorder which is either asymptomatic or may be associated with arthritis of the weight bearing joints.
• Amylase	• Pancreatitis • Perforated or penetrating peptic ulcer • Acute ethanol ingestion • Salivary gland disease (mumps, inflammation) • Obstruction of pancreatic duct • Severe renal disease • Macroamylasaemia • Stone in the common bile duct • Spasm of the common bile duct	• Extensive destruction of the pancreas • Marked hepatic insufficiency
• Aspartate Aminotransferase (AST), (SGOT)	• Hepatocellular destruction e.g. viral hepatitis, toxic hepatitis, shock liver, alcoholic hepatitis • Myocardial infarction • In vivo hemolysis • Musculoskeletal disease • Pulmonary infarction • Post-hepatic obstructive biliary tract disease	• End stage liver disease • Chronic haemodialysis • Pregnancy

• Bilirubin total	• Liver insufficiency • Extra-hepatic obstruction • Haemolysis • Neonate - from a variety of causes including neonatal physiological hyperbilirubinemia • Gilbert's disease	
• Calcium	• Hyperparathyroidism • Myeloma • Metastatic carcinoma • Sarcoidosis • Thiazide therapy • Thyrotoxicosis	• Hyperparathyroidism • Hypoalbuminaemia • Chronic renal disease • Acute pancreatitis • Prolonged or severe vitamin D deficiency (rickets, osteomalacia) • Sprue steatorrhea • Malnutrition
• Chloride	• Prolonged diarrhoea • Renal tubular disease • Hyperparathyroidism • Dehydration	• Prolonged vomiting • Burns • Salt-losing renal disease • Over hydration • Thiazide therapy
• Cholesterol	• Familial (hereditary) hypercholesterolemia • Biliary obstruction • Nephrotic syndrome • Hypothyroidism • Pregnancy	• Severe liver insufficiency • Malnutrition • Hyperthyroidism • Chronic anemia • Waldenstrom'smacroglobulinae-mia • Thyroiditis
• Creatinine Kinase (CK), (CPK)	• Myocardial infarction/injury • Progressive muscular dystrophy • Dermatomyositis • Rhabdomyolysis due to drug ingestion, hyper osmolality, autoimmune disease etc. • Delirium tremens • Convulsions (status epilepticus) • Crush syndrome • Hypothyroidism • Surgery • Severe exercise • Intramuscular injection	• Physical inactivity • Decreased muscle mass
• Creatinine	• Impaired renal function. Serum creatinine is a more specific indicator of renal diseases than is serum urea.	• Rare and not clinically significant.
• CO_2 content (HCO_3+H_2CO_3)	• Primary metabolic alkalosis e.g. vomiting gastric suction, diuretic therapy, hypokalaemia • Primary respiratory acidosis e.g. chronic pulmonary disease, airway obstruction, respiratory center depression,pulmonary emphysema	• Primary metabolic acidosis e.g. diabetic ketoacidosis, uremia, starvation, lactic acidosis, alcoholic ketosis, salicylate ingestion, psychogenic hyper ventilation arterial hypoxemia. 3rd trimester of pregnancy.

• Glucose	• Diabetes mellitus a. maturity-onset (Adult) b. Growth-onset (juvenille) • Pancreatitis • Endocrine disorders e.g. Acromegaly, Cushing's syndrome, thyrotoxicosis, phaeochromocytoma,hyperaldostero-nism • Drugs e.g. steroids, thiazides, oral con-traceptives • Chronic renal failure • Stress • IV glucose infusion • Post prandial	• Insulinoma • Adrenal cortical insufficiency • Hypopituitarism • Extra-pancreatic neoplasms • Severe disease • Ethanol ingestion • Drugs, e.g. sulfonylurea, salicylates, metformin, insulin etc. • Reactive hypoglycaemia a. Functional hypoglycemia b. Pre-diabetic hypoglycemia c. Alimentary hypoglycemia • Glycogen storage disease of the newborn (type 1)
• Glutamyl-transferase (GGT)	• .Hepatobiliary disease • History of ethanol ingestion • History of drug ingestion e.g. phenytoin, phenobarbital	
• Iron	• Haemochromatosis • Haemosiderosis of excessive iron intake • Decreased formation of RBC's e.g. thalassemia pyridoxine deficiency anaemia, a plastic anaemia • Haemolytic anaemias • Acute liver damage • Iron ingestion	• Iron deficiency anaemia • Late pregnancy • Chronic infections • Menstruation • Nephrosis • Kwashiorkor
• Lactase dehydrogenase (LD)(LDH)	• Haemolytic disease e.g. megaloblastic anaemia, sickle cell anaemia • Myocardial infarction • Malignancy e.g. leukaemia, lymphoma, metastatic carcinoma • Pulmonary infarction • Infectious mononucleosis • Progressive muscular dystrophy • Deliriumtremens • Trauma • Shock liver • Hepatic disorders	
• Lithium	• Lithium medication given in excessive amounts	
• Magnesium	• Renal disease • MgSO4 injection	• Alcoholism • Malnutrition • Malabsorption • Severe diarrhoea

• Phosphorus (inorganic)	• Renal insufficiency • Hypoparathyroidism • Childhood • Excess vitamin D intake • High calcium diet	• Hypoparathyroidism (primary) • Vitamin D deficiency • Malabsorption • Administration of glucose (hyperalimentation) • Hyperinsulinism • Loss of phosphate in urine e.g. Fanconi's syndrome • Insulin treatment of diabetes acidosis • Rickets • Sprue steatorrhea
• Potassium	• Renal glomerular disease • Adrenal-cortical insufficiency • Diabetic ketoacidosis • Excessive intravenous potassium therapy • Sepsis • Panhypopituitarism	• Renal tubular disease • Hyperaldosteronism • Malnutrition • Treatment of diabetic ketoacidosis • Hyperinsulinism • Metabolicalkalosis (with a compensatory intracellular acidosis) • Diuretic therapy • Gastro-intestinal loss
• Sodium	• Dehydration • Diabetes insipidus (loss of dilute urine) • Loss of hypotonic gastrointestinal fluids • Salt poisoning • Peritoneal dialysis with hypertonic glucose solution • Selective depression of sense of thirst • Skin losses (burns) sweating • Hyperaldosteronism • CNS disorders e.g. meningitis, stroke, trauma, neuro surgical procedures	• Dilutional hyponatremia e.g. cirrhosis, CHF, water overload • Depletional hyponatremia e.g. salt-losing nephritis, tubular lesions, diuretic therapy, adrenal insufficiency, gastro-intestinal loss, sweating • Syndrome of inappropriate ADH secretion • Delusional hyponatremia e.g. profound hyper-lipidaemia
• Total protein	• Dehydration • Chronic inflammation • Myeloma (IgG, IgA) • Sarcoidosis	• Overhydration • Hepatic insufficiency • Malnutrition • Myeloma (light chain) • Agammaglobulinemia • Malignancy • Nephrosis enteropathy • Protein losing enteropathy
• Trigicerides	• Familial hypertriglyceridaemia • Nephrotic syndrome • Diabetes mellitus • Pancreatitis • Pregnancy • Liver disease (Zieve's syndrome) • Excessive alcohol intake	• Malnutrition • Congenital a-beta-lipoproteinaemia

• Uric acid (urate)	• Gout • Renal failure • Ketoacidosis • Lactate excess e.g. after ethanol ingestion • Diuretic therapy • Lesch-Nyhan syndrome • Chronic lead nephropathy • Polycystic kidneys • Leukaemia, lymphoma, polycythemia • Diet (high-protein, high nucleoproteins) • Drugs (anti-uricosuric drugs, e.g. diuretics)	• Administration of ACTH • Administration of uricosuric drugs (high doses of salicylates, probeacid, cortisone, allopurinol, coumarins) • Renal tubular defects (Fanconi's syndrome, Wilson's disease)
• Urea nitrogen	• Impaired renal functions • Pre-renal azotemia, e.g. congestive heart failure shock etc. • Post renal-azotemia • Gastro-intestinal bleeding • High protein diet • Drugs:corticosteroids, tetracycline	• Pregnancy (third trimester) • Severe liver insufficiency • Overhydration • Malnutrition especially decreased protein intake • Lipoid nephrosis

5.6 TOXICOLOGY AND THERAPEUTIC DRUG MONITORING

For overdoses and drug screen; Samples of toxicology are analyzed at the Government Chemists laboratory while drug screens are done at a referral laboratory in Nairobi.

A request for a drug screen requires:

- At least 20 mL of urine
- Detailed patient information
- Clinical condition e.g. coma grade, fitting
- Current known prescribed drugs
- Overdose drug or toxin if known
- Urine specimen type i.e. voided or catheter

Blood samples taken for a drug screen/toxicology are of little use unless there is prior knowledge of the agent ingested!

5.6.1 A GUIDE TO DRUG OVERDOSE/TOXICOLOGY SCREENING

TEST	SPECIMEN/NOTES
ALCOHOL (ETHANOL)	1 yellow topped tube
AMPHETAMINE	20 mL urine in urine bottle with no preservative – send to lab immediately

THEOPHYLLINE	1 yellow-topped tube taken pre-dose (trough levels) or 8-12 hours post-dose (peak levels). Patients on IV infusions should be monitored in the first 12 hours. (Plus a baseline level if there is a likelihood of prior administration). Ideally the infusion should be stopped for 15 minutes before sampling. Patients with an acute overdose (serum levels >100 mg/L) or a chronic overdose require earlier intervention. Repeat serum theophylline and potassium levels every 2-3 hours
ANTIEPILEPTICS **PHENOBARBITONE** **PRIMIDONE** **ETHOSUXIMIDE** **VALPROATE** **PHENYTOIN** **CARBAMAZEPINE** **LAMOTRIGINE**	1 yellow-topped tube, except Lamotrigine (must be 1 red-topped tube_. Samples for all drugs taken pre-dose.
AMINOGLYCOSIDES **GENTAMICIN**	1 yellow-topped tube sent to Clinical Biochemistry
BARBITURATE SCREENING	1 red-topped tube
BENZODIAZEPINES **DIAZEPAM** **LORAZEPAM** **NITRAZEPAM** **NORDIAZEPAM** **TEMAZEPAM**	1 red-topped tube – contact lab for other benzodiazepines that can be assayed
BENZODIAZEPINE SCREENING	1 red-topped tube and 20 mL of urine collected at the same time with the serum
CANNABINOIDS	20 mL urine in preservative-free bottle
DRUGS OF ABUSE SCREENING **CANNABINOIDS** **COCAINE METABOLITES** **LSD** **OPIATES**	20 mL urine in preservative-free bottle. If there is a delay in sending sample to lab, check urine pH and record it on the request form. (Note: test the pH of a separate urine aliquot from that sent for lab analysis)
METHANOL	1 green-topped tube. Treatment: ethanol IV, or if level >0.5 g/L, haemodialysis
OPIATES	20 mL urine in preservative
ACETAMINOPHEN (PARACETAMOL)	1 yellow-topped tube. Sample at >4 hours after OD. If 24 hours have elapsed since OD paracetamol will likely have disappeared from serum except in massive OD. Transaminases and prothrombin time should be measuredin this case, although the effect on these is usually not maximal until day 3-4
SALICYLATE	1 yellow-topped tube. Sample at >4 hours after overdose. The prognosis for acute salicylate poisoning cannot be determined from serum concentration alone; clinical features, particularly impaired consciousness and the arterial pH must be considered. Acidaemia, regardless of class of acid-base abnormality, has a poor prognosis. Haemodialysis is the treatment of choice for severe intoxication (>750 mg/L) or less severe intoxication if there are complications.

5.7 BLOOD AND URINE ALCOHOL; SUBSTANCES OF ABUSE TESTING

Blood Alcohol:

Blood specimens for blood alcohol testing will be collected at the laboratory. The quantitative measurement of ethanol in blood is the preferred test in chronic alcoholism or in an acutely intoxicated patient. Note that the Laboratory does not undertake the analysis of samples for legal purposes relating to the Road Traffic/ Transport Acts.

Urine alcohol:

The result of a urine ethanol assay is expressed qualitatively as "positive" or "negative".

Testing for substances of abuse:

These are also known as Drug screen; Drug test; Substance abuse testing; Major uses of drug testing are to detect the presence of drugs prohibited by law such as cannabis, cocaine or heroin. Another major use of these tests is to detect the presence of enhancing steroids in sports. The sample usually requested is a random urine test, less often a venous blood sample. Drug-testing in blood samples is considered to be the most accurate way of telling if a person is intoxicated. Blood drug tests are not used very often because they need specialized equipment making it a costlier testing method.

The detection windows depend upon multiple factors such as the class of drug, the amount and frequency of use, age, overall health, body mass, metabolic rate and urine ph. In testing, the parent drug and metabolites are incorporated into the test. This is because the parent drug may only be detectable for a few hours after use (e.g. heroin and cocaine) but their metabolites are detectable for a several days in urine. In these situations, reporting of the longer detection times of the metabolites is done.

Requests for testing for substances of abuse must be clearly written. If blood is to be specifically tested this should be stated and the limitations above should have been carefully considered. The pattern of abuse changes from time to time and a few patients may abuse unusual or exotic drugs. The Laboratory staff will do their best to offer advice or help in circumstances which are not covered by these notes.

5.7.1 Approximate Detection Times of Some Common Drugs of Abuse in Urine*

Drug/Substance	Urine detection times
Alcohol	up to 1 day
Amphetamines (including MDMA, MDA)	1-3 days
Barbiturates	1-3 days
Benzodiazepines	1-3 days
Cannabis	up to 14 days
Cocaine	1-3 days
Codeine	1-2 days
Cyclizing	1-2 days
Dihydrocodeine	1-2 days
Heroin (morphine)	up to 1 day
Methadone	1-3 days
6-MAM	up to 1 day

***Note that detection times are only very approximate and are very dependent upon the dose, its frequency, route of administration and urine excretion/dilution.**

Important Reminders

- **Indicate which drugs you suspect.**
- **Submit both blood and urine.** (Some drugs are detected only in urine; others are detected only in blood. If you submit just one fluid, you will miss some important drugs)
- **For drugs whose absorption might be delayed, get a second serum specimen 2-3hours later.** (Such drugs include acetaminophen, salicylate, tricyclic antidepressants)

CHAPTER 6-MICROBIOLOGY AND INFECTIOUS DISEASES

6.1 GENERAL GUIDELINES FOR SPECIMEN COLLECTION

It is of critical importance that clinicians use the following guidelines for the proper collection and transport of specimens. All the diagnostic information from the microbiology laboratory is contingent on the quality of specimen received. Consequences of a poorly collected and/or poorly transported specimen include;

- Failure to isolate the causative microorganism
- Recovery of contaminants or normal microbial flora

This may be misleading, and results in improper treatment of the patient.To minimize contamination:

- Use strict aseptic technique when collecting the specimen
- Collect the specimen prior to initiation of antimicrobial therapy
- Collect the specimen from anatomic sites most likely to yield pathogens and least likely to yield contaminants

Tissue or fluid submitted for culture is always superior to material on swabs.

Provide complete information on specimen requisition forms including:

- Demographic details of patient
- Name of the clinician (contact details are very useful)
- Clinical diagnosis
- Type of specimen
- Specific site from which the specimen was collected
- Time of specimen collection
- Specific pathogens that are being sought (to optimize conditions for culture)
- Methods by which specimens were collected
- Whether the patient may be infected by pathogens known to be hazardous to lab personnel (please notify the lab prior to sending the specimens).

Deliver specimens to the lab as soon as possible.

Specimens that is unacceptable for microbiology testing:

- Unlabeled or Improperly labeled specimen.
- Specimens received in leaking, cracked or broken containers.
- Specimens with obvious (visually apparent) contamination.
- Specimens not appropriate for a particular test.
- Unpreserved specimen received > 12 hours after collection.

If a specimen is to be collected through intact skin, cleanse the skin first with 70% alcohol and wait for 60 seconds, followed by a second cleansing with a new 70% alcohol swab which is applied for 60 seconds before the specimen is collected.

6.1.1 GUIDELINES FOR PROPER SPECIMEN TRANSPORT

- Collect specimen in steady, sterile, screw-cap, leak-proof, containers with lids that do not create an aerosol when opened.
- All specimens should be transported to the lab promptly. Failure to do this may result in the death of fastidious organisms and in overgrowth by more hardy bacteria.
- If prompt delivery is not possible specimens should be refrigerated at 4- 8°C, with the following exceptions:
- CSF specimens should be kept at room temperatures unless they are to be cultured for viruses. If viral

cultures are requested an aliquot should be removed aseptically and stored at 4 – 8°C. Specimens that may harbor temperature sensitive organism such as Neisseria species should be left at room temperature.

6.1.2 SPECIMEN CONTAINERS

Bijous bottles; These are sterile, disposable culture collection and transport systems consisting of a plastic tube/vial containing transport medium to prevent drying of bacteria and maintain ph.

Calcium Alginate swabs (calgswabs); can be toxic for some strains of HSV, and may be toxic for some cell cultures. Useful for collection of cultures. Some lots of calcium alginate may be toxic to certain gonococcalstrains. Cotton swabs may also be used; however, some brands of cotton contain fatty acids that may be inhibitory for gonococci. Therefore, calcium alginate and cotton swabs should be used only if the specimen is inoculated directly onto growth media or transport in non-nutritive media containing charcoal to absorb or neutralize inhibitory material.

Cotton swabs; Residual fatty acids may inhibit some bacteria and spp. If cotton is glued or spun to wooden applicator stick, wooden stick may inactivate HSV and interfere with some identification tests.

Dacron swabs; Useful for collection of viral and group A streptococcus specimens, and for the isolation of Neisseria gonorrhoeae.Nasopharyngeal and urethrogenital swabs. Flexible wire shafts and small tips provide easier specimen collection.

Sterile screw-cap jars: Useful for collection of urine, sputum, stool, bronchi alveolar lavage, and biopsy specimens. If biopsy specimen is small, add small amount of sterile 0.85% NaCl to jar. Never place biopsy specimen in formalin, or wrap in gauze.

Sterile Petri dishes/slides: Use for hair or scraping specimens. Tape Petri dish/slides securely prior to transport.

Sterile tubes; Screw-cap glass, plastic tubes, or sterile vacutainer tubes without additives

Useful for collection of sterile fluids, bronchoalveolar lavage, drainage, or brush specimens.

COLLECTION AND TRANSPORT OF SPUTUM

REQUIREMENTS

Wide necked, lean proof, screw capped container
Gloves – Lab coats, request forms, 10% Jik 5% Phenol, marker pen

PROCEDURE

a. Give the patient a clean (need not to be sterile): dry, wide-necked, leak proof container, and request him or her to cough deeply to produce a sputum specimen.

CAUTION: When sputum, specimen is being collected, adequate safety precautions should be taken to prevent the spread of infection and organisms.
NOTE: The specimen must be sputum not saliva, sputum is best collected as spot and the following morning soon after the patient wakes and before any mouth wash is used. If the patient is a young child it is not possible to obtain sputum, gastric washings can be used for isolation of M.tuberculosis but not for other respiratory pathogens. Gastric washing should be neutralized as soon as possible after collection.

b. Label the container and fill in the request form as described in 7.0of this manual.
c. If pneumonia or bronchopneumonia is suspected deliver the sputum to the laboratory with little delay as possible as it requires culturing as soon as possible. Mark the specimen HIGH RISK. If the specimen is for the isolation of M.tuberculosis, ensure it is delivered to the laboratory within 2 hours or kept at 4 degree Celsius until delivery is possible. Refrigeration slows down the multiplication of commensals. If MDR is suspected-sputum should be collected in a sterile container.

COLLECTION AND TRANSPORT OF PUS FROM WOUNDS, ABSCESSES, BURNS AND SINUSES

a. Using a sterile technique, aspirate or collect from a drainage tube up to 5ml of pus.Transfer to leak-proof sterile container. If pus is not being discharged, uses a sterile cotton wool swab to collect a sample from the infected site.

b. Label the specimen and as soon as possible deliver it with a request form to the laboratory.

COLLECTION AND TRANSPORT OF FAECES (STOOL SPECIMENS).

a. Give the patient clean, dry disinfectant free bedpan or suitable wide-necked container in which to pass a specimen. The container needs not to be sterile. Ask the patie3nt to avoid contaminating the feaces with the urine.

b. Transfer a portion (about a spoon full) of the specimen especially that which contains mucus, or blood into a clean, dry leak-proof container.

NOTE: If the specimen contains worms or tapeworm segments transfer those to separate those to a separate container and send them for identification.

c. Write on the color of the specimen and whether it is formed, semi formed, uniformed or fluid. Report also if blood .mucus, worms or tapeworms segments are present. Rectal swabs: if is not possible to obtain faeces,collect a specimen by inserting a cotton wool swab into the rectum for about 10 seconds care should be taken to avoid contamination of specimen with bacteria from the anal skin.

d. Label the specimen and send it with a request from to reach the laboratory within 2 hours. If the specimen contains blood, or amoebic, dysentery is suspected. Deliver to the laboratory as soon as possible. The specimen must be fresh to demonstrate actively motile amoebae NOTE. If there is delay and salmonella, shigella or vibrio is suspected, insert the swab in a container of sterile early – Blair transport medium, breaking of the swab stick to allow the bottle to be replaced tightly. The species survive well in early Blair medium for up to 48 hours. If cholera is suspected, transfer about 1ml of specimen into 10ml of stile alkaline peptone water, label and send to reach the laboratory within 8 hours of collection.

6.1.3 VIRAL TRANSPORT SYSTEM

The diagnosis of viral infections by culture relies on the collection of proper specimens, proper care to protect the virus in the specimens from environmental damage, and use of an adequate transport system to maintain virus activity. Collection of specimens with swabs that are toxic to either virus or cell culture should be avoided. A variety of transport media have been formulated, beginning with early bacteriological transport media. Certain swab-tube combinations have proven to be both effective and convenient. Of the liquid transport media, sucrose-based and broth-based media appear to be the most widely accepted and used. Viral transport media allow for more effective use of viral culture and culture enhancement techniques for the diagnosis of human viral infections. The viral transport media provides room temperature transport of viruses, chlamydiae, and mycoplasma and urea plasma. The need for refrigerated storage and transport on ice is eliminated, decreasing the prevalence of specimen rejection and eliminating messy specimen transport. Other advantages are; clinical pathogens are maintained, while growth of contaminants is inhibited; Suppression of bacterial and fungal contamination by incorporating antibiotics in the medium; preserves viruses and chlamydia for long-term frozen storage; utilizes a phenol red pH indicator to ensure medium integrity at the time of specimen collection.

6.1.4 BLOOD CULTURES

Blood culture is the gold standard of microbiological diagnosis of septicemia, infective endocarditis and PUO (pyrexia of unknown origin). Once positive results are available, use of antimicrobial agents can be directed accordingly. Every precaution should be taken to minimize the percentage of contaminated blood cultures. The phlebotomist should wear disposable gloves and prepare a sterile pack prior to performing the procedure. To reduce the chance of contaminating organisms from the skin, the following guidelines are provided:

Site selection

- Select a different site for each blood sample.
- Avoid drawing blood through indwelling intravenous or intra- arterial catheters.

Site Preparation

Vigorously cleanse the venipuncture site with 70%isopropyl or ethyl alcohol; Cleanse the skin with 70% alcohol and wait for 60 seconds, followed by a second cleansing with a new 70% alcohol swab, which is applied for 60 seconds before the specimen is collected. Allow alcohol solution at the venipuncture site to dry. Do not touch the venipuncture site after preparation and prior to phlebotomy.

Preparation of Blood Culture Bottles

Remove the lid without touching the top of the bottle. If contamination occurs during removal of the lid, alcohol can be used to disinfect the top. Do not use iodine on the tops of the Bract/Alert bottle.

Collection of Blood

- Withdraw blood using a needle and syringe into the vein, and withdraw blood. The closed system consisting of a vacuum bottle and double needle collection tube can be used. Do not change needles before injecting the blood into the culture bottle.
- After the blood is inserted into the blood culture system, mix well to avoid clotting.
- Use a new needle if the veins are missed initially.
- Add sufficient volume of blood to attain a 1:10 ratio of blood to medium. Each bottle can accommodate maximum 10 mL; this volume is also recommended for optimal recovery of organisms.

Specimen Volume

The volume of blood is critical because the number of organisms in most bacteraemias is low, especially if the patient is on antimicrobial therapy.

- Children: Infants and children have a lower total blood volume. 3 to 5 mL of blood should be drawn per venipuncture; the minimum requirement is 1 mL per blood culture.
- Adults: 20 mL of blood distributed between two aerobics bottles. If clinical diagnosis indicates an aerobic infection the initial volume drawn should be 20 mL of blood, which is divided equally between the aerobics and anaerobic bottle.

Optimizing Blood Cultures

Blood obtained from one venipuncture site defines one blood culture, regardless of the number of bottles filled.

- Timing: Blood must be cultured as early as possible in the course of a febrile episode, and before administration of antibiotics.
- Number of sites sampled: One venipuncture is rarely sufficient. Sampling from two venipuncture sites is sufficient. It may be of benefit to collect sets over 3 consecutive days in FAN bottles (with charcoal which will absorb antibiotics) in patients on antimicrobial therapy or if fungaemia is suspected. Multiple blood cultures should never be drawn from the same venipuncture site. Sampling of two or three blood cultures should be spaced at intervals of at least 20 minutes, to detect transient or intermittent bacteremia.

The HACEK group of bacteria (Haemophilus aphrophilus, Actinobacillus actinomycetemcomitans, Cardobacterium hominies, Eikenella corrodens and Kingella kingae) are associated with bacterial endocarditis. Brucella spp. and Bartonella henselae can also cause blood stream infections.

For isolation of fastidious organisms, blood culture bottles must be held for 21 days. Please note blood for fungal culture should be collected directly into a FAN BacT/Alert blood culture bottle. Blood for serology assays (e.g. antigen detection) should be collected in a red vacutainer tube without additives. If blood/bone marrow is sent for tubercle bacilli submit 5 mL blood/50 mL 13A Bactec media (obtainable from microbiology laboratory)

This culture allows for positive culture in 10-14 days.

6.2 TISSUE BIOPSIES, ASPIRATES/SWABS OF ABSCESSES, WOUND SWABS AND FLUID SAMPLES

General principles:

The character of the lesions being cultured may limit the usefulness of these cultures. Lesions connecting with skin, mucosal surfaces or the GIT will be encumbered by the presence of indigenous microflora. Therefore, for meaningful cultures results, surgically obtained tissue sample, aspirates of closed abscesses and aliquots of pus/fluid are preferred rather than swabs.

Swabs of superficial skin ulcers, the skin surface of sinus tract and from open abscesses often yield mixed bacterial flora, which do not reflect the organisms of true infectious significance. Therefore, every effort should be made to sample from deeper aspects of the lesion with careful avoidance of the contaminated tissue surface.

When anaerobic bacteria are expected, the specimen should be inoculated into anaerobic transport medium (available on request from the laboratory) at the bedside and promptly transported to the lab. When looking for fastidious or unusual organisms, please notify the lab so that the cultures can be appropriately set up and held for prolonged incubation as needed.

Providing the lab with the location of the type of the wound/abscesses/tissue as well as clinical diagnosis is useful because it can hasten recognition of specific pathogens associated with the type of infection. Specimens can be transported to the laboratory using a sterile container (e.g. green lidded container or red top tube).

6.2.1 ASPIRATES

- Decontaminate the collection site with 70% alcohol and then with 10% povidone –iodine solution and allow to dry.
- Aspirate the deepest part of the lesion using a 3-5 mL syringe and a 22-23-gauge needle.
- If the initial aspirate fails to yield material, inject non-bacteriostatic NaCl subcutaneously and repeat the aspiration attempt.
- The aspirating syringe can be used as the transport container if the needle is removed and the needle is capped.
- Alternatively, transfer the aspirate into a sterile container and transport to the laboratory promptly.

6.2.2 SWABS

- Obtain the swab from deep within the wound (separate the wound margins with sterile gloves)
- Dry swabs are unacceptable specimens. The swabs should be delivered to the laboratory without delay.

6.2.3 DEEP LESIONS

- Disinfect the surface as above
- Aspirate the deepest part of the lesion
- If the specimen is collected in theatre, submit a portion of the wall of the lesion
- Submit the specimen in a sterile container and deliver to the laboratory without delay

6.2.4 TISSUE BIOPSIES

- Specimen collected at the time of surgery/endoscopy should be submitted in sterile containers without formalin added. The specimen may be kept moist by addition of non-bacteriostatic saline.

6.2.5 ULCERS

- Clean the surfaces as previously described and allow to dry
- Remove overlying debris
- Curette the base of the ulcer and collect exudates from the ulcer using a syringe or swab

6.2.6 STERILE FLUIDS

Sterile fluids (e.g., synovial, culdocentes and serous cavity fluids) should be collected so as to avoid contamination with commensal flora. Submit the fluid in one of the following:

- A capped syringe (with the needle removed)
- Sterile tubes
- Blood culture bottles (in the same sample to medium ratio as recommended for blood, that is, 10 mL for adult bottles and 3 mL for pediatric bottles)

Swabs are the least desirable sample for culture of fluids and are discouraged.

6.3 SPECIMEN COLLECTION FOR MYCOLOGY

6.3.1 TISSUE:

- Collect tissue biopsies aseptically
- Put the specimen in a sterile container and do not add saline
- Transport to the laboratory immediately or else refrigerate at 4-8°C.

6.3.2 PURULENT EXUDATES AND FLUIDS

- Pus should be aspirated aseptically from closed lesions where possible
- Transport to the lab rapidly, in the syringe used for aspiration with the needle removed to avoid accidental inoculation, or in a sterile container
- Wound dressing can be examined for granules in suspected mycetoma or actinomycosis
- Pus swabs are generally inferior specimens and should be avoided
- Other swabs (e.g. vaginal, penile, throat, oral) should be transported to the laboratory as soon as possible
- Aspirate bone marrow into a sterile heparinized tube or into a FAN BacT/Alert blood culture bottle. Also, make two smears for special fungal stains and submit this together with the culture specimen
- Pleural, peritoneal and joints fluids should be collected aseptically in a clean sterile container without or with heparin to prevent clotting

6.3.3 OCULAR SPECIMENS

6.3.3.1 GENERAL CONSIDERATIONS

A variety of different techniques are used to collect specimens from different parts of the eye by an ophthalmologist as described below.These specimens should be inoculated into culture/transport media and slides prepared at the time of collection (available from the lab). A conjunctiva swab should accompany a specimen collected by invasive method, serving as a control (the conjunctiva is constantly contaminated by environmental/ocular adnexal flora). Send prepared smears/inoculated media promptly to the lab.

6.3.3.2 SPECIAL CONSIDERATIONS:

- **N. gonorrhea:** use a Dacron swab to collect the specimen; inoculate directly onto New York City agar
- Fungi: inoculate into appropriate media
- Anaerobes: inoculate into anaerobic transport media/directly onto appropriate media
- Parasites: consult the lab for further instructions on specimen submission
- Viruses: use Dacron /cotton swab with non-wood shafts and send specimen in appropriate viral culture transport media
- Chlamydia: use calcium alginate swabs for specimen collection
- Mycobacteria: consult the lab for further instructions on specimen submission

6.4.4 SPECIMEN COLLECTION

6.4.4.1 CONJUNCTIVA/LID MARGINS

INOCULATION OF MEDIA

- Roll appropriate swab over conjunctiva before topical medicines are applied
- A separate swab should be used for each eye
- Inoculate media directly and send promptly to the lab

SMEAR PREPARATION

- Instill 1-2 drops of local anaesthetic
- Use a spatula to scrape across lower tarsal conjunctiva
- Smear the material in a circular area (1cm diameter) on at least 2 clean glass slides and immerse in 95% methyl alcohol for 5 minutes.
- Repeat on the other eye.

6.4.4.2 CORNEA

- Obtain conjunctival specimen (as above) prior to obtaining corneal scrapings
- Instill 1-2 drops of local anesthetic
- Obtain 3-5 scraping/cornea, using an appropriate spatula, from areas of ulceration (the advancing edge of the ulcers)/suppuration
- Inoculate appropriate media and then prepare 2-3 slides, as described above
- Transport promptly to the lab

6.4.4.3 INTRAOCULAR FLUIDS

- Use a needle aspiration technique to collect fluid
- Inoculate appropriate media and then prepare 2 smears by spreading a drop of fluid over clean glass slides
- Conjunctival swabs should be collected along with the fluid to assist in interpretation of the culture results

6.4.4.4 ORBITAL OR PRESEPTAL CELLULITIS

- Clean the skin with alcohol and iodine
- Collect purulent material with a syringe and needle from an open wound or make a stab incision in the absence of an open wound to collect pus
- Inoculate media and then prepare slides
- Transport promptly to the lab

6.4.4.5 LACRIMAL GLAND

Collect swab specimen of purulent discharge

6.4.4.6 LACRIMAL SAC

Express or needle aspirate purulent material for media inoculation and smear preparation.

6.4.4.7 LACRIMAL CANALICULUS

Compress the inner aspect of the eyelid to express pus and inoculate media/prepare slides

6.4.4.8 MYCOLOGY SPECIMENS

Corneal scrapings must be transferred onto an agar plate (blood, dextrose). Smears of the scraped material should also be prepared on clean alcohol-flamed slides.

6.4.5 STOOL SPECIMENS

6.4.5.1 GENERAL PRINCIPLES

- Specimen should be collected in sterile containers that will not leak; green cap universal containers are suitable providing the lids are adequately closed.
- Containers should be clean and dry. The presence of water and urine can result in inaccurate interpretation.
- Patient's clinical details should be provided. This is essential especially where special techniques are required to culture. E.g. cholera or opportunistic parasites in HIV.
- Stool specimen should be transported to the lab as soon as possible,within 1 hour.
- 2-3 specimens can be submitted on separate days to improve isolation of pathogens and identification of parasites (see parasitology section).
- Do not submit more than 1 stool specimen for a particular patient in a 24-hour period.
- Stool specimen should be preferably being sent in the first days of admission. Diarrhea developing after this period is often nosocomial in origin e.g. antibiotic associated diarrhea. In these cases, a Clostridium difficile toxin assay may be more appropriate.

6.4.5.2 CHOLERA

- As this is notifiable disease, early diagnosis is essential.Please state clearly that cholera is suspected, as this requires special culture media.If stool is not immediately available, rectal swabs can be submitted in transport medium (see below).Direct liaison with the lab will ensure an early diagnosis.

6.4.5.3 RECTAL SWABS

Rectal swabs may be submitted where stool cannot be obtained or when screening for carriage is required.These swabs must be delivered promptly to the lab.

Method of collection:

- Moisten the swab in sterile transport medium
- Insert swab gently into the rectal sphincter 2-3 cm and rotate to sample anal crypts. Remove swab and check for visible faeces
- Deliver to the lab promptly

6.4.5.4 RECTAL BIOPSY

Specimen requiring microbiological culture must NOT be submitted in formalin. Submit these specimens in a sterile, sealed container in normal saline to prevent drying.

6.4.5.5 CLOSTRIDIUM DIFFICILE TOXIN ASSAY

Patients suspected of antibiotic-associated diarrhea should have stool submitted for C.difficile toxin assay. Please stipulate the request clearly. Stool samples must be freshly collected and kept refrigerated or on ice.

6.4.5.6 PARASITES

- Request for parasite examination must be clearly stated
- Specimen for amoeba must reach the lab within 30 min of sampling
- Substances such as bismuth, antibiotics and anti- motility agents can interfere with parasite detection
- Please provide clinical information. Opportunistic parasites in HIV require special staining procedure
- Several specimens may be needed to identify the presence of parasites due to cyclical variations. Usually up to 3 specimens on different days can be submitted.
- Sellotape - for pinworm, apply sellotape to perianal area, detach and apply to glass slide.

6.4.6 URINE SPECIMENS

6.4.6.1 GENERAL PRINCIPLES

Urine is normally a sterile body fluid; if specimens are not collected adequately, contamination by normal flora of the perineum, urethra and vagina can occur.

Please stipulate the following: Mode of specimen collection e.g. suprapubic aspirate, catheter; Date, Time and Clinical diagnosis.

- Specimen> 8 hours old will be rejected unless refrigeration has occurred
- Early morning urine specimens have the best yield
- Do not force fluids prior to urine collection as this will dilute the colony counts and result in potential misinterpretation
- Specific instructions must be given to patients for collection of clean-catch midstream urine

6.4.6.2 FEMALES

- Wash hands well
- Cleanse the urethral opening and vaginal vestibule with sterile gauze pads soaked in normal saline.
- Hold the labia apart during voiding.

6.4.6.3 MALES

- Cleanse the penis, retract the foreskin and wash with normal saline.
- Circumcised men are not required to cleanse the peri-urethral area before voiding.
- Keep the foreskin retracted while voiding.

6.4.6.4 BOTH MALES AND FEMALES

- Should pass a few milliliters of urine into the toilet bowl, DO NOT STOP THE FLOW, and then collect the midstream portion of urine into a sterile container.
- Never submit the urine in bedpan or urinal.
- No dipstick testing should be done on urine specimens that are to be submitted for culture.
- Urine should be delivered promptly to the lab, within 2 hours. Urine should be refrigerated if delay in transportation is unavoidable.

6.4.7 ADDITIONAL SPECIMENS (IN SITUATIONS WHERE MSU IS NOT AVAILABLE OR APPROPRIATE)

6.4.7.1 CATHETER SPECIMENS

STRAIGHT CATHETERS

- These may be used to obtain urine directly from the bladder. This is not routine as organisms can be introduced during the procedure. If performed, aseptic technique must be followed.
- Suprapubic aspirate may be required in infants.
- Urine collected from indwelling catheters is often contaminated. Use 70% alcohol to clean the catheter port. Aspirate with a sterile needle and syringe then transfer to a sterile container.
- The lab will not process Foley catheter tips, as these have no value due to extensive contamination by colonizing flora.

PARASITES AND CASTS

- Request for parasites and/or casts must be clearly stated.
- These specimens must arrive promptly.
- Urine for Schistosomahaemotobium is best collected from midday to 2pm when egg excretion is highest.

RENAL TUBERCULOSIS

- Urine specimens for AFB (acid fast bacilli) and TB culture may be useful adjunct in the diagnosis of renal TB.
- Up to 5 early morning specimens may be required.
- Microscopy for AFB is not a sensitive tool. However, TB culture has a better yield providing sufficient specimens are submitted; 80% - 90% sensitive if at least three early morning specimens are cultured.
- Culture may take 4-6 weeks and clinicians should ensure the follow up of these results.

6.4.8 URINE AND RECTAL SWABS FOR MYCOLOGY

Collect urine in a sterile container and send to the lab without delay. Stool specimens for fungal culture are rarely useful as the significance of their presence in such a contaminated material is controversial.

6.4.9 CSF SPECIMENS

- CSF is collected for the diagnosis of meningitis and is usually obtained by lumbar puncture. Subdural taps and ventricular aspiration may also be used.
- The lumber puncture should be performed using an aseptic technique. The patient is appropriately draped and an area overlying the lumber spine is disinfected using the same skin preparation as required for collection of blood cultures (see section on blood culture specimen collection).
- All microbiological CSF specimens should be collected in red-topped vacutainer tubes.
- Each bacteriological and fungal test requires at least 0.5 mL of CSF, although larger volumes are preferred. The specimen should be transported to the lab promptly and processed as soon as possible. If delay in processing is unavoidable, the specimen should be kept at room temperature. When requesting latex testing on CSF, please ensure that you have specified whether bacterial or cryptococcal latex detection is required.
- Processing CSF for mycobacteria: the number of mycobacteria present in CSF is small;therefore, a large volume of specimen (3-5 mL) is necessary for growth of the organisms in culture.
- Specimensfor viral serological assays require at least 0-2 mL (molecular assays 0.5 mL and viral at least 2 mL). Please do not request viral studies unless a good clinical history including age of patient is provided. Requests for specific virus testing are preferred. If delay in processing is unavoidable, specimens should be ideally stored at 4oC.
- For CSF queries relating to biochemistry, please contact the biochemistry lab.

6.4.10 NASOPHARYNGEAL AND RESPIRATORY TRACT SPECIMENS

6.4.10.1 COLLECTION OF THROAT/PHARYNGEAL SPECIMENS

Routinely used for the diagnosis of group a streptococcal pharyngitis. Please notify the lab if looking for any other organisms, e.g.Neisseria gonorrhoeae.Do not attempt any throat swabs if the epiglottis is inflamed, as sampling may cause serious respiratory obstruction.

- Depress tongue gently with tongue depressor.
- Extend sterile swab back and forth over tonsillar pillars and behind the uvula; avoid touching the cheeks, tongues, uvula, or lips
- Sweep the swab back and forth over the posterior pharynx, tonsillar areas, and any inflamed or ulcerated areas to obtain a sample.
- Send the swab to the lab in suitable transport media, as soon as possible.

6.4.10.2 NASAL SWABS

Submitted primarily for the detection of nasal staphylococcal carriers.

- Insert a sterile swab into the nose until resistance is met at a level of the turbinate.
- Rotate the swab against the nasal mucosa.
- Repeat the processes on the other side.

6.4.10.3 NASOPHARYNGEAL SWABS

Submitted primarily for the detection of carriers of n. meningitides and to diagnose B. pertussis. Carefully insert a flexible wire calcium alginate – tipped swab through the nose into the posterior nasopharynx, and rotate the swab (keep the swab near the floor of the nose).

6.4.10.4 NASOPHARYNGEAL ASPIRATES

- Nasopharyngeal aspirates are appropriate specimens for the detection of Bordetella pertussis, Corynebacteriumdiptheriae, Neisseriagonorrhoeae, and N. meningitidis. The lab must be alerted before the specimen is submitted.

6.4.10.5 SINUS ASPIRATES

If anaerobic infection is suspected, then kindly submit the specimen in anaerobic transport media.

Collection and Transport of Effusions (Synovial, Plural, Pericardial, Ascetic and Hydrocele fluids

After aspiration, a aseptically dispense the fluid as follows.

2-3ml into a dry sterile screw-cap tube or bottle. 9ml into a screw-cap bottle which contains 1ml of sterile trisodium citrate 3g/1 solution mix the fluid with the anticoagulant.

Label and as soon as possible deliver the samples with a request form to the laboratory.

6.4.10.6 ORAL CULTURES

Used for the detection of yeasts or fusospirochetal disease

- Rinse mouth with sterile saline.
- Wipe the lesion with sterile gauze.
- Swab or scrape areas of exudation or ulceration.

6.4.10.7 SPECIMEN SUBMITTED FOR VIRAL DETECTION

- Swabs and tissue specimen for viral culture should be placed in viral transport medium.
- Separate specimens from the same site are needed if bacterial or fungal cultures are also required.
- Liquid specimens; bronco-alveolar lavage fluid or nasopharyngeal aspirates, do not need to be submitted in viral transport media.
- For swabbing, only fiber-tipped swabs (e.g. cotton, Dacron polyester) should be used. Calcium alginate swabs are contra-indicated.

6.4.10.8 UNUSUAL PHARYNGEAL PATHOGENS

Specimen Collection and Transport for Isolation of Gonococcal Pharyngitis

- Swabs of tonsillar and the posterior pharynx should be collected.
- Swabs should only be used if there is a very short interval between collection and plating.
- The swab should be placed in a non-nutrient transport medium such as Stuarts transport medium.
- If a delay in transport to the lab is inevitable, the specimen should be inoculated to New York City agar at the time of collection and be put into a 5-10% environment.
- Candle jars or commercial systems that provide a capnoeic atmosphere are recommended.

SPECIMEN COLLECTION AND TRANSPORT FOR ISOLATION OF BORDETELLA PERTUSSIS

- The preferred specimen is a nasopharyngeal swab, or ideally aspirates of the nasopharynx or bronchus.
- Small-tipped calcium alginate or Dacron swabs are suitable for collection of specimens.
- Rayon or cotton swabs should be avoided, as they contain fatty acids that are toxic to B. pertussis.
- Ideally the specimen should be inoculated at the bedside into charcoal-based media, and two slides should be prepared.
- If this is not feasible, the swabs should be sent to the lab in transport medium such as Amies' medium with charcoal.
- All specimens should be delivered to the lab as soon as possible.

6.4.10.9 SPECIMENS FOR LOWER RESPIRATORY TRACT INFECTIONS

Specimens include sputum, tracheal aspirates, bronchial washings, bronchial brushes, bronchial biopsy specimens, bronchoalveolar lavage fluid, trans-tracheal aspirate, lung aspirate and lung biopsy specimens.

Sputum should preferably be collected early in the morning, before the patient has eaten or taken medication.The patient should be given clear instructions on how to produce a specimen:

- The patient should rinse his/her mouth with water, then take two deep breaths, holding the breath for a few seconds after each inhalation and then exhaling slowly. This should produce a specimen from deep in the lungs.
- The patient should hold the container to the lower lip and gently release the specimen from the mouth into the container and avoid spills. An adequate specimen should contain at least 5 mL of sputum.
- The specimen container must be tightly capped and clearly labeled. This should be done by the health care professional requesting the specimen, and should include the relevant clinical information as well as the diagnostic tests required. If for example, TB microscopy and culture are required, then this should be noted clearly; otherwise only microscopy will be done.
- The specimen should be transported to the lab as soon as possible after collection. If there is a delay in transport to the lab, for example from an outside clinic, then specimens should be refrigerated. DO NOT FREEZE SPECIMENS!
- Aerosols containing TB bacilli may be produced during collection of a sputum specimen, and is advisable to collect specimens away from other people and in a well-ventilated area.
- The person supervising the sputum collection should stand behind the patient to avoid breathing in any aerosols that may be created when the patient coughs.
- If infection with Legionella pneumophilla is suspected, a urine specimen should be submitted for urine antigen testing.
- Sputum induced with hypertonic saline has a greater sensitivity for the diagnosis of Pneumocystis jiroveci than expectorated sputum.
- If bacterial pneumonia is suspected, concomitant blood cultures are recommended.

SPECIMENS FOR MYCOLOGY

Collect specimens appropriately, as described, into sterile containers without additive and send to the lab for processing.If delay is suspected, refrigeration is not advisable for suspected histoplasmosis.

6.4.11 MYCOLOGY SPECIMENS

6.4.11.1 SKIN

- Clean the area of the skin from which a specimen is to be collected with 70% alcohol.
- Scrape the active peripheral edge of the lesion with a scalpel or the end of the microscope slide.
- Place the scales in a sterile Petri-dish/container or between two slides taped together and send to the lab.

6.4.11.2 NAILS

- Cleanse nails with alcohol wipe.
- Scrape deeply using a blunt scalpel to obtain recently invaded nail tissue.
- Initial scrapings are usually contaminated and thus should be discarded.
- Clippings of whole thickness of affected nail can be obtained using nail clippers.
- Place the scrapings and nail clippings in a sterile petri-dish/container and send to the lab.

6.4.11.3 HAIR AND SCALP

Send basal portions of the infected hair and scalp skin scrapings of the affected areas to the lab in a dry, sterile container to prevent the growth of contaminating fungi.

6.4.12 COMMON MICROBIOLOGICAL TESTS

Test	Tests performed	Special instructions
Blood		
Blood culture aerobic	Daily	10ml in BacT/Alert bottle
Blood culture anaerobic	Daily	10ml in BacT/Alert bottle
Blood culture paediatric	Daily	20ml in BacT/Alert bottle
Blood culture FAN	Daily	10ml in BacT/Alert bottle
CSF		
CSF MC & S Microscopy (cell count, Gram stain) Culture and sensitivity	Daily	Red top tube
CSF bacterial latex	Daily	Red top tube
Sputum		
Sputum MC & S	Daily	Sterile universal container
Sputum microscopy (Ziel Nelsen stain)	Daily	Sterile universal container
ENT		
Ear MC & S	Daily	Sterile swab

Throat and nose swab culture	Daily	Sterile swab
Tracheal aspirate MC & S	Daily	Sterile universal container
Bronchial washing MC & S	Daily	Sterile universal container
Bronchial alveolar lavage MC & S	Daily	Sterile universal container
PUS		
Pus or fluid MC & S	Daily	Sterile swab/Sterile universal container
Tissue MC & S	Daily	Sterile universal container
Genital		
Genital microscopy and culture	Daily	Sterile swab
Urine		
Urine MC & S	Daily	Sterile universal container
Dysmorphic cells/casts in urine	Daily	Sterile universal container
Faeces		
Faeces MC & S	Daily	Sterile universal container
Faecesparasite investigation	Daily	Sterile universal container
Clostridium difficile toxins A & B	Outsourced	Sterile universal container
Mycology		
Blood culture FAN	Daily	10ml in BacT/alert bottle
Pus or fluid microscopy and culture	Daily	Sterile universal container
Tissue	Daily	Sterile universal container
Sputum microscopy and culture	Daily	Sterile universal container
Bronchial washing and tracheal aspirate microscopy and culture	Daily	Sterile universal container
Bronchial alveolar lavage microscopy and culture	Daily	Sterile universal container
CSF microscopy (India ink)and culture	Daily	Red top tube
CSF Cryptococcal latex	Daily	Red top tube
Mucormycosis tissue specimen/nasal scraping	Daily	Sterile universal container

Skin scrapings	Daily	Sterile universal container
Serology		
Rapid HIV	Daily	Clotted blood in red top tube
RPR	Daily	Clotted blood in red top tube
TPHA	Outsourced	Clotted blood in red top tube
VDRL	Daily	Clotted blood in red top tube
TMX	Daily	Clotted blood in red top tube
Parasitology		
Urine bilharzia	Daily	Sterile universal container
Malaria smear	Daily	EDTA tube
Malaria antigen	Daily	EDTA tube/fingerpick

6.4.13 COLLECTION OF SPECIMENS TO TEST FOOD PRODUCTS, SURFACES AND UTENSILS FOR THE PRESENCE OF FOOD POISONING ORGANISMS

6.4.13.1 COLLECTION OF SPECIMEN AND SAMPLING PLAN

FOOD

Samples should be transported to the lab on ice in a cooler bag. The samples should where possible be freshly prepared food or in the case of an outbreak all the foods that were eaten by the people at the time. Store the samples for a minimum practicable period under such conditions that changes in composition are prevented or kept to a minimum. Samples must be placed in suitable sterile containers or plastic bags

ENVIRONMENTAL SWABS

Samples are collected by swabbing from a group of four articles or components of the same kind, and must not include samples taken from any article or components of any other kind. If the number of articles or components of one kind sampled is less than four, the sample is collected from such lesser number.

- Sterile absorbents cotton wool-tipped swabs are used for collection of bacteriological specimens.
- McCartney bottles containing 10 mL of quarter-strength Ringer's solution are required.
- For the purpose of sampling, two 10 mL bottles of Ringer's solution are used for each article of group of articles or component or group of components.

The following procedures are followed during sample collection:

AREA TO BE SWABBED

- For cups, glasses and other drinking utensils the samples are collected from the exterior and interior surface to a depth of at least 12mm from the top of the rim.
- For spoons and ice-cream scoops the samples are collected from the entire inner and outer surface of the bowl.
- For plates, saucers, bowls and the like, over an area of as nearly as possible 2500mm2 of the surface as this comes in contact with food.
- For all other articles or components, samples are collected from all parts of the surface likely to come into contact with the food.

METHOD OF SWABBING

The defined area or areas of each article or component or group of articles or components from which a sample is to be taken shall be swabbed as follows:

- Firstly, with a swab moistened with Ringer's solution from one of the McCartney bottles, after the excess moisture has been expressed from the swab from the inside of the bottle before removal; swab an area of 50x 50mm. After sampling the swab is immediately immersed in the same bottle and the protruding portion of the stick above the neck of the bottle is broken off and the screw top replaced;
- Secondly and immediately afterwards, over the same defined area or areas with a dry swab which shall be placed in the second bottle of ringer's solution.

In each case the bottles are suitably marked to identify the sample or component from which the sample was taken and to distinguish the wet from the dry swab.The person who collects this samples must, at the time of sampling, record in duplicate the name and address of the premises, the number of articles or components in the group sampled, the time of taking sample and the identification mark on each bottle.

As soon as possible after sampling, the bottles containing the swabs together with the duplicate copy of the particulars mentioned in paragraph above must be delivered to the lab within 3 hours after the collection of samples. Where this is not possible, samples must be maintained at a temperature not higher than 5oC.

6.4.14 CULTURE OF HAEMODIALYSIS WATER

6.4.14.1 TEST FREQUENCY AND TIMING

Routine: Collect samples for monitoring purposes for at least monthly.
Repeat: Collect repeat samples when bacterial counts exceed the allowable levels. If culture growth exceeds permissible standards re-culture the water system, usually weekly, until acceptable results are obtained.
Ad hoc: Collect when clinical indication suggests pyrogenic and/or septicaemia complications following a specific request by the clinician and/or the infection control practitioner.

Timing
If the water system has been treated with formaldehyde or chlorine for sanitization, flush the system completely before collecting samples. Drain and flush storage tanks and water lines (15-minute flush time recommended) before collecting samples. Culture water and dialysis fluid monthly unless a greater frequency is dictated by historical data at your institution.

6.4.14.2 SPECIMEN COLLECTION

DIALYSIS WATER

- Sample at a point immediately past the water production system (e.g. reverse osmosis system, deionization units etc.)
- Sample wherever storage of water occurs e.g. storage tank used to store water from the water treatment system.
- Sample just before the water enters the dialysis machine or central portion

DIALYSATE

- Following dialysis, collect fluid from dialyzer.
- Single-pass system: Sample where dialysate leaves dialyzer.
- Re-circulatory system: Sample at the periphery of the re-circulation chamber containing the coil dialyzer.

VOLUME
Collect a minimum of 5 mL from each sample point into sterile containers.

METHOD
At each point of collection, open the valve and allow the water to flow for a minimum of 2 minutes before the sample is collected.

TRANSPORT AND STORAGE

Culture dialysis and dialysate within 30 min of collection, or store at 4-5oC and culture within 24 hours.

6.4.15 CULTURE OF CONTINUOUS AMBULATORY PERITONEAL DIALYSIS FLUID

A. Culture dialysis and dialysate within 30 min of collection, or store at 4-5oC and culture within 24 hours.

6.4.15.1 SPECIMEN COLLECTION AND TRANSPORT

Enclose dialysate bag in a larger plastic bag, into a disposable plastic pan, and transport it to the lab.

Transport:

- For immediately delivery transport at room temperature
- For delayed delivery (> 1hr after collection) refrigerate but do not freeze.

6.4.16 CULTURE OF INTRAVASCULAR DEVICES

6.4.16.1 SPECIMEN COLLECTION

To prevent contamination by skin microorganisms and antibiotic ointment clean the skin insertion site with an iodophore and alcohol prior to the removal of the cannula. Remove the cannula in an aseptic manner once the alcohol has dried. If purulence of the catheter exit site is evident, send pus for culture and gram stain.

LONG CATHETERS

Two portions of these catheters should be sent for culture: the distal intravascular tip and the proximal transcutaneous segment. Each segment should be approximately2-3 cm long.

SHORT CATHETERS

The cannula is cultured in its entirety following removal of the hub. To remove the hub, use sterile scissors or snap off steel needle with a sterile hemostat.

6.4.16.2 SPECIMEN TRANSPORT

Transport catheter tips in a sterile container. If tips are cut into a proper length; there is no need to bend them for insertion into transport container. Culture tips within 2 hr. of collection to prevent desiccation of microorganisms.

6.4.17 SURVEILLANCE CULTURES FOR IMMUNOCOMPROMISED HOSTS

6.4.17.1 SELECTIVE CULTURE OF FUNGI

Collect urine, stool, and or pharyngeal specimens for routine cultures

6.4.17.2 GENERAL SURVEILLANCE CULTURE

A selection of moist sites, e.g. hand, maxilla, throat, rectum, vagina, and the central line site, is chosen by agreement of the care givers involved with the project. All swab specimens are submitted in the transport media that are routinely employed for other swab cultures.

6.4.17.3 SELECTIVE BOWEL CULTURE

A rectal swab or stool specimen is collected and submitted as if for routine culture.

6.4.18 INFORMATION AND SAMPLES REQUIRED TO DIAGNOSE INFUSATE RELATED SEPSIS

INFORMATION

- The time of administration of the infusate
- The diagnosis of the patient's underlying condition
- The blood culture result from the patient taken at the time of suspected infusate related sepsis
- If any additives were administered with the infusion or added directly to the bag by the pharmacist
- How the bag was stored prior to administration
- The procedure followed when setting up the bag: how the skin was disinfected and if the operator was wearing gloves
- Was the bag inspected for leaks?

SPECIMENS

The following specimens are required and must be sent to the lab in separate sealed plastic bags:

- The full IV set and infusate in use at the time when the patient spiked a temperature
- Any Add-in-line bags
- A blood culture taken through the line
- A blood culture taken from a remote venipuncture site
- A swab from each catheter hub
- A swab from the skin site
- The transcutaneous segment of the catheter
- The distal portion of the catheter
- Any other bags with the same batch number.

6.4.19 TESTING OF DISINFECTANTS

Kindly furnish us with the following information:

- Nature of the disinfectant (e.g. QAC, aldehyde, hypochlorite, etc.)
- Test(s) required (e.g. in-use disinfectant test)/range of organisms required for the challenge.
- Dilution factor
- Contact times
- Volume must be appropriate; 100ml/1 gram.

6.4.20 COLLECTION OF MILK SAMPLES

- Samples are collected using aseptic technique and are transferred to sterile samples containers taking precautions to prevent the contamination of samples.
- Sample containers are surrounded by crushed ice or other suitable refrigerant which comes to contact with the container and is capable of reducing the temperature and maintaining it until delivered to the lab at a temperature not exceeding 5-6oC.
- For milk or cream it must be clearly stated whether it is pasteurized, certified, raw or ultra light treatment (UHT)
- Volume of the sample must be sufficient i.e. 100ml.

6.4.21 COLLECTION AND SENDING OF INFUSATES

- Aerobic BacT/Alert bottles must be used.
- The plastic cap on the BacT/Alert bottle must be removed, and the rubber band must be disinfected with a preptic/webcol swab containing 70% isopropyl alcohol into which the infusate will be inoculated.
- When alcohol has evaporated, under strictly aseptic conditions, inoculate 10 mL of infusate sample into the BacT/alert bottle.
- Each bottle must be accompanied with its own slip containing: all relevant data regarding the sender (name, address, contact numbers, account details); the batch number, the date, and the bar code sticker, which must be removed from the bottle.
- The slips must be attached to the corresponding bottle.
- The bottle must be sent to the lab in a safe container to prevent breakage of BacT/alert bottles

6.4.22 COLLECTION OF WATER SAMPLES FOR LEGIONELLA CULTURES

SAMPLE CONTAINERS

Samples of water shall be collected in glass, polythene or similar containers able to hold a volume of 1000-2000 mL. If used previously they shall be cleaned, rinsed with distilled or mains tap water then pasteurized either with flowing hot water (>70oC) or steam for a period of not less than 5minutes or autoclaved at 121 +/- 1oC for 15 min. Small sterile containers with a capacity of 100-250 mL may be used to collect slime, deposit or sediment. Access to sample points may be difficult, which can make the use of glass containers unsafe because of possible breakage.

SAMPLING IN THE PRESENCE OF BIOCIDE

If the water sampled contains or is thought to contain an oxidizing biocide, then add an excess of an inactivating agent to the container before or at the time of sampling. Chlorine and other oxidizing biocides are inactivated by the additional of potassium thiosulfate or sodium thiosulfate to the container at a concentration of 100mg per liter. For other biocides, the addition of a universal neutralizing agent is not practicable.

SAMPLING FREQUENCY

There are no standards as yet, but it is suggested that each system should be sampled at least annually and preferably every 6 months. If a cooling system has not been in use during the winter months it should be tested before start-up..

SAMPLE VOLUME

A minimum of 500 mL is required, but 2000 mL is preferred if the water is clear.

TRANSPORT TO THE LAB

Samples should be transported to the lab at less than 18°C but not less than 6°C, and be protected from heat and sunlight. Deliver the samples to the lab as soon as possible, preferably within 1 working day but not more than 2. The maximum time interval between taking the sample and processing is 5 days. If the sample to the lab is delayed the specimens should be stored at 6°C.

If delay in transport to the lab is unavoidable the samples may be frozen preferably at -70°C, samples must not be repeatedly frozen and thawed as this will affect the viability of the organisms in the sample.

SAMPLE LABELING

Each sample must be labeled with an indelible marking pen. A list of the samples, sender's details plus an order number must be included in the consignment to facilitate the clerking and processing of the samples. The time and date the sample is taken, plus if a biocide is present should be stated on the sample list.

6.4.23 COLLECTION OF DOMESTIC POTABLE WATER SAMPLE

A sterile container should be used.These containers are obtainable from the Infection Control lab. Alternatively; a glass bottle may be used that has been boiled for 5 minutes and then placed in an oven to air dry at 100oC.

- Allow the water to run for 5 minutes.
- Collect 100 – 200 mL in a sterile container.
- Ensure that the lid is securely in place and that the sample is not leaking.
- Place the sample in cooler bag with ice packs and transport to the laboratory without delay.
- Fill in laboratory test request document
- Supply address, phone and fax numbers.
- Samples may be delivered Mondays to Fridays from 0730 to 1500 hr.

Tests performed for microbiological portability are total plate count, coliforms and E.coli.

CHAPTER 7 - IMMUNOLOGY

The Immunology section of the lab provides a wide range of services for the investigation of autoimmunity, allergy, immunodeficiency, and infectious disease serology.

7.1 GENERAL GUIDE TO THE USE OF IMMUNOLOGICAL TESTS FOR COMMON CLINICAL SCENARIOS

Interpretation of results of immunology tests should be made in the context of clinical findings. Indiscriminate use of the tests will lead to a low positive predictive value e.g. if they are performed on patients who have little or no real clinical evidence of relevant disease. For example, do not request for autoantibodies, if you do not think there are real clinical grounds for suspecting the patient to have an autoimmune disease, as the results will not be helpful!

7.1.1 AUTOIMMUNITY

Rheumatoid arthritis (RA)

Rheumatoid factor (RF) is absent in 30-40% of patients with RA, and most patients with Juvenile Idiopathic Arthritis (JIA). The decision to refer or treat is made primarily on clinical grounds. High levels are associated with more severe disease or complications. However, RF cannot be used to monitor disease activity. Anti-CCP antibodies are a more specific test for RA but may be negative in 30% of patients.

Connective tissue disease (CTDs)

Anti-nuclear antibodies (ANA) are used to screen for CTDs; a negative result suggests that the diagnosis is unlikely. Positive results should be further analyzed for double stranded DNA antibodies (anti-dsDNA – More specific test for SLE) or antibodies to extractable nuclear antigens (ENAs), e.g. Ro, La, Sm, RNP etc.

7.1.2 IMMUNODEFICIENCY

Primary Immunodeficiency (PID) is rare but should be considered in patients with:

- Severe infection e.g. severe chicken pox
- Prolonged, unusual or difficult to treat infections
- Unusual infection and opportunistic infection e.g. staphylococcal pneumonia, atypical tuberculosis, pneumocystis
- Recurrent deep abscesses of the organs or skin
- Recurrent infection e.g. pneumonia, sinusitis, otitis media
- Infants under 6 months with failure to thrive

The most common PID is associated with antibody deficiency; patients usually present with recurrent respiratory tract infections caused by pneumococcus and Haemophilus influenza. Serum immunoglobulin levels and lymphocyte counts will be sufficient to exclude PID in many of these patients.

7.1.3 ALLERGY

The diagnosis of allergy is primarily clinical. Allergen-specific IgE testing should be used to support clinical findings; it cannot exclude or definitely confirm a diagnosis.

We use the UniCAP (ImmunoCAP) method for the allergen-specific IgE assay. 5 mL of blood in a plain vacutainer is required for each panel (food and environmental allergens). Results are semi quantitative; reported as classes 0 to 6 (from no specific antibodies detected, to very high antibody titre).

7.1.4 INFECTIOUS DISEASE SEROLOGY

We provide a comprehensive infectious disease serology service, that includes the detection of antibodies to viruses and bacteria as well as viral and bacterial antigens in serum. Most routine serology screening assays are performed by ELISA assay technology on a fully automated open platform system using Mini Vidas system, enabling the samples to be processed within 48 hours of receipt. Most of the assays are confirmed in- house using other automated or semi-automated procedures such as Gemini ELISA system.

Specific IgM can give information early, which may influence management. An IgM response is usually present early in the clinical course of some infections. Serology also gives a retrospective diagnosis, by detecting a rise in antibodies to the infecting agent. This may be particularly helpful when post infectious complications occur.

For detection of antibody in blood, 5 mL should be collected into a plain vacutainer as early as possible in the illness (acute phase), and another specimen collected during convalescence 10 to 14 days later, to detect a rise in antibody titre.

For viral serology requests not processed at GCH, please adequately fill the request forms by providing complete clinical information and date of onset of symptoms; otherwise the specimen will not be processed.

Routine screening is available for the following:

- Viral antibodies and antigens: Hepatitis A (IgG and IgM), Hepatitis B surface antigen (HBsAg), Hepatitis C antibody, (Measles IgG and IgM), HIV I & II antibodies
- Bacterial antibodies and antigens: Treponema pallidum (antibody), Toxoplasma gondii (total antibody and IgM), antistreptolysin/ASOT, Clostridium difficile toxins (A & B); H. pylori antigen tests, Weil Felix serology, Salmonella serology (antigen tests in stool and serum)

Special serology: Tests for other bacterial and viral antigens and antibodies which are not available in house are sent to various reference laboratories; e.g. Brucella serology, Lyme disease serology, Dengue fever serology, Yellow fever serology, CMV (IgM and IgG), EBV, HSV (I&II), Mumps, Rubella (IgM and IgG), VZV (IgG), Influenza A and B, RSV, Adenovirus, Hepatitis B core antibody (anti-HBc) and antibody to Hepatitis B surface antigen (anti-HBS - Immune status following vaccination)

7.2 USE AND INTERPRETATION OF COMMONLY REQUESTED IMMUNOLOGY TESTS

Test	Rationale
Anti-Nuclear Antibodies (ANA)	Screening for lupus or connective tissue disease; a negative result suggests that the diagnosis is unlikely. However, the test has low specificity for SLE. ANA can be detected in other connective tissue diseases such as RA, chronic active hepatitis, juvenile arthritis, and in systemic infections. All positives should be further analysed for antibodies to ds DNA and ENAs
Complement C3 and C4	Monitoring conditions associated with immune complexes, e.g. SLE, sub-acute bacterial endocarditis, systemic vasculitis and post-streptococcal glomerulonephritis. C3 and C4 raised – acute phase response Low C3, normal C4 – post-streptococcal glomerulonephritis, gram negative sepsis, Membranoproliferative glomerulonephritis, C3 nephritic factor Low C4, normal C3 – active SLE, cryoglobulinaemia C3 and C4 low – immune complex disease e.g. SLE, sepsis, severe liver disease (reduced synthesis)
C-Reactive Protein (performed in Clinical Chemistry)	Raised in bacterial infections and inflammatory diseases e.g. rheumatoid arthritis, vasculitides. It is not raised in active SLE unless there is associated serositis. Useful in distinguishing active SLE from infection (unless they co-exist), diagnosing infection in neutropaenic patients and for monitoring inflammatory conditions such as bacterial infections or systemic vasculitides
Cyclic citrullinated peptide	Present in about 70% of patients with RA. More specific than RF

Double stranded DNA antibodies (anti-dsDNA)	Should be performed if ANA is positive. Positive results support the diagnosis of SLE. A negative test does not exclude SLE. Levels may correlate with disease activity in SLE.
Extractable Nuclear Antigens (ENA)	Follow-up test to positive ANA Sm– suspected CTD/SLE; specific for SLE but found in only 20-30% of SLE patients. RNP – Suspected CTD/SLE; Found in mixed CTD & SLE. Anti-U1RNP abs have increased association with SLE in absence of anti-RNP70 abs; whereas high titre anti-RNP70 abs in the absence of anti-U1RNP are found in MCTD Ro/SS-A – Suspected CTD/SLE; Investigation of congenital heart block; Associated with Sjögren's syndrome. Also found in variants of SLE including subacute cutaneous lupus and neonatal lupus with congenital heart block La/SS-B– suspected CTD/SLE; Investigation for congenital heart block; found with anti-Ro in Sjögren's syndrome and SLE Jo-1 – Associated with inflammatory muscle disease (Polymyositis, dermatomyositis) Scl-70 – Suspected CTD/SLE; Foundin 20-40% of patients with systemic sclerosis
Immunoglobulins (IgG/M/A)	Suspected primary or secondary immunodeficiency. Polyclonal elevation of IgG may occur in HIV infection, connective tissue disease Polyclonal elevation of IgG and IgA may occur in chronic respiratory (TB) and gastrointestinal infection, rheumatoid arthritis, inflammatory liver disease Polyclonal elevation of IgG, IgA and IgM may occur in chronic infection Polyclonal elevation of IgA may occur in acute infection, inflammatory liver disease, rheumatoid arthritis Polyclonal elevation of IgM may occur in acute viral infection. Congenital infection Low immunoglobulin levels may be found in premature infants, acute infection and lymphoproliferative disease Patients with low immunoglobulin levels and recurrent or unusual infections should be referred to an Immunologist to exclude immunodeficiency * IgG may be lost via renal tract or gut. Check albumin is normal before interpreting IgG.
Test	**Rationale**
IgE (total)	Suspected atopic disease. Normal total IgE does not exclude atopy/allergy. Levels should be considered when interpreting specific IgE results. High levels also found in parasitic infestations
Insulin IgG antibodies	Predictor for type I diabetes. The test is of no diagnostic value by itself; should be used in conjunction with clinical findings
Islet tyrosine phosphatase 2 (IA2) antibodies	Foundin 50-75% of type I diabetes patients at and prior to disease onset. More prevalent in younger patients. Associated with rapid progression to onset disease

Allergen-Specific IgE/RAST	Suspected allergic reactions. Allergens/panels should be selected based on history. Screening is not recommended, as there is a significant false positive rate. Most useful in association with a clear history of an allergic reaction, to identify the specific cause.
Lymphocyte cell markers (CD3/CD4/CD8)	Monitoring disease progression and response to treatment in known HIV infection Declining CD4 count is seen in progressive HIV infection with increasing viral load
Pancreatic islet cell antibodies	Type I diabetes mellitus (IDDM). Antibodies are present in up to 70% of new-onset diabetes, but are transient, often disappearing once the islets have been destroyed. Can be useful intesting 1st degree relatives of patients with IDDM as the presence of high titre islet cell antibodies confers a risk for development of IDDM
Procalcitonin	To aid in: Diagnosis & risk stratification of bacterial sepsis. Diagnosis of renal involvement in children with UTI. Distinguishing bacterial from viral infections. Monitoring of therapeutic response to antibacterial therapy. Diagnosis of systemic secondary infection after surgery and in severe trauma, burns and multi-organ failure
Rheumatoid factor (RF)	RF is a poor screening test for JIA. Positive in RA and in ~33% of older children with polyarticular arthritis. Levels do not correlate with disease activity. Also found in many other connective tissue and inflammatory diseases.
Thyroid peroxidase antibodies (TPO)	Suspected autoimmune thyroid disease. High levels in Grave's disease and Hashimoto's disease. Indicates the need to monitor thyroid function
TSH receptor antibody	Hyperthyroidism in Grave's disease
Quantiferon TB	Blood test for detection of Mycobacterium tuberculosis infection, whether active or latent TB. Cannot distinguish between active TB and latent disease; should be used in conjunction with clinical risk assessment, radiography, microbiology and other medical and diagnostic evaluation for the diagnosis of clinically significant infection

CHAPTER 8 - HISTOPATHOLOGY AND CYTOLOGY

8.1 HISTOPATHOLOGY

8.1.1 SPECIMEN SUBMISSION

Biopsies may be submitted to the laboratory at any time during the day. The biopsies must be accompanied by a duly completed request form containing the following information:

- Name of Patient
- Age
- Sex
- Date specimen obtained
- Ward
- IP No/Kranium UHID Number

The clinician should provide adequate/operative notes, specimen type and legible signature.

Small biopsies must be fixed immediately and should be covered by at least 10-fold volume of 10% formalin. Big biopsies can be dissected into two or three segments that can be easily reconstructed by the grossing pathologist and again volumes of formalin as above should be used for fixation.

8.1.2 SPECIAL TESTS/HANDLING

Biopsies requiring specific handling or special tests should be notified to the laboratory. Examples are biopsies that require immunohistochemical staining, etc.

8.1.3 TURN-AROUND TIME (TAT)

The expected TAT is 5-10days. However, if a histology report is required urgently then the lab must be informed beforehand so the process can be expedited. Certain specimens may take a longer time than usual to process e.g. bone biopsies

8.1.4 QUERIES

All queries concerning any histopathology report may be directed to the pathologist on duty

8.2 CYTOPATHOLOGY

8.2.1 PROCEDURES

Fine needle aspiration biopsy (FNAB)

This procedure can easily be booked to be carried out by the pathologist. The laboratory should be called in order for the appointment to be made. If a clinician opts to perform the procedure; the smears must be immediately fixed in 95% alcohol. This can be obtained from the laboratory if the procedure is to be performed within the hospital.

At least two air dried smears must be prepared in addition for suspected lymphoma, leukemia or TB infection. The smears must be appropriately labeled and accompanied by a correctly completed request form indicating the following details:

- Name of Patient
- Age
- Sex
- Date of FNA
- Ward
- IP No/Kranium UHID No

The clinician should provide adequate notes indicating the clinical history, site of FNA, gross findings, and legible name and signature.
The expected TAT is 24-36hrs.

Pap smear (Cervical smear)

i. The cervical smear must be properly taken using the recommended procedure by the clinician/ nurse/technologist. The smear must be fixed immediately; this can be wet fixation (95% alcohol) or spray fixation.

ii. The slides should remain in fixative for at least 30minutes and then submitted to the laboratory in a slide carrier with a completed request form

Other specimens for cytology

a. **Sputum**

An early morning, deeply coughed specimen (before breakfast and teeth brushing) is recommended. Saliva specimens must be avoided. Chest physiotherapy may aid in a better sample collection. Induced sputum collection may also be performed.

The specimen should be collected in a clean container with a well-fitting cap.Three consecutive samples are recommended as this increases the sensitivity-fold. Specimens must be submitted to the laboratory immediately on collection and if there is any reason for delay, specimen should be refrigerated for not more than 24 hours. If the specimen is to be transported for a long distance, an equal volume of 95% alcohol should be added.

b. **Bronchial Lavage/Washings**

These should be submitted to the laboratory in a clean container with a well-fitting cap. The same points as above regarding number of samples and preservation apply.

c. **Urine**

Whole output samples greater than 50 mL are required. The first morning sampling should be avoided because of rapid overnight degeneration of cells. A clean container with a well-fitting cap is required for collection.
The same points as above regarding number of samples and preservation apply.

d. **Pleural/Pericardial/Ascitic/Cerebrospinal fluids**

Aspiration should be performed carefully to avoid traumatic haemorrhagic samples. The same points as above regarding number of samples and preservation apply.

e. **Smear imprints**

Smear imprints may be taken after removal of tissue specimens at surgery.Once an imprint has been taken, it must be fixed immediately in 95% alcohol for at least 3 minutes. Air-dried smears should also be prepared if necessary.

CHAPTER 9 - OTHER TESTS

9.1 PATERNITY TESTING

9.1.1 EXPLANATION OF THE TEST

This test is used to provide genetic proof of whether the putative father is the biological father of an individual. The principle of the test is based on the fact that a child inherits half of his or her DNA from each biological parent. A comparison of the DNA reveals whether the child could have inherited DNA from the putative father. When individuals are biologically related as parent and child, their DNA profiles show predictable patterns of genetic inheritance. Ideally blood from the child, mother and putative father should be tested. However, blood from the child and putative father may be sufficient in most instances to exclude paternity. Although paternity tests are more common than maternity tests, there may be circumstances in which there is uncertainty regarding maternity such as in adopted children seeking biological mothers, hospital mix-ups and in-vitro fertilization cases where laboratory mix-up is suspected.

The technique used for paternal testing is DNA profiling and it can exclude an individual who is falsely alleged to be a biological parent of a child. Paternity tests may be required for legal purposes and therefore a strict chain of custody procedure should be observed if the results are to be legally defensible.

9.1.2 REQUESTING A PATERNITY TEST.

The laboratory should be informed at least 24 hours before requesting the test. The blood sample is collected by the phlebotomist. In legal cases the lawyer/representative of child/child's mother should be present to identify the child. Note that in order to proceed with a DNA test, consent must be provided by all the tested parties. If a child below 18 years is tested, a parent or somebody with parental responsibility over the child needs to give consent for the child's DNA to be tested.

9.1.2.1 TEST RESULTS

The paternity test will be completed within 21 working days from the time all the necessary samples are taken at the laboratory. A paternity test has two possible results. Each result is indicated by the Probability of Paternity listed on the result report.

A Probability of Paternity of 99.999% or higher means the alleged father is not excluded as the biological father of the child. A calculated probability of paternity greater than 99.800% is considered proof of paternity.

A Probability of Paternity of 0% means the alleged father is excluded as the biological father of the child. In other words, he is not the father.
The method of analysis will be indicated in the results.

9.2 CHROMOSOME STUDIES (KARYOTYPE)

9.2.1 DESCRIPTION OF TEST:

Karyotypes describe the number of chromosomes, and what they look like under a light microscope. For humans, white blood cells are used most frequently because they are easily induced to divide and grow in tissue culture. The metaphase spreads are then stained – usually with Giemsa stain - and the chromosomes described. Many laboratories use automated computer programmes to analyse the karyotype. This test is performed at a referral laboratory.

A special tube for collection of the blood containing heparin anticoagulant is used. The laboratory should be informed beforehand if this test is to be performed on a patient so that the logistics of transport can be planned. If required, fluorescent in-situ hybridization (FISH) may be requested. This is performed when a specific marker or chromosome abnormality is being investigated for, such as a translocation like t(9;22) as seen in chronic myeloid leukaemia, or t(15;17) seen in acute promyelocytic leukaemia or other disorders.

Clinical information MUST always be provided with the request form. This should include the age and suspected sex of the patient, the indication for the test and clinical features, a signed consent form from the patient/parent and guardian MUST be attached.

9.2.2 REQUESTING THE TEST

The following details should be provided in the requisition form:

- Name of Patient
- Age
- Sex
- Date
- Ward
- IP No/Kranium UHID Number (for inpatients)
- The clinician should provide adequate details indicating the clinical history (including the reason for performing the test), legible name and signature.

9.2.3 TEST RESULTS:

These will be available after a minimum of 14 days. The report will indicate the number of cells counted and analysed, the number of cells karyotyped and the karyotype of the cells. Further clinical interpretation is provided depending on the clinical setting..

9.3 BARR BODY TESTING (SEX CHROMATIN TEST

9.3.1 EXPLANATION OF THE TEST

A Barr body is a condensed mass of compacted chromatin that represents an inactivated X chromosome. It is observed in special preparations of nucleated somatic cells as a small densely heterochromatic body localized at the nuclear periphery. It corresponds to the drumstick appendage in the nucleus of neutrophils seen in females. In men and women with more than one X chromosome, the number of Barr bodies visible is always one less than the total number of X chromosomes.

Barr body determination is used to evaluate abnormal sexual development such as ambiguous genitalia, delayed menarche and delayed puberty. The gender of some of these chromosomal karyotypes and syndromes cannot be correctly identified using the Barr body technique and therefore this test has been largely replaced by karyotyping.
The sample used for Barr body testing is most often a buccal smear.

Some different possible combinations of X and Y chromosomes in humans.

Condition	No Barr Bodies	Sex chromosomes
Phenotypic female	One	XX
Phenotypic female	None	XO
Phenotypic female	Two	XXX
Phenotypic male	None	XY
Phenotypic male	One	XXY

9.3.2 REQUESTING THE TEST.

Clinical information should always be provided with the request form. This should include the age and suspected sex of the patient, the indication for the test and clinical features.

CHAPTER 10 - MORTUARY SERVICES

10.1 STORAGE

The mortuary provides optimal storage of bodies at temperatures of 4-8°C. Embalming is performed on request and at an extra fee, provided it is not a forensic case. Autopsy (post mortem) is performed on request by the attending clinician, or parent/guardian or in the case of a forensic investigation in which case the police will be involved.

10.2 AUTOPSY SERVICES (POST MORTEM)

The autopsy should be booked with the pathologist by calling the laboratory, which in turn will get hold of the pathologist to perform the postmortem. The file of the deceased or a clinical summary and a signed consent form shall be sent to the laboratory before the autopsy is to be performed.

The death notification booklet may also be submitted if the attending clinician has not completed it. The pathologist will then complete it after performing the autopsy. A first degree relative must be available on the day of the autopsy to identify the body of the deceased and if necessary, to witness the autopsy.

In medico-legal cases, the police pathologist will be notified as he/she is the one supposed to carry out the post mortem. The hospital will be represented by the hospital pathologist. In certain cases, the family/relatives may request an independent pathologist to be present at the time of autopsy.

A summarized report of the autopsy findings shall be made available within the same day of the autopsy. A complete report with histopathology/toxicology findings should be available within two weeks to three months' time once all the results are available.

APPENDIX 1-TAT/SAMPLE TYPE/ REFFERENCE RANGE/STAT TEST

SERUM OR PLASMA SEROLOGY								
Test Name	**Sample**	**Container (Top Color)**	**Units**	**RR Male**	**RR Female**	**Reference Ranges**	**TAT**	**(S) STAT availability**
11-Desoxycor-tisol	serum	RED with Clot Separator	µg/l	< 8	< 8	< 8	2 Weeks	
17-OH-Pro-gesterone	serum	RED with Clot Separator	no/L		0.6- 10.3	3.0- 7.3 (male) 0.6- 10.3 (female)	2 Weeks	
ACE (An-giotensin converting enzyme)	serum	RED with Clot Separator	mU/ml	12 - 68 patients with sar-coidosis 45 - 135	12 - 68 patients with sar-coidosis 45 - 135	2 - 68 45 – 135(patients with sarcoidosis)	2 Weeks	
Acetylcholine receptor abs.	serum	RED with Clot Separator	nmol/l	up to 0.2	up to 0.2	up to 0.2	2 Weeks	
Acetylsalicylic acid	serum	RED with Clot Separator		see report	see report	see report and interpre-tation	2 Weeks	
Adenovirus IgG abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 Weeks	
Adenovirus IgM abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 Weeks	
ADH	ED-TA-plasma frozen serum frozen	LAVENDER	pg/ml	0.8 - 4.5	0.8 - 4.5	0.8 - 4.5	2Weeks	
Adrenaline (Epinephrine)	2 ml ED-TA-plasma frozen	LAVENDER	pg/ml	up to 85	up to 85	up to 85	2 Weeks	
Adrenaline abs	serum	RED with Clot Separator		< 1: 10	< 1: 10	< 1: 10	2 Weeks	
Adrenocor-ticotrophic hormone (ACTH)	ED-TA-plasma frozen	RED with Clot Separator	pg/ml	< 46	< 46	< 46	2 Weeks	
Aflatoxin B1	serum	RED with Clot Separator		< 0.2	< 0.2	< 0.2		
AFP	amniotic fluid	PLAIN TUBE	pg/ml		see report	See report and interpre-tation	2 Weeks	
ALANINE AMINO TRANSFER-ASE (SGPT / ALT)	serum	RED with Clot Separator, or dark GREEN top	U/l				90 minutes	
Premature						0-28		

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
0-1 days						0-31		
2-5 days						0-49		
6-180 days						0-56		
7-12 months						0-54		
1-3 yrs						0-33		
4-6 yrs						0-29		
7-12yrs						0-39		
13-15 yrs						0-24		
>15 yrs						0-35 (female) 0-50 (male)		
Alkaline phosphatase	serum or plasma	RED with Clot Separator, or dark GREEN top	U/l	40 - 129	35 - 104	up to 462	90 minutes	
1-30 days			U/l			48-406		
1-12 months						124-341		
1-3 yrs						108-317		
4-6 yrs						96-297		
7-9 yrs						69-325		
10-12 yrs						51-332		
13-15 yrs						50-162		
16-18 yrs						47-119		
>19 yrs						30-120		
Alpha-1-Anti-trypsine	serum	RED with Clot Separator	g/l	0.9 - 2.0	0.9 - 2.0	0.9 - 2.0	2Weeks	
Alpha-1-Feto-protein		RED with Clot Separator					2Weeks	
<30 d	serum		ng/mL			500-100,000		
1–3 months	serum		ng/mL			40.0-1000		
4 months–18 yr	serum		ng/mL			<12.0		
Pregnancy w 14	serum		ng/mL			<27.9		
median) w 15	serum		ng/mL			<30.9		
week 16	serum		ng/mL			<36.1		
week 17	serum		ng/mL			<40.4		
week 18	serum		ng/mL			<48.3		

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
week 19	serum		ng/mL			<54.8		
Adults	serum		ng/mL	<7.0	<7.0	<7.0		
alpha-1-Glycoprotein	serum	RED with Clot Separator	g/l	0.50 - 1.20	0.50 - 1.20	0.50 - 1.20		
Alpha-1-Microglobulin	serum	RED with Clot Separator	mg/l	50 - 85	50 - 85	50 - 85	2 Weeks	
alpha-2-Antiplasmin		RED with Clot Separator	%	80-120 %	80-120 %	130 - 300	2 Weeks	
Alpha-2-Macroglobulin	serum	RED with Clot Separator	mg/dl	130 - 300	130 - 300	150 - 450 (Newborn)	2 Weeks	
Alprazolam	serum	RED with Clot Separator	µg/l	5 - 50	5 - 50	5 - 50	2 Weeks	
Aluminium	serum	RED with Clot Separator	ng/ml	up to 20	up to 20	130 - 300	2 Weeks	
Alveolar basal membrane abs	serum	RED with Clot Separator		see report	see report	see report		
AMA (antimitochondrial abs)	serum	RED with Clot Separator		1:<40	1:<40	1:<40	2 Weeks	
AMA-M2 (M2-PDH-abs.)	serum	RED with Clot Separator		negative	negative	negative	2 Weeks	
Amino acid	serum (with empty stomach), to the prevention of artifacts, sample should be frozen immediately	RED with Clot Separator		see report	see report	see report	2 Weeks	
Aminophylline (Theophylline)	serum	RED with Clot Separator	µg/ml	10 - 20	10 - 20	10 - 20	90 minutes	S
Amiodarone	serum	RED with Clot Separator	mg/l	0.50-2.50	0.50-2.50	0.50-2.50	2 weeks	
Amitriptyline	serum	RED with Clot Separator	ng/ml	50 - 250	50 - 250	50 - 250	2 weeks	
Ammonia	EDTA-plasma 2 ml shipping frozen	LAVENDER	µg/dl	18 - 60	18 - 60	18 - 60	2 weeks	
Amoebiasis serology (antibodies)	serum	RED with Clot Separator		negative	negative	Positive or negative	1 Week	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Amphet-amines	serum	RED with Clot Separator	ng/ml				2 weeks	
Amylase	serum	RED with Clot Separator, or dark GREEN top	U/l	up to 115	up to 115	up to 115	90 minutes	S
ANA (antinuclear abs)	serum	RED with Clot Separator		1:<40	1:<40	1:<40	2 weeks	
Androstene-dione	serum	RED with Clot Separator	ng/ml	0.7 - 2.7	0.4 - 3.5	0.7-2.7(male) 0.4-3.5 (fe-male)	2 weeks	
Anti DNA(ss)	serum	RED with Clot Separator		negative	negative	Positive/ negative	2 weeks	
Antibodies to centromeres	serum	RED with Clot Separator		1:<40	1:<40	1:<40		
Anticardiolipin igG abs	serum	RED with Clot Separator	U/ml	< 14	< 14	< 14	2 weeks	
Anticardiolipin igM abs	serum	RED with Clot Separator	U/ml	< 10	< 10	< 10	2 weeks	
Anti-DNA (ds)	serum	RED with Clot Separator	IU/ml	< 15	< 15	< 15	2 weeks	
Antimicroso-mal abs (TPO)	serum	RED with Clot Separator	ng/dl	< 60	< 60	< 60	2 weeks	
Anti-MOG-IgA	serum	RED with Clot Separator		negative (1:<100)	negative (1:<100)	negative (1:<100)	2 weeks	
Anti-MOG-IgG	serum	RED with Clot Separator		negative (1:<100)	negative (1:<100)	negative (1:<100)	2 weeks	
Anti-MOG-igM	serum	RED with Clot Separator		negative (1:<100)	negative (1:<100)	negative (1:<100)	2 weeks	
Antiplatelet antibodies	serum	RED with Clot Separator		see report	see report	see report and interpre-tation	2 weeks	
Antistaphylol-ysin	serum	RED with Clot Separator	IU/ml	< 2.0	< 2.0	< 2.0	2 weeks	
Antistrep-tococcal D-NASE B	serum	RED with Clot Separator	U/ml	up to 200	up to 200	up to 75 (children) -up to 200 (adults)	2 weeks	
Antistreptol-ysin	serum	RED with Clot Separator	IU/ml	< 200	< 200	< 200	2hrs	
Antithrom-bin III	1ml citrate plasma frozen	SKY BLUE	%	80-120	80-120	80-120		

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
APC resistance	1ml citrate plasma frozen	SKY BLUE	Ratio	< 2.4 results < 2.2 indicates an abnormal risk of thrombosis	< 2.4 results < 2.2 indicates an abnormal risk of thrombosis	< 2.4 results < 2.2 indicates an abnormal risk of thrombosis	2 weeks	
APC resistance with anticoagulants	1ml citrate plasma frozen	SKY BLUE	Ratio	> 0.8	> 0.8	> 0.8	2 weeks	
Apolipoprotein A1	serum	RED with Clot Separator	mg/dl	102 - 215	102 - 215	102 - 215	2 weeks	
Apolipoprotein A2	serum	RED with Clot Separator	mg/dl	25 - 50	25 - 50	25 - 50	2 weeks	
Apolipoprotein B	serum	RED with Clot Separator	mg/dl	59 - 155	59 - 155	59 - 155	2 weeks	
Apolipoprotein B 100-Mutation	EDTA-blood	LAVENDER		see report	see report	see report and interpretation	2 weeks	
Apolipoprotein C II	serum	RED with Clot Separator		see report	see report	see report and interpretation	2 weeks	
Apoprotein E	serum	RED with Clot Separator	mg/dl	2.3 - 6.3	2.3 - 6.3	2.3 - 6.3	2 weeks	
Arginine (Amino acid)	erum (with empty stomach), to the prevention of artifacts sample frozen	RED with Clot Separator	umol/l	15 - 150	15 - 150	15 - 150	2 weeks	
Arsenic	serum	RED with Clot Separator	µg/l	< 10	< 10	< 10	2Weeks	
ASMA (anti-smooth muscle abs)	serum	RED with Clot Separator		1:<40	1:<40	1:<40	2Weeks	
ASPARTATE AMINO TRANSFERASE (SGOT / AST)	serum	RED with Clot Separator, or dark GREEN top					90 minutes	
Premature		RED with Clot Separator				0-64		
0-1 days		RED with Clot Separator				0-109		

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
2-5 days		RED with Clot Separator				0-97		
6-180 days		RED with Clot Separator				0-77		
7-12 months		RED with Clot Separator				0-82		
1-3 yrs		RED with Clot Separator				0-48		
4-6 yrs		RED with Clot Separator				0-36		
7-12yrs		RED with Clot Separator				0-47		
13-15 yrs		RED with Clot Separator				0-27		
>15 yrs		RED with Clot Separator			0-35			
>15 yrs		RED with Clot Separator			0-35			
Aspergillus abs (Elisa)	serum	RED with Clot Separator	Index	< 0.5	< 0.5	< 0.5	2 Weeks	
Aspergillus abs (Immuno-diffusion)	serum	RED with Clot Separator		negative	negative	negative	2 Weeks	
Aspergillus antigen Latex	serum	RED with Clot Separator		negative	negative	negative		
Aspergillus precipitation	serum	RED with Clot Separator		negative	negative	negative	2 Weeks	
Atenolol-Lev-el	serum	RED with Clot Separator	ug/ml	0.05 - 1.0	0.05 - 1.0	0.05 - 1.0	2 Weeks	
Atomoxetin (LCMS)	serum	RED with Clot Separator	ng/ml	see report	see report	see report and interpre-tation	2 Weeks	
Azathioprine a. 6-Mercapt.	serum	RED with Clot Separator	ng/ml	40 - 300	40 - 300	40 - 300	2 Weeks	
Baclofen	serum	RED with Clot Separator	µg/ml	0.1 - 0.6	0.1 - 0.6	0.1 - 0.6		
Barbiturates	serum	RED with Clot Separator		see report	see report	see report and interpre-tation	2 Weeks	
Bartonella henselae IgG abs	serum	RED with Clot Separator		<1:64	<1:64	<1:64	2 Weeks	
Bartonella henselae IgM abs	serum	RED with Clot Separator		<1:20	<1:20	<1:20	2 Weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Benzhexol (Trihexyphe-nidyl) HPLC	serum	RED with Clot Separator		see report	see report	see report and interpre-tation	2 weeks	
Benzodiaze-pine i. serum	serum	RED with Clot Separator		see report	see report	see report	2 weeks	
Beta-2-Glyco-protein I	serum	RED with Clot Separator	U/ml	< 8	< 8	< 8	2 weeks	
Beta-2-Mi-croglobulin i. serum	serum	RED with Clot Separator	µg/l	0.6 - 2.1	0.6 - 2.1	0.6 - 2.1	2 weeks	
Beta-Caro-tene	serum	RED with Clot Separator	ng/ml	150 - 1250	150 - 1250	150 - 1250	2 weeks	
Beta-HCG	serum	RED with Clot Separator	mIU/ml	0-3	0-5	0-3(male) 0-5(female)	36 HRS	S
Post meno-pausal					0 - 10	0 - 10		
Pregnancy 4 - 5 weeks					1500 - 23000	1500 - 23000		
Pregnancy 5 - 6 weeks					3400 - 135300	3400 - 135300		
Pregnancy 6 - 7 weeks					10500 - 161000	10500 - 161000		
Pregnancy 7 - 12 weeks					11500 - 289000	11500 - 289000--		
Pregnancy 12 - 16 weeks					18300 - 137000	18300 - 137000		
Pregnancy 16 - 29 weeks					1400 - 53000	1400 - 53000		
Pregnancy 29 - 41 weeks					940 - 60000	940 - 60000		
Bilharziosis: S. mansoni (IHA)	serum	RED with Clot Separator		1:<16	1:<16	1:<16	1 Week	
BILIRUBIN Total	serum	RED with Clot Separator, or dark GREEN top	µmol/l				90 minutes	
Premature day 0-1						17-187		
Premature day 1-2						103-205		
Premature day 3-5						171-240		
0-1 days						34-103		
1-2 days						103-171		
3-5 days						68-137		

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
>15 yrs				0-21	0-21			
Bilirubin direct		RED with Clot Separator, or dark GREEN top	μmol/l	<3.4	<3.4	<3.4	90 minutes	S
Blood Urea Nitrogen (BUN)	serum	RED with Clot Separator	mmol/l				90 minutes	S
Premature						1.1-8.9		
1-30 days						1.4-4.3		
1-12 months						1.8-6.4		
1-15 yrs						1.8-6.4		
15-60 yrs				2.1-7.1	2.1-7.1	2.1-7.1		
60 yrs				2.9-8.2	2.9-8.2	2.9-8.2		
BNP (Brain natriuretic. Peptide)	ED-TA-blood	LAVENDER	pg/ml	< 100	< 100	< 100	2 weeks	
Bordetella pertussis IgG	serum	RED with Clot Separator	Index	< 8.5	< 8.5	< 8.5	2 weeks	
Br. abortus	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Bromazepam	serum	RED with Clot Separator	ng/ml	50 - 200	50 - 200	50 - 200	2 weeks	
Bromide	serum	RED with Clot Separator	μg/ml	up to 3000 μg/ml	up to 3000 μg/ml	up to 3000 μg/ml	2 weeks	
Bromocriptine	serum	RED with Clot Separator	μg/l	9 - 54	9 - 54	9 - 54	2 weeks	
Brucella abs	serum	RED with Clot Separator		negative	negative	negative	24 HRS	
Buspirone	serum	RED with Clot Separator	ng/ml	1 - 10	1 - 10	1 - 10	2 weeks	
C1 Ester-ase-inhibitor concentration	serum	RED with Clot Separator	mg/ dl	15 - 35	15 - 35	15 - 35	2 weeks	
C1q-comple-ment	serum	RED with Clot Separator	mg/ dl	5.0 - 25.0	5.0 - 25.0	5.0 - 25.0	2 weeks	
C2-comple-ment	serum	RED with Clot Separator	mg/l	10 - 30	10 - 30	10 - 30	2 weeks	
C3 comple-ment	serum	RED with Clot Separator, or dark GREEN top	g/l	0.9 - 1.8	0.9 - 1.8	0.9 - 1.8	36 HRS	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
C4 complement	serum	RED with Clot Separator, or dark GREEN top	g/l	0.10 - 0.40	0.10 - 0.40	0.10 - 0.40	36 HRS	
CA 125	serum	RED with Clot Separator	U/ml	< 35	< 35	see report and interpretation	2 weeks	
CA 15-3	serum	RED with Clot Separator	U/ml	< 32.4	< 32.4	see report and interpretation	2 weeks	
CA 19-9	serum	RED with Clot Separator	U/ml	< 34	< 34	see report and interpretation	2 weeks	
CA 50	serum	RED with Clot Separator	U/ml	up to 25	up to 25	up to 25	2 weeks	
CA 549	serum	RED with Clot Separator		see report	see report	see report and interpretation	2 weeks	
CA 72-4	serum	RED with Clot Separator	U/ml	< 5.0	< 5.0	< 5.0	2 weeks	
Cadmium i. edta	EDTA-blood	LAVENDER	ng/ml	< 4ng/ml BAT 15ng/ml	< 4ng/ml BAT 15ng/ml	< 4ng/ml BAT 15ng/ml	2 weeks	
Calcitonin	serum frozen	RED with Clot Separator	pg/ml	up to 8.4	up to 5.0	up to 11.5(children) up to 8.4(male) up to 5.0(female)	2 weeks	
Calcium , serum	serum	RED with Clot Separator, or dark GREEN top	mmol/L				2 weeks	
0-10 days		RED with Clot Separator	mmol/L			1.9-2.66	90 minutes	S
2-15 yrs		RED with Clot Separator	mmol/L			1.9-2.6		
>15 yrs		RED with Clot Separator	mmol/L			2.1-2.66		
Calprotectin , stool	stool	Clean Container	mg/kg	< 50	< 50	< 50	2 weeks	
Campylobacter ab (CFR)	serum	RED with Clot Separator		1:<20	1:<20	1:<20	2 weeks	
c-ANCA	serum	RED with Clot Separator		1:<2	1:<2	1:<2	2 weeks	
Cancer assoc. serum antigen (CASA)	serum	RED with Clot Separator	U/ml	< 4.0	< 4.0	< 4.0	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Candida antigen	serum	RED with Clot Separator		see report	see report	see report and interpretation	2 weeks	
Candida IgG abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Candida IgM abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9		
Cannabinoid EIA	serum	RED with Clot Separator	ng/ml	see report	see report	see report and interpre-tation	2 weeks	
Carbamaze-pine	serum	RED with Clot Separator	µg/ml	4.0 - 10.0	4.0 - 10.0	4.0 - 10.0	90 minutes	S
Carbimacol (Thiamacol)	serum	RED with Clot Separator	ng/ml				2 weeks	
Carboxy he-moglobin	ED-TA-blood	LAVENDER	%	smoker: 1.5 - 5.0 % Hb non smoker: < 1.5 %	smoker: 1.5 - 5.0 % Hb non smoker: < 1.5 %	smoker: 1.5 - 5.0 % Hb non smoker: < 1.5 %	2 weeks	
Carcinoem-bryonic anti-gen (CEA)	serum	RED with Clot Separator	ng/ml			see report and interpretation	2 weeks	
Non-smokers (20-69 yrs)		RED with Clot Separator		0-3.8	0-3.8	0-3.8		
Non-smokers >70 yrs)		RED with Clot Separator		0-5	0-5	0-5		
Smokers (20-69 yrs)		RED with Clot Separator		0-5.5	0-5.5	0-5.5		
Non-smokers >70 yrs)		RED with Clot Separator		0-6.5	0-6.5	0-6.5		
Carnitine, free	serum	RED with Clot Separator	µmol/l	24.6 - 51.0	17.9 - 45.5	24.6 - 51.0 (Male) 17.9 - 45.5 (female)	2 weeks	
Carnitine, total	serum	RED with Clot Separator	µmol/l	29.0 - 58.2	22.9 - 53.3	29.0 - 58.2(Male) 22.9 - 53.3(female)	2 weeks	
CH 50	serum	RED with Clot Separator	%	70 - 140	70 - 140	70 - 140	2 weeks	
Chikungunya-fever (IIF)	serum	RED with Clot Separator		<1:10	<1:10	<1:10	2 weeks	
Chl. psittaci	serum	RED with Clot Separator		1:<20	1:<20	1:<20	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Chlamydia antigen trachomatis (PCR) ejaculate	ejaculate			negative	negative	negative	2 weeks	
Chlamydia antigen trachomatis (PCR) smear	smear	RED with Clot Separator		negative	negative	negative	2 weeks	
Chlamydia igA abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Chlamydia igG abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Chlamydia pneumonia igA	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Chlamydia pneumonia igG	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Chloride	serum	RED with Clot Separator					90 minutes	S
Premature	serum	RED with Clot Separator	mmol/l			97-122	90 minutes	S
1-30 days	serum	RED with Clot Separator				95-116	90 minutes	S
1-12 months	serum	RED with Clot Separator				93-112	90 minutes	S
>1 year	serum	RED with Clot Separator				98-115	90 minutes	S
Chloroquine	serum	RED with Clot Separator		see report	see report	see report	2 weeks	
Chlorphenox-amine	serum	RED with Clot Separator	ng/ml	see report	see report	see report	2 weeks	
Cholesterol		RED with Clot Separator, or dark GREEN top					90 minutes	
Total Cholesterol			mmol/l	3.2-5.2	3.2-5.2	3.2-5.2		
HDL			mmol/l	0.3-5.0	0.3-5.0	0.3-5.0		
LDL			mmol/l	2.6-3.36	2.6-3.36	2.6-3.36		
Triglycerides			mmol/l	0.3-1.7	0.3-1.7	0.3-1.7		
Chlorproma-zine	serum	RED with Clot Separator	ng/ml	30-250	30-250	30-250	2 weeks	
Cholinester-ase (CHE)	serum	RED with Clot Separator, or dark GREEN top	U/l	5300 - 12900	3700 - 12900	5300 - 12900	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Chromogranin A	serum	RED with Clot Separator	ng/ml	< 100	< 100	< 100	2 weeks	
Circ. immuno-complexes	serum	RED with Clot Separator		see report	see report	see report	2 weeks	
Citalopram	serum	RED with Clot Separator	ng/ml	30 - 130	30 - 130	30 - 130	2 weeks	
CKBB	serum	RED with Clot Separator	U/l	up to 5	up to 5	up to 5	2 weeks	S
CKMM	serum	RED with Clot Separator	U/l	up to 174	up to 140	up to 174	2 weeks	S
Clobazam	serum	RED with Clot Separator	ng/ml	100 - 400	100 - 400	100 - 400	2 weeks	
Clomipramine	serum	RED with Clot Separator	ng/ml	20 - 140	20 - 140	20 - 140	2 weeks	
Clonazepam	serum	RED with Clot Separator	ng/ml	ther. range: 15- 60	ther. range: 15- 60	ther. range: 15- 60	2 weeks	
Clozapine	serum	RED with Clot Separator	ng/ml	ther. range 50-700	ther. range 50-700	ther. range 50-700	2 weeks	
CMV-early antigen (pp65)	EDTA-blood	LAVENDER		negative	negative	negative	2 weeks	
Cocaine i. serum	serum	RED with Clot Separator	ng/ml	see report	see report	see report		
Coenzyme Q-10	serum	RED with Clot Separator	ng/ml	500 - 1100	500 - 1100	500 - 1100	2 weeks	
Ceruloplasmin	serum	RED with Clot Separator	g/l	0.2 - 0.5	0.2 - 0.5	see report	2 weeks	
Caffeine	serum	RED with Clot Separator	µg/ml	2 - 10	2 - 10	2 - 10		
Collagen-Telopeptide i.S.	serum	RED with Clot Separator	nMBCE/l	7.7 - 19.3	7.7 - 19.3	7.7 - 19.3	2 weeks	
Connective tissue abs	serum	RED with Clot Separator		1:<40	1:<40	1:<40	2 weeks	
Copper i. serum	serum	RED with Clot Separator	µg/dl	70 - 150	70 - 150	70 - 150	2 weeks	
Cortisol Serum 0070-1000 Hrs	serum	RED with Clot Separator	µg/dL	6.2-19.4	6.2-19.4	6.2-19.4		
Cortisol Serum 1600-2000Hrs	serum	RED with Clot Separator	µg/dL	2.3-11.9	2.3-11.9	2.3-11.9		

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Cotinine i.serum	serum	RED with Clot Separator	ng/ml	Non smoker: up to 15 smoker: 15 - 50 strong smoker: > 50	Non smoker: up to 15 smoker: 15 - 50 strong smoker: > 50	Non smoker: up to 15 smoker: 15 - 50 strong smoker: > 50		
Coxsackie B1 IgG	serum	RED with Clot Separator		< 1:16	< 1:16	< 1:16	2 weeks	
Coxsackie B1 IgM	serum	RED with Clot Separator		< 1:16	< 1:16	< 1:16	2 weeks	
Coxsackie B2 IgG	serum	RED with Clot Separator		< 1:16	< 1:16	< 1:16	2 weeks	
Coxsackie B2 IgM	serum	RED with Clot Separator		< 1:16	< 1:16	< 1:16	2 weeks	
Coxsackie B3 IgG	serum	RED with Clot Separator		< 1:16	< 1:16	< 1:16		
Coxsackie B3 IgM	serum	RED with Clot Separator		< 1:16	< 1:16	< 1:16	2 weeks	
Coxsackie B4 IgG	serum	RED with Clot Separator		< 1:16	< 1:16	< 1:16	2 weeks	
Coxsackie B4 IgM	serum	RED with Clot Separator		< 1:16	< 1:16	< 1:16	2 weeks	
Coxsackie B5 IgG	serum	RED with Clot Separator		< 1:16	< 1:16	< 1:16	2 weeks	
Coxsackie B5 IgM	serum	RED with Clot Separator		< 1:16	< 1:16	< 1:16	2 weeks	
Coxsackie B6 IgG	serum	RED with Clot Separator		< 1:16	< 1:16	< 1:16	2 weeks	
Coxsackie B6 IgM	serum	RED with Clot Separator		< 1:16	< 1:16	< 1:16	2 weeks	
C-Peptide i. serum	serum	RED with Clot Separator	ng/ml	1.1 - 5.0	1.1 - 5.0	1.1 - 5.0	2 weeks	
C-Reactive Protein i. serum	serum	RED with Clot Separator, or dark GREEN top	ng/ml	< 0.5	< 0.5	< 0.5	90 minutes	
C-reactive protein (highly sensitive) hsCRP	serum	RED with Clot Separator, or dark GREEN top	mg/L	TO	BE	REVIEWED	1 day	
1-28 days						<4.1		
1-12 months						<2.8		
1-15 yrs						<2.8		

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
15-49 yrs				<3.0	<3.0			
50-64yrs				<7.9	<8.5			
>65 yrs				<6.8	<6.6			
Creatinine i. serum	serum	RED with Clot Separator, or dark GREEN top					90 minutes	S
1-30 days		RED with Clot Separator	µmol/l			44-90		
1-12 months		RED with Clot Separator				18-65		
1-15 yrs		RED with Clot Separator				27-62		
15-50 yrs		RED with Clot Separator		74-110				
>50 yrs		RED with Clot Separator		72-127				
>15 yrs		RED with Clot Separator			58-96			
Creatineki-nase (CPK) (37˚C)	serum	RED with Clot Separator	u/l	46-171	34-145		90 minutes	S
Creatinineki-nase MB Frac-tion (CKMB)	serum	RED with Clot Separator	u/l	0-24	0-24		90 minutes	S
Cryoglobulins	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Cyanide	ED-TA-blood	RED with Clot Separator	µg/l	< 50 , tox. >200	< 50 , tox. >200	< 50 , tox. >200	2 weeks	
Cycl. citrull. peptide abs (CCP)	serum	RED with Clot Separator	U/ml	< 25.0: negative 25.1-50.0: border-line > 50.0 : positive	< 25.0: negative 25.1-50.0: borderline > 50.0 : positive	< 25.0: negative 25.1-50.0: borderline > 50.0 : positive	2 weeks	
Cyfra 21-1	serum	RED with Clot Separator	ng/ml	<3.3	<3.3	<3.3	2 weeks	
Cyst. cellulo-sae (BLOT)	serum	RED with Clot Separator		negative	negative	negative	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Cystin, free (Amino acid) i. serum	serum (with empty stomach), to the pre-vention of artifacts sample frozen	RED with Clot Separator		< 140	< 140	< 140	2 weeks	
Cytomegaly DNA	ED-TA-blood	LAVENDER	cop/ml	< 600	< 600	< 600	2 weeks	
Cytomegaly igG abs i. serum	serum	RED with Clot Separator	U/ml	< 15 U/ml - negative > 15 U/ml - pos-itive	< 15 U/ml - negative > 15 U/ml - positive	< 15 U/ml - negative > 15 U/ml - positive	2 weeks	
Cytomegaly igM abs i. serum	serum	RED with Clot Separator	Index	< 0.40 - nega-tive 0.40 - 0.59 - border-line > 0.60 - positive	< 0.40 - negative 0.40 - 0.59 - border-line > 0.60 - positive	< 0.40 - negative 0.40 - 0.59 - borderline > 0.60 - positive	2 weeks	
Dengue fever igG abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Dengue fever igM abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Dextroprop-oxyphen	serum	RED with Clot Separator	ng/ml	50-300, tox. <1000	50-300, tox. <1000	50-300, tox. <1000	2 weeks	
DHEA-Sul-phate	serum	RED with Clot Separator	µg/dl	80 - 560	35 - 430	35 - 430	2 weeks	
Diazepam	serum	RED with Clot Separator	ng/ml	200-500	200-500	200-500		
Dihydrotes-tosterone	serum	RED with Clot Separator	ng/ml	16.0 - 110.0	5.6 - 20.0		2 weeks	
Dipropylac-etate	serum	RED with Clot Separator	mg/ml	50 - 100	50 - 100	50 - 100	2 weeks	
Dopamine i. plasma	ED-TA-blood	LAVENDER	pg/ml	up to 50	up to 50	up to 50	2 weeks	
Duloxetin	serum	RED with Clot Separator	ng/ml	20 - 80	20 - 80	20 - 80	2 weeks	
EBV igG abs (WB) i. serum	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
EBV igM abs (WB) i. serum	serum	RED with Clot Separator		negative	negative	negative	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
EBV Nucleus antigen	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
EBV PCR	ED-TA-blood	LAVENDER		negative	negative	negative	2 weeks	
EBV-VCA igG abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
EBV-VCA igM abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Echinococcosis (IHA)	serum	RED with Clot Separator		< 1:32	< 1:32	< 1:32	2 weeks	
Elastase i. serum	serum	RED with Clot Separator	ng/ml	< 3.5	< 3.5	< 3.5	2 weeks	
ENA (abs extractable nucl. ag)	serum	RED with Clot Separator		negative	negative	negative	1 Week	
Endomysium igA abs/ Transglutaminase	serum	RED with Clot Separator	U/ml	< 7 - negative 7 - 10 - borderline > 10 - positive	< 7 - negative 7 - 10 - borderline > 10 - positive	< 7 - negative 7 - 10 - borderline > 10 - positive	2 weeks	
Entamoeba histolytica igA abs (IFT)	serum	RED with Clot Separator		1:<40	1:<40	1:<40	2 weeks	
Entamoeba histolytica igG abs (IFT)	serum	RED with Clot Separator		1:<40	1:<40	1:<40	2 weeks	
Epinephrine (Adrenalin) i. plasma	2 ml ED-TA-plasma frozen	LAVENDER	pg/ml	up to 85	up to 85	up to 85	2 weeks	
Erythropoetin	serum	RED with Clot Separator	mIU/ml	3.7 - 29.5	3.7 - 29.5	3.7 - 29.5	2 weeks	
Estradiol (E2)		RED with Clot Separator					3days	
1–10 yr f	serum		pmol/L			22.0–99		
1–10 yr m	serum		pmol/L			18.4–73		
Follicular phase	serum		pmol/L		46–607			
Ovulatory phase	serum		pmol/L		315–1828			
Luteal phase	serum		pmol/L		161–774			
Postmenopause	serum		pmol/L		18.4–201			
Pregnancy, 1st trimester	serum		pmol/L		789–15780			
Male	serum		pmol/L	28–156				

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Estriol (E3) in Pregnancy		RED with Clot Separator						
28-30 weeks	serum		nmol/L		132–486			
31-32 weeks	serum		nmol/L		121–1145			
33-36 weeks	serum		nmol/L		167–1215			
37-40weeks	serum		nmol/L		330–1596			
Ethanol Level	serum the recep-tacle must be totally filled and airtight	RED with Clot Separator	mg/dl	negative	negative	negative	90 minutes	
Ethosuximide	serum	RED with Clot Separator	µg/ml	40 - 100	40 - 100	40 - 100	2 weeks	
Fibrinogen	citrated blood 1:10	SKY BLUE	mg/dl	180 - 350	180 - 350	180 - 350	2 weeks	
Felbamate/ Felbatol	serum	RED with Clot Separator					2 weeks	
Ferritin	serum	RED with Clot Separator	ng/mL	22 - 322	10 - 291		3days	
<1 yr						12- 327		
1- 3 yr						6- 67		
4- 6 yr						4- 67		
7- 12 yr,						7- 84		
Male						14- 124		
13- 17 yr, f						13- 68		
Male						14- 152		
17- 60 yr, female					13- 150			
18- 60 yr, Male				30- 400				
Fibronectin	ED-TA-plasma frozen	LAVENDER	mg/dl	25 - 40	25 - 40	25 - 40	2 weeks	
Filaria (D.immitis)	serum	RED with Clot Separator		see report	see report	see report	2 weeks	
Flavi virus IgM	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Flecainide level	serum	RED with Clot Separator	µg/ml	max. 0.75 - 1.25 min. 0.45 - 0.9	max. 0.75 - 1.25 min. 0.45 - 0.9	max. 0.75 - 1.25 min. 0.45 - 0.9	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Fluoride i. serum	serum	RED with Clot Separator		up to 30 ng/ml ther. range: 80 - 200 ng/ml	up to 30 ng/ml ther. range: 80 - 200 ng/ml	up to 30 ng/ ml ther. range: 80 - 200 ng/ ml	2 weeks	
Fluoxetine	serum	RED with Clot Separator	ng/ml	60 - 450	60 - 450	60 - 450	2 weeks	
Flupentixol	serum	RED with Clot Separator	ng/ml	1 - 15	1 - 15	1 - 15	2 weeks	
Fluphenazine	serum	RED with Clot Separator	ng/ml	1-10	1-10	1-10		
Fluvoxamine	serum	RED with Clot Separator	ng/ml	30 - 300	30 - 300	30 - 300	2 weeks	
Folate/Folic acid	serum	RED with Clot Separator	ng/ml	7.2-15.4	7.2-15.4		36 HRS	
Adults Deficient				<3.0	<3.0			
Adults Bor-derline				3.0 - 7.2	3.0 - 7.2			
Folic acid in red cells	ED-TA-blood	LAVENDER	ng/ml	175 - 700	175 - 700	175 - 700	2 weeks	
free Androgen index, free	serum	RED with Clot Separator		20 - 40 years 45 - 70 40 - 50 years 35 - 50 50 - 70 years 25 - 35	4 - 6	see report		
FSH	serum	RED with Clot Separator					36 HRS	
Follicular phase		RED with Clot Separator	mIU/L		3.5–12.5			
Ovulatory phase		RED with Clot Separator	mIU/L		4.7–21.5			
Luteal phase		RED with Clot Separator	mIU/L		1.7–7.7			
Postmeno-pause		RED with Clot Separator	mIU/L		25.8–134.8			
Male		RED with Clot Separator	mIU/L	1.5–12.4				
FSME igG abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
FSME igM abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
FT3 (free Triiodithyrox-ine)	serum	RED with Clot Separator	pg/mL				36 HRS	
Adults		RED with Clot Separator						
Male				2.0- 4.4				
f On contraceptive					2.6- 4.5			
1st trimester					2.5- 3.9			
2nd trimester					2.1- 3.6			
3rd trimester					2.0- 3.3			
Newborn						1.73- 6.30		
6 d - 3 mth						1.95- 6.04		
4- 12 mth						2.15- 5.83		
1- 6 yr						2.41- 5.50		
7- 11 yr						2.53- 5.22		
12- 20 yr						2.56- 5.01		
FT4 (free Thyroxine)	serum	RED with Clot Separator	ng/dL				36 HRS	
Adult Male				1.0- 1.7				
Adult female					1.0- 1.6			
Pregnancy 1st trimester					0.9- 1.5			
2nd trimester					0.8- 1.3			
3rd trimester					0.7- 1.2			
Newborn						0.86- 2.49		
6d–3 mth						0.89- 2.20		
4–12 mth						0.92- 1.99		
1–6yr						0.96- 1.77		
7–11 yr						0.97- 1.67		
12–20 yr						0.98- 1.63		
FTA-Abs. IFT	serum	RED with Clot Separator					2 weeks	
G6-PDH (Glu-cose-6-PDH)	EDTA-blood	LAVENDER	mU/ Mrd. Ery	221 - 570	221 - 570	221 - 570	2 weeks	
Gabapentin	serum	RED with Clot Separator	µg/ml	2 - 10	2 - 10	2 - 10	2 weeks	
GAD-ab	serum	RED with Clot Separator	U/ml	< 0.9	< 0.9	< 0.9	2 weeks	
Galactose	sodium-fluoride-blood	GREY	mg/dl	< 4.5	< 4.5	newborn < 10	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Gamma Glutamyl Transferase (GGT)	serum	RED with Clot Separator, or dark GREEN top	U/l			0-132		
1-3 months						1-39	2 weeks	
6-12 months						3-22	2 weeks	
1-12 yrs							2 weeks	
13-18 yrs			2-42	4-24			2 weeks	
>19 yrs			0-60	0-35			2 weeks	
Gangliosid abs profile	serum	RED with Clot Separator		see report	see report	see report	2 weeks	
Gastric abs.	serum	RED with Clot Separator		see report	see report	see report	2 weeks	
Gastrin	serum frozen	RED with Clot Separator	pg/ml	40 - 210 - fast-ingmuch higher after protein rich meal	40 - 210 - fast-ingmuch higher af-ter protein rich meal	40 - 210 - fastingmuch higher after protein rich meal	2 weeks	
Gentamycin	serum	RED with Clot Separator	ng/ml	lower peak value: 1 - 2 upper peak val-ue: 4- 8	lower peak val-ue: 1 - 2 upper peak val-ue: 4- 8	lower peak value: 1 - 2 upper peak value: 4- 8	2 weeks	
Giardia lamblia abs	serum	RED with Clot Separator		see report	see report	see report	2 weeks	
Gliadin igA abs	serum	RED with Clot Separator	rel. U/ ml	< 25	< 25	> 4 Jahre < 25	2 weeks	
Gliadin igG abs	serum	RED with Clot Separator	rel. U/ ml	< 25	< 25	< 25	2 weeks	
Glucose							90 minutes	S
Glucose, Fasting (reference intervals not diagnostic criteria)	Plasma	GREY	mmol/L					
Cord						2.5-5.3		
Premature						1.1-3.3		
Neonate						1.7-3.3		
1 day						2.2-3.3		
>1 day						2.8-4.5		
Child						3.3-5.6		

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Adult				4.1-5.6	4.1-5.6			
>60 yrs				4.6-6.4	4.6-6.4			
>90 yrs				4.2-6.7	4.2-6.7			
Glucose, whole blood	Plasma	GREY TOP. If DARK GREEN top used sample should be in the lab within 30min of with drawing) (indicate time of withdrawal)						
Adult				3.5-5.3				
Diabetes Diagnostic Criteria (ADA)		GREY						
AIC*	Plasma		%	>6.5				
FPG*	Plasma		mmol/L	7	7			
2 hr plasma glucose, post OGTT*	Plasma		mmol/L	11.1	11.1			
Random Plasma Glucose (plus classic symptoms of hyperglycemia or hyperglycemic crisis)			mmol/L	11.1	11.1			
*(in the absence of unequivocal hyperglycemia, confirm by repeat testing)								
Gestational Diabetes Mellitus(ADA)±								
100g glucose Load			mmol/L					
Fasting				5.3				
1 h				10				
2 h				8.6				
3 h				7.8				
75g glucose Load			mmol/L					

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Fasting				5.3				
1h				10				
2h				8.6				
±(Two or more of the venous plasma concentrations must be met or exceeded for a positive diagnosis). The test should be done in the morning after an overnight fast of between 8 and 14 h and after at least 3 days of unrestricted diet (150 g carbohydrate per day) and unlimited physical activity.)								
Glomerular basal membrane abs	serum	RED with Clot Separator		1:<40	1:<40	1:<40	2 weeks	
Gonococcal abs (cfr)	serum	RED with Clot Separator		1:<5	1:<5	1:<5	2 weeks	
Gonorrhea, DNA PCR	smear	RED with Clot Separator		negative	negative	negative		
Haemochromatosis - PCR	EDTA-blood	LAVENDER		see report	see report	see report	2 weeks	
Haemophilus antibodies	serum	RED with Clot Separator	ug/ml	protection due to vaccination 0.15 - 1.0 ng/ml short time	protection due to vaccination 0.15 - 1.0 ng/ml short time	protection due to vaccination 0.15 - 1.0 ng/ml short time	2 weeks	
Haloperidol	serum	RED with Clot Separator	ng/ml	2 - 50	2 - 50	2 - 50		
Hantavirus igG abs (Immunoblot)	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Hantavirus IgM abs (Immunoblot)	serum	RED with Clot Separator		negative	negative	negative	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Haptoglobin	serum	RED with Clot Separator	g/l	0.30 - 2.00	0.30 - 2.00	0.30 - 2.00	2 weeks	
HBA1C	EDTA-blood	LAVENDER	%	4.5 - 6.2	4.5 - 6.2	4.5 - 6.2		
HbA2 (Hb-electro-phoresis)	EDTA-blood	LAVENDER		see report	see report	see report	2 weeks	
HBDH	serum	RED with Clot Separator, or dark GREEN top	U/l	90 - 180	90 - 180		90 minutes	S
Hb-electro-phoresis	EDTA-blood	RED with Clot Separator		see report	see report	see report	2 weeks	
HbF (Hb-elec-trophoresis)	EDTA-blood	LAVENDER		see report	see report	see report	2 weeks	
HCV-RNA quant.	EDTA-blood	LAVENDER	IU/ml	< 12	< 12	< 12	2 weeks	
HDL-Cholesterol	serum	RED with Clot Separator	mg/dl	> 40	> 40		90 minutes	
Helicobacter igA abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Helicobacter igG abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Heparin (Anti-Xa-activity)	citrated plasma frozen	SKY BLUE	U/ml	ther. range: 0.4 - 0.8	ther. range: 0.4 - 0.8	ther. range: 0.4 - 0.8		
Hepatitis A igG abs	serum	RED with Clot Separator		negative	negative	negative	36 HRS	
Hepatitis A igM abs.	serum	RED with Clot Separator		negative	negative	negative	36 HRS	
Hepatitis B c abs	serum	RED with Clot Separator		negative	negative	negative	36 HRS	
Hepatitis B c igM abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Hepatitis B DNA quant.	EDTA-blood	LAVENDER	IU/ml	< 12	< 12	< 12	2 weeks	
Hepatitis B e abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Hepatitis B e antigen	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Hepatitis B s abs	serum	RED with Clot Separator	mIU/ml	< 10	< 10	< 10	36 HRS	
Hepatitis B Surface antigen	serum	RED with Clot Separator		negative	negative	negative	36 HRS	
Hepatitis C abs	serum	RED with Clot Separator	Index	< 0.8 negative	< 0.8 neg-ative	< 0.8 neg-ative	36 HRS	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Hepatitis C Genotype	EDTA-blood	LAVENDER		see report	see report	see report	2 weeks	
Hepatitis Delta antigen	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Hepatitis Delta igG abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Hepatitis Delta igM abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Hepatitis E igG abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Hepatitis E igM abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Herpes simplex type I igG abs	serum	RED with Clot Separator		< 0.9	< 0.9	< 0.9	1 Week	
Herpes simplex type I igM abs	serum	RED with Clot Separator		< 0.9	< 0.9	< 0.9	1 Week	
Herpes simplex type II igG abs	serum	RED with Clot Separator		< 0.9	< 0.9	< 0.9	1 Week	
Herpes simplex type II igM abs	serum	RED with Clot Separator		< 0.9	< 0.9	< 0.9	1 Week	
Human Chori-onic Gonado-trophin	serum	RED with Clot Separator					1 day	S
Premeno-pausal f	serum				<1 U/L			
non-pregnant f								
Postmeno-pausal f					<7 U/L			
m				<2 U/L				
HGH (human growth hormone)	serum	RED with Clot Separator	ng/ml	0.1 - 3.5 (day)	0.1 - 3.5 (day)	0.1 - 7.5 (day)	2 weeks	
Histamine	Heparin plasma frozen	DARK GREEN	ng/ml	see report	see report	see report	2 weeks	
Histoplasma abs (CFR)	serum	RED with Clot Separator		< 1 : 5	< 1 : 5	< 1 : 5	2 weeks	
HIV I + II	serum	RED with Clot Separator		negative	negative	negative	48 HRS	
HIV I Western blot	serum	RED with Clot Separator		negative	negative	negative	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
HIV Virus-load RT-PCR Abbott	EDTA-blood	LAVENDER	Copies/ml	< 40	< 40	< 40	3 Days	
HLA B*2701-B*2711	EDTA-blood	LAVENDER		see report	see report	see report	2 weeks	
HLA B27	EDTA-blood	LAVENDER		negative	negative	negative	2 weeks	
HLA-51	EDTA-blood	LAVENDER		see report	see report	see report	2 weeks	
Homocysteine	serum	RED with Clot Separator	µmol/l	< 13.9	< 13.9	< 13.9	36 HRS	
HPV DNA (high risk)	smear	RED with Clot Separator		negative	negative	negative	2 weeks	
HPV DNA (low risk)	smear	RED with Clot Separator		negative	negative	negative	2 weeks	
HSV viral culture	virus transport medium	KIT SPECIFIC		negative	negative	negative	2 weeks	
h-Thyroglob-ulin	serum	RED with Clot Separator	µg/l	up to 70	up to 70	up to 70	2 weeks	
Human Herpes Virus 6 igG abs i. serum	serum	RED with Clot Separator		see report	see report	see report	2 weeks	
Human Herpes Virus 8 (HHV8) DNA	EDTA-blood	LAVENDER		negative	negative	negative	2 weeks	
Ibuprofen	serum	RED with Clot Separator	µg/ml	15 - 30	15 - 30	15 - 30		
Imipramine	serum	RED with Clot Separator	ng/ml	50 - 150	50 - 150	50 - 150	2 weeks	
Immunofix-ation	serum	RED with Clot Separator		see report	see report	see report	2 weeks	
Immunoglob-ulin A (igA)	serum	RED with Clot Separator	mg/dl			see report		
<1 yr						<81		
1- 3 yr						16- 98		
4- 6 yr						27- 190		
7- 9 yr						33- 298		
10- 11 yr						52- 199		
12- 13 yr						57- 350		
14- 15 yr						46- 243		
16- 19 yr						60- 339		

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Adults						70- 400		
Immunoglob-ulin D (igD)	serum	RED with Clot Separator	IU/ml	up to 153	up to 153	up to 153	2 weeks	
Immunoglob-ulin E (igE)	serum	RED with Clot Separator	IU/ml					
Neonates						<1.5		
Children, adolescents 1 yr						<1.5		
2- 5 yr						<60		
6- 9 yr						<90		
10- 15 yr						<200		
Adults						<100		
Immunoglob-ulin G (igG)	serum	RED with Clot Separator	mg/dl					
Neonates						227-1378		
1- 3 yr						442-895		
4- 6 yr						492-1430		
7- 9 yr						559-1439		
10- 11 yr						681-1523		
12- 13 yr						741-1513		
14- 15 yr						699-1671		
16- 19 yr						536-1547		
Adults						700-1600		
Immunoglob-ulin G1	serum	RED with Clot Separator	mg/ml	2.80 - 8.00	2.80 - 8.00	see report	2 weeks	
Immunoglob-ulin M (igM)	serum	RED with Clot Separator	g/l					
<1 yr						<1.21		
1- 3 yr						0.16- 1.22		
4- 6 yr						0.20- 1.76		
7- 9 yr						0.26- 1.74		
10- 11 yr						0.26- 1.50		
12- 13 yr						0.29- 2.00		
14- 15 yr						0.19- 1.57		
16- 19 yr						0.20- 2.17		
Adults						0.4- 2.3		
Influenza A igA abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Influenza A igG abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Influenza B igA abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Influenza B igG abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Inhibin-b	serum	RED with Clot Separator	pg/ml	100 - 400	100 - 400	100 - 400	2 weeks	
Inorganic Phosphate (see phosphate)		RED with Clot Separator, or dark GREEN top						
Insulin (fasting)	serum frozen	RED with Clot Separator	µIU/ml	2.6-11	2.6-11			
Insulin growth factor	serum	RED with Clot Separator	ug/ml	see report	see report	see report	2 weeks	
Insulin igG abs (IAA)	serum	RED with Clot Separator	U/ml	< 1.0	< 1.0	< 1.0	2 weeks	
Insulin-receptor igA abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Interleukin 1	serum frozen	RED with Clot Separator	pg/ml	< 5.0	< 5.0	< 5.0	2 weeks	
Interleukin 6	serum frozen	RED with Clot Separator	pg/ml	< 9.70	< 9.70	< 9.70	2 weeks	
Intrinsic-factor abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Iron	serum	RED with Clot Separator	mmol/L					
1- 30 d, Female						5.2- 22.7 mmol/L		
Male						5.7- 20.0 mmol/L		
1- 12 mth, Female						4.5- 22.6 mmol/L		
Male						4.8- 19.5 mmol/L		
1- 3 yr						4.5- 18.1 mmol/L		
Male						5.2- 16.3 mmol/L		
4- 6 yr						5.0- 16.7 mmol/L		

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Male						4.5- 20.6 mmol/L		
7- 9 yr						5.4- 18.6 mmol/L		
Male						4.8- 17.2 mmol/L		
10- 12 yr, f						5.7- 18.6 mmol/L		
Male						5.0- 20.0 mmol/L		
13- 15 yr, f						5.4- 19.5 mmol/L		
Male						4.7- 19.7 mmol/L		
16- 18 yr, f						5.9- 18.3 mmol/L		
Male						4.8- 24.7 mmol/L		
Adults				11- 28 mmol/L	6.6- 26 mmol/L			
Total Iron binding capacity	serum	RED with Clot Separator, or dark GREEN top	ug/dl	300 - 400	250 - 350			
Islet cell abs (IFT)	serum	RED with Clot Separator		< 1:10	< 1:10	< 1:10	2 weeks	
Itraconazole	serum	RED with Clot Separator	µg/ml	0.4 - 2.0	0.4 - 2.0	0.4 - 2.0	2 weeks	
JO-1 antigen	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Jodid i.serum	serum	RED with Clot Separator	µg/l	46.0 - 70.0	46.0 - 70.0	46.0 - 70.0		
Cobalt	serum	RED with Clot Separator	ng/ml	limit of detection 3ng/ml	limit of detection 3ng/ml	limit of detection 3ng/ml	2 weeks	
Lactate	Fluoride Plasma	RED with Clot Separator, or dark GREEN top	mg/dl	5 - 20	5 - 20	5 - 20		
Lactate Dehydrogenase	Serum	RED with Clot Separator, or dark GREEN top	u/l	0-248	0-248		90 minutes	
Lamictal level	serum	RED with Clot Separator	µg/ml	2.0 - 10.0	2.0 - 10.0	2.0 - 10.0	2 weeks	
Lanoxin	serum	RED with Clot Separator	ng/ml	0.8 - 2.0	0.8 - 2.0	0.8 - 2.0	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
LDL-Receptor PCR	EDTA-blood	LAVENDER		see report	see report	see report	2 weeks	
Lead	EDTA-blood	LAVENDER	µg/l	< 100 (BLW < 400)	< 100 (BLW < 450)		2 weeks	
Legionella se-rovare 1,4,6,8	serum	RED with Clot Separator		< 1:128	< 1:128	< 1:128	2 weeks	
Legionella se-rovare 2,3,5,7	serum	RED with Clot Separator		< 1:128	< 1:128	< 1:128	2 weeks	
Leishmaniasis abs (igG)	serum	RED with Clot Separator		1:<40	1:<40	1:<40	2 weeks	
Leishmaniasis abs (igM)	serum	RED with Clot Separator		1:<20	1:<20	1:<20	2 weeks	
Leishmaniasis abs (IHA)	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Leishmaniasis PCR	EDTA-blood	RED with Clot Separator		see report	see report	see report	2 weeks	
Leptospira canicola	serum	RED with Clot Separator		< 1 : 10	< 1 : 10	< 1 : 10	2 weeks	
Leptospira grippotyphosa	serum	RED with Clot Separator		< 1 : 10	< 1 : 10	< 1 : 10	2 weeks	
Leptospira ictero haemorrh.	serum	RED with Clot Separator		< 1 : 10	< 1 : 10	< 1 : 10		
Leptospira igG abs (EIA)	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Leptospira igM abs (EIA)	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Leptospira pomona	serum	RED with Clot Separator		< 1 : 10	< 1 : 10	< 1 : 10	2 weeks	
Leptospira sejroe	serum	RED with Clot Separator		< 1 : 10	< 1 : 10	< 1 : 10		
Luteinizing hormone (LH)	serum	RED with Clot Separator	mIU/mL				36 HRS	
f Follicular phase					2.4- 12.6			
f Ovulatory phase					14- 95.6			
f Luteal phase					1.0- 11.4			
f Postmeno-pause					7.7- 58.5			
Male				1.7- 8.6				
Levetirace-tam/Keppra	serum	RED with Clot Separator	µg/ml	2 - 45	2 - 45	2 - 45	2 weeks	
Levomepro-macin	serum	RED with Clot Separator	ng/ml	10 - 140	10 - 140	10 - 140	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Lidocaine	serum	RED with Clot Separator	µg/ml	1.5-5.0	1.5-5.0	1.5-5.0	2 weeks	
Lipase	serum	RED with Clot Separator, or dark GREEN top						
Neonates						<34 U/L (<0.57 mkat/L)		
<12 yr						<31 U/L(<0.52 mkat/L)		
16- 18 yr						<55 U/ L(<0.92 mkat/L)		
Adults				<60 U/L (<1 mkat/L)	<60 U/L (<1 mkat/L)		2 weeks	
Lipoprotein a	serum	RED with Clot Separator	mg/dl	< 30	< 30	< 30		
Listeriosis type 1 (H-antigen)	serum	RED with Clot Separator		1:< 40	1:< 40	1:< 40	2 weeks	
Listeriosis type 1 (O-antigen)	serum	RED with Clot Separator		1:< 40	1:< 40	1:< 40	2 weeks	
Listeriosis type 4 (o-antigen)	serum	RED with Clot Separator		1:< 40	1:< 40	1:< 40	2 weeks	
Listeriosis type 4 (H-antigen)	serum	RED with Clot Separator		1:< 40	1:< 40	1:< 40	2 weeks	
Lithium	serum	RED with Clot Separator	mmol/l	0.4-1.6	0.4-1.6	0.4-1.6	2 HRS	S
Liver-cytosol abs. type 1 (LC1)	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Liver-kidney-microsom. abs.	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Lupus-anticoagulant-diagnostic	6 ml citrated plasma							
frozen	SKY BLUE		see report	see report	see report	2 weeks		
Lyme borrelio-sis igG abs	serum	RED with Clot Separator	Index	< 20	< 20	< 20	2 weeks	
Lyme borrelio-sis igM abs	serum	RED with Clot Separator	Index	< 20	< 20	< 20	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
M.tropica (Pl. falciparum) igG abs	serum	RED with Clot Separator		1:<40	1:<40	1:<40	2 weeks	
M.tropica (Pl. falciparum) igM abs	serum	RED with Clot Separator		1:<10	1:<10	1:<10		
M2-Pyruvate Kinase	EDTA-plasma frozen	LAVENDER	U/ml	<15	<15	<15	2 weeks	
Magnesium i. serum	serum	RED with Clot Separator, or dark GREEN top	mmol/l	0.70 - 1.10	0.70 - 1.10	0.60 - 1.00	90 min-utes	
Malaria anti-gen	EDTA-blood	LAVENDER		negative	negative	negative	4 HRS	
Manganese i. EDTA	EDTA-blood	LAVENDER	µg/l	7.1 - 10.5 (BAT: 20.0)	7.1 - 10.5 (BAT: 20.0)	7.1 - 10.5 (BAT: 20.0)	2 weeks	
Manganese i. serum	serum	RED with Clot Separator	µg/l	0.3 - 0.9	0.3 - 0.9	0.3 - 0.9	2 weeks	
Maprotilin	serum	RED with Clot Separator	ng/ml	50-250 ng/ml	50-250 ng/ml	50-250 ng/ml	2 weeks	
Mast cell Tryptase	serum	RED with Clot Separator	µg/l	<11.4	<11.4	<11.4	2 weeks	
Measles igG abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Measles igM abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Melatonin	serum	RED with Clot Separator	pg/ml	during the day: up to 30.0 during the night: up to 150	during the day: up to 30.0 during the night: up to 150	during the day: up to 30.0 during the night: up to 150	2 weeks	
Melperon	serum	RED with Clot Separator	ng/ml	100-200	100-200	100-200	2 weeks	
Meningococ-cal ag	serum	RED with Clot Separator		negative	negative	negative	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Meningococcal antibodies	serum	RED with Clot Separator	RE/ml	< 31 RE/ml	< 31 RE/ml	< 31 RE/ml		
Mercury i. EDTA	EDTA-blood	LAVENDER	ng/ml	< 2.0	< 2.0	< 2.0	2 weeks	
Methadone i. serum	serum	RED with Clot Separator	ng/ml	see report	see report	see report	2 weeks	
Methaemoglobin	EDTA-blood	LAVENDER	% Hb	up to 1	up to 1	up to 1	2 weeks	
Methaqualon	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Methotrexate	serum	RED with Clot Separator	µmol/l	High dosage therapy:after 24 hrs. < 10after 48 hrs. < 1.0after 72 hrs. < 0.1<Low dosage therapy(10-20 mg once the week):after 2 h 0.2 - 0.9after 4 h 0.1 - 0.4after 24 h ca. 0.1	High dosage therapy:after 24 hrs. < 10after 48 hrs. < 1.0after 72 hrs. < 0.1<Low dosage therapy(10-20 mg once the week):after 2 h 0.2 - 0.9after 4 h 0.1 - 0.4after 24 h ca. 0.1	High dosage therapy:after 24 hrs. < 10after 48 hrs. < 1.0after 72 hrs. < 0.1<Low dosage therapy(10-20 mg once the week):after 2 h 0.2 - 0.9after 4 h 0.1 - 0.4after 24 h ca. 0.1	2 weeks	
Methyl-Tetrahydrofolate (MTHFR)PCR	EDTA-blood	LAVENDER		see report	see report	see report		
Midazolam	serum	RED with Clot Separator	ng/ml	20-200	20-200	20-200	2 weeks	
Mirtacapin	serum	RED with Clot Separator	ng/ml	10 - 100	10 - 100	10 - 100	2 weeks	
Mumps igG abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Mumps igM abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Muscle spec. Tyrosinkinase	serum	RED with Clot Separator	Ak-Ratio	< 10	< 10	< 10	2 weeks	
Mycophenolat-Mofetil	serum	RED with Clot Separator	µg/ml	see report	see report	see report	2 weeks	
Mycoplasma igA abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Mycoplasma igG abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Mycoplasma igM abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Myelin abs	serum	RED with Clot Separator	U/ml	< 1000	< 1000	< 1000	2 weeks	
Myelin associ-ated glyco-protein abs i. EDTA-blood	EDTA-blood	LAVENDER	Elisa-LU/ml	< 1000	< 1000	< 1000	2 weeks	
Myelin associ-ated glyco-protein abs i. serum	serum	RED with Clot Separator	Elisa-LU/ml	< 1000	< 1000	< 1000	2 weeks	
Myeloperoxi-dase EIA	serum	RED with Clot Separator	U/mL	< 5	< 5	< 5	2 weeks	
Myoglobin i. serum quant.	serum	RED with Clot Separator	ng/mL	28- 72	25- 58		36 HRS	
Myosin abs. (EIA)	serum	RED with Clot Separator	AK-Ratio	< 10	< 10	< 10	2 weeks	
Neopterin	serum	RED with Clot Separator	ng/ml	< 2.5	< 2.5	< 2.5	2 weeks	
Neuronal abs. (Hu-ANNA type 1)	serum	RED with Clot Separator		<1:20	<1:20	<1:20	2 weeks	
Niacin	serum	RED with Clot Separator	µg/l	8-52	8-52	8-52	2 weeks	
Niacin (Nicotinamide, Vitamin PP)	serum	RED with Clot Separator	µg/l	8.0 - 52.0	8.0 - 52.0	8.0 - 52.0	2 weeks	
Nickel	serum	RED with Clot Separator	µg/l	up to 3.3	up to 3.3	up to 3.3	2 weeks	
Norepineph-rine i. plasma	EDTA-plasma frozen	LAVENDER	pg/ml	up to 400	up to 400	up to 400	2 weeks	
NSE (neuron-spec. Enolase)	serum	RED with Clot Separator	µg/l	<18.3	<18.3	<18.3	2 weeks	
NT-p.BNP (n-term.p.Brain N.Peptide)	Heparin plasma	DARK GREEN	pg/ml	see report	see report	see report	2 weeks	
Oestriol free	serum	RED with Clot Separator	ng/ml		see report		2 weeks	
Oestriol total (E3)	serum	RED with Clot Separator	ng/ml		see report		2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Oestrone	serum	RED with Clot Separator	pg/ml	20 - 80	50 - 100 follicle phase100 - 300 luteal phase 10 - 60 meno-pausal			
Olanzapine	serum	RED with Clot Separator	ng/ml	5 - 75	5 - 75	5 - 75	2 weeks	
Opiates i. serum	serum	RED with Clot Separator	ng/ml	see report	see report	see report	2 weeks	
Osmolality i. serum (calcu-lated)	serum	RED with Clot Separator	mos-mol/ kgH20	280 - 300	280 - 310	280 - 310	90 minutes	
Osmolality i. serum (measured)	serum	RED with Clot Separator	mos-mol/ kgH20	280 - 300	280 - 310	280 - 310	2 weeks	
Ostase (bone AP)	serum	RED with Clot Separator	U/l	15.0 - 41.3	< 50 years: 11.6 - 30.6	> 50 years: 14.8 - 43.4	2 weeks	
Osteocalcin	ED-TA-plasma frozen	LAVENDER	ng/ml	> 50 Jahre: up to 26.3	premeno-pausal: up to 31.2	postmeno-pausal: up to 41.3	2 weeks	
Oxazepam (efficient metabolite)	serum	RED with Clot Separator	ng/ml	500-2000	500-2000		2 weeks	
p53 antibodies	serum	RED with Clot Separator	mg/l	< 0.4	< 0.4	< 0.4	2 weeks	
PAI (Plasminogen-activity-inhibitor)	citrate plasma (1:10) deep-frozen	SKY BLUE	ng/ml	4 - 43	4 - 43	4 - 43	2 weeks	
Pancreas islet cell abs IFT	serum	RED with Clot Separator		<1:10	<1:10	<1:10	2 weeks	
Paracetamol	serum	RED with Clot Separator	µg/ml	5 - 25	5 - 25	5 - 25	2 weeks	
Parathormone	serum frozen	LAVENDER	pg/ml	14 - 72	14 - 72	14 - 72	2 weeks	
Parathyroid Hormone (PTH)		RED with Clot Separator	ng/L				36 HRS	
2- 4 yr					f	3.6- 32		
					m	5.7- 34		
5- 6 yr					f	1.0- 13		
					m	4.4- 16		
7- 8 yr					f	2.7- 25		
					m	2.5- 27		

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
9- 10 yr					f	2.0- 30		
					m	4.6- 34		
11- 12 yr					f	4.3- 34		
					m	2.5- 25		
13- 14 yr					f	1.6- 37		
					m	1.4- 26		
15- 16 yr					f	1.2- 39		
					m	4.5- 36		
				12- 50	12- 50			
Parotis abs	serum	RED with Clot Separator		negative	negative	negative		
Paroxetin	serum	RED with Clot Separator	ng/ml	30-500	30-500	30-500	2 weeks	
Parvovirus B 19 DNA	EDTA-blood	LAVENDER		negative	negative	negative	2 weeks	
Parvovirus igG abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Parvovirus igM abs	serum	RED with Clot Separator		negative	negative	negative		
Perazin	serum	RED with Clot Separator	ng/ml	50 - 250	50 - 250	see report	2 weeks	
Pethidine	serum	RED with Clot Separator	ng/ml	100 - 500	100 - 500	100 - 500	2 weeks	
Phenylalanine (Amino acid)	serum,to the pre-vention of artifacts-sample frozen	RED with Clot Separator	mg/dl	0.8 - 2.0	0.8 - 2.0	1 - 12 months: up to 2.0under PKU-thera-py:1-10 years: 0.7 - 4.011-16 years: 0.7 - 15.0more than 16 years: < 20	2 weeks	
Phenytoin	serum	RED with Clot Separator	µg/ml	10 - 20	10 - 20	10 - 20	90 min-utes	
Phosphate (inorganic)	serum	RED with Clot Separator	mmol/L				90 min-utes	
1-15 yrs		RED with Clot Separator				1.29-2.26		
>15 yrs		RED with Clot Separator			0.81-1.45			
>15 yrs		RED with Clot Separator		0.81-1.45				
Phosphorus	serum	RED with Clot Separator	mg/dl	2.7 - 4.5	2.7 - 4.5	dependent on age	90 min-utes	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Picorna virus abs (CFR)	serum	RED with Clot Separator		1:<20	1:<20	1:<20	2 weeks	
Piracetame	serum	RED with Clot Separator	ug/ml	20 - 50	20 - 50	20 - 50	2 weeks	
Plasmin-inhibitor activity	citrate plasma (1:10) deep-frozen	SKY BLUE	%	75 - 120	75 - 120	75 - 120	2 weeks	
Pneumococcal antibodies	serum	RED with Clot Separator	mg/l	see report	see report	see report		
Pneumocystis jirovecii DNS (PCR)	bronchial lavage	Clean Container		negative	negative	negative	2 weeks	
Polio type 1, 2, 3 (NT)	serum	RED with Clot Separator		1:<8	1:<8	1:<8	2 weeks	
Porphyrin tot., serum	serum	RED with Clot Separator	ug/dl	< 2.0 ug/dl	< 2.0 ug/dl	< 2.0 ug/dl		
Potassium i. serum	serum	RED with Clot Separator	mmol/l				90 minutes	S
1-30 days		RED with Clot Separator				3.7-5.9		
1-12 months		RED with Clot Separator				4.1-5.3		
1-15 yrs		RED with Clot Separator				3.4-4.7		
>15 yrs		RED with Clot Separator		3.3-5.5	3.3-5.5			
Prec. abs. chicken's feathers	serum	RED with Clot Separator	mg/l	reference region < 30	reference region < 30	reference region < 30	2 weeks	
Pregabalin	serum	RED with Clot Separator	ng/ml	0.5 - 16	0.5 - 16	0.5 - 16	2 weeks	
Pregnancy Associated Plasma Prot. A	serum	RED with Clot Separator	mU/L		see report		2 weeks	
Primidone	serum	RED with Clot Separator	µg/ml	5.0 - 15.0	5.0 - 15.0	5.0 - 15.0	2 weeks	
Progesterone	serum	RED with Clot Separator	ng/mL					
Adult female Follicular Phase					0.25 - 2.61			
F Luteal Phase					3.28 - 38.63			
F Post Menopausal					0.2 - 0.82			

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
F 1st Trimester					12.3 - 81.8			
F 2nd Trimester					11.1 - 81.4			
F 3rd Trimester					39.3 - 387.8			
Male				0.2-3.37				
Prolactin	serum	RED with Clot Separator	ng/mL				36 HRS	
1- 12 mth					F	0.2-29.9		
					M	0.3-28.9		
1- 3 yr					F	1.0-17.1		
					M	2.3-13.2		
4- 6 yr					F	1.6-13.1		
					M	0.8-16.9		
7- 9 yr					F	0.3-12.9		
					M	1.9-11.6		
10- 12 yr					F	1.9-9.6		
					M	0.9-12.9		
13- 15 yr					F	3.0-14.4		
					M	1.6-16.6		
16- 18 yr					F	2.1-18.4		
					M	2.7-15.2		
Adults				4.6-21.4				
Promethazine	serum	RED with Clot Separator	ng/ml	50 - 400	50 - 400	50 - 400	2 weeks	
Propranolol	serum	RED with Clot Separator	ng/ml	50 - 300	50 - 300	50 - 300	2 weeks	
Protein C	citrated plasma							
frozen	SKY BLUE	%	69 - 134	70 - 140	69 - 134	2 weeks		
Protein S - activity	citrated plasma							
frozen	SKY BLUE	%	75 - 130	60 - 140		2 weeks		
Protein, total i. serum	serum	RED with Clot Separator, or dark GREEN top	g/l				90 minutes	
Premature						36-60		
1-30 days						46-70		
1-12 months						51-73		

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
1-15yrs						60-80		
>15 yrs				60-85	60-85			
Protoporphy-rin erythr.	Heparin blood	DARK GREEN	µg/dl	< 50	< 50	< 50	2 weeks	
Prostate specific antigen, total (taps) male		RED with Clot Separator	ng/mL				36 HRS	
<40 yr				<4.0				
40–50 yr				<2.5				
51–60 yr				<3.5				
61–70 yr				<4.5				
>70 yr				<6.5				
PSA free/ total ratio (male)		RED with Clot Separator		<0.10			36 HRS	
50-59 yrs			SI Units	49.20%	26.90%			
60-69 yrs			SI Units	57.50%	33.90%			
>70 yrs			SI Units	64.50%	40.80%			
				male	male			
				0.19–0.25	>0.25			
50-59 yrs			Conv. Units	18.30%	9.10%			
60-69 yrs			Conv. Units	23.90%	12.20%			
>70 yrs			Conv. Units	29.70%	15.80%			
Q-fever	serum	RED with Clot Separator		1:<10	1:<10	1:<10	2 weeks	
Quetiapine	serum	RED with Clot Separator	ng/ml	70 - 170	70 - 170	70 - 170	2 weeks	
Quinidin	serum	RED with Clot Separator	µg/ml	1.0-5.0	1.0-5.0	1.0-5.0	2 weeks	
Rabies abs	serum	RED with Clot Separator	IU/ml	immunity: > 0.5	immunity: > 0.5	immunity: > 0.5	2 weeks	
Ramiprilat	serum	RED with Clot Separator	µg/l	see report	see report	see report	2 weeks	
Renin	EDTA-plasma frozen	LAVENDER	ng/l	horizontal: 3.0 - 16 vertical: 3.0 - 33	horizontal: 3.0 - 16 vertical: 3.0 - 33	horizontal: 3.0 - 16 vertical: 3.0 - 33	2 weeks	
Reticulin abs	serum	RED with Clot Separator		< 1:10	< 1:10	< 1:10	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Rickettsia conori (IFT)	serum	RED with Clot Separator		< 1 : 40	< 1 : 40	< 1 : 40	2 weeks	
Rickettsia mooseri (IFT)	serum	RED with Clot Separator		< 1 : 40	< 1 : 40	< 1 : 40	2 weeks	
Rifampi-cin-Level	serum	RED with Clot Separator	µg/ml	ther.range: max. 4 - 10 min. 0.1 - 1.0 toxic: max. > 12 min. > 2	ther.range: max. 4 - 10 min. 0.1 - 1.0 toxic: max. > 12 min. > 2	ther.range: max. 4 - 10 min. 0.1 - 1.0 toxic: max. > 12 min. > 2	2 weeks	
Risperidone	serum	RED with Clot Separator	µg/l	see report	see report	see report	2 weeks	
RNP antigen	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
RSV igG abs	serum	RED with Clot Separator		1: <40	1: <40	1: <40	2 weeks	
RSV igM abs	serum	RED with Clot Separator		1: <40	1: <40	1: <40	2 weeks	
Rubella (HaH)	serum	RED with Clot Separator		1:<8	1:<8	1:<8	2 weeks	
Rubella igG abs	serum	RED with Clot Separator	IE/ml	see report	see report	see report	36 HRS	
Rubella igM abs	serum	RED with Clot Separator	Index	see report	see report	see report	36 HRS	
Salicylate	serum	RED with Clot Separator	mg/l	therap. range: 20 - 200	therap. range: 20 - 200	therap. range: 20 - 200	2 weeks	
SCCA	serum	RED with Clot Separator	ng/ml	< 1.5	< 1.5	< 1.5	2 weeks	
SCL 70 antigen	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Serotonin i. EDTA	EDTA-blood	LAVENDER	ng/ml	20 - 240	20 - 240	20 - 240	2 weeks	
SHBG	serum	RED with Clot Separator	nmol/l	13 - 71	18 - 114 under oral contracep-tives 56 - 159		2 weeks	
Single stranded DNA (ss-DNA)	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Sirolimus	EDTA-blood	LAVENDER	ng/ml	5 - 30	5 - 30	5 - 30	2 weeks	
Smith antigen	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Sodium i. serum	serum	RED with Clot Separator	mmol/l	135 - 150	135 - 150	dependent on age	3 Hours	S

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Soluble-liver antigen-abs. (SLA)	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Somatomedin C (IgF-1)	serum	RED with Clot Separator	ng/ml	see report	see report	see report	2 weeks	
Sotalol	serum	RED with Clot Separator	µg/ml	1.0 - 3.0	1.0 - 3.0	1.0 - 3.0	2 weeks	
Sperm abs	serum	RED with Clot Separator	U/ml	< 20	< 20	< 20	2 weeks	
Spironolac-tone	serum	RED with Clot Separator	ng/ml	50 - 200	50 - 200	50 - 200	2 weeks	
SSA (Ro) antigen	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
SSB (La) antigen	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
ß-crosslaps	serum	RED with Clot Separator	ng/ml	30 - 50 years: up to 0.58450 - 70 years: up to 0.704> 70 years: up to 0.854	premeno-pausal: up to 0.588post-meno-pausal: up to 1.008	see report	2 weeks	
Striated mus-cle abs.	serum	RED with Clot Separator		1:<40	1:<40	1:<40	2 weeks	
Sultiam	serum	RED with Clot Separator	µg/ml	1.5 - 10.0	1.5 - 10.0	1.0 - 6.0	2 weeks	
Tacrolimus (Prograf/FK 506)	ED-TA-blood	LAVENDER	ng/ml	5 - 20	5 - 20	see report	2 weeks	
Teicoplanin	serum	RED with Clot Separator	mg/l	5.0 - 60.0	5.0 - 60.0	5.0 - 60.0	2 weeks	
Temazepam	serum	RED with Clot Separator	ng/ml	200-800	200-800	200-800	2 weeks	
Testosterone	serum	RED with Clot Separator	ng/ml				36 HRS	
<1 yr						0.12- 0.21		
1- 6 yr						0.03- 0.32		
7- 12 yr						0.03- 0.68		
13- 17 yr						0.28- 11.1		
					0.06- 0.82			
				2.8- 8.0				

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Testosterone, free	serum	RED with Clot Separator	pg/ml	20 - 39 yr.: 8.8 -27 40 - 59 yr.: 7.2 - 23 > 59 yr. : 5.6 - 19	20 - 39 yr.: 2.6 40 - 59 yr.: < 2.0 > 59 yr. : < 1.5	see report	2 weeks	
Tetanus abs	serum	RED with Clot Separator	IE/ml	< 0.01 - no immunity 1.1 - 3.0 - immunity	< 0.01 - no immunity 1.1 - 3.0 - immunity	< 0.01 - no immunity 1.1 - 3.0 - immunity	2 weeks	
Tg recovery	serum	RED with Clot Separator	%	70 - 130	70 - 130	70 - 130	2 weeks	
Thallium i. serum	serum	RED with Clot Separator		< 5.0	< 5.0	< 5.0	2 weeks	
Theophylline (Aminophylline)	serum	RED with Clot Separator	µg/ml	10 - 20	10 - 20	10 - 20	90 minutes	
Thiocyanate	serum	RED with Clot Separator	mg/l	nonsmoker: < 4.6 ther. range: 5.0 - 30.0 toxic: < 50.0	nonsmoker: < 4.6 ther. range: 5.0 - 30.0 toxic: < 50.0	nonsmoker: < 4.6 ther. range: 5.0 - 30.0 toxic: < 50.0	2 weeks	
Thioridacine	serum	RED with Clot Separator	ng/ml	100 - 2500	100 - 2500	100 - 2500	2 weeks	
Thymidin-Kinase	serum	RED with Clot Separator	U/l	up to 6	up to 6	up to 10	2 weeks	
Thyroglobulin abs	serum	RED with Clot Separator	U/ml	< 60	< 60	< 60	2 weeks	
TSH (Reference ranges May change depending on the method of analysis)	serum	RED with Clot Separator	µU/L				36 HRS	
Newborns						0.70- 15.2		
6 d - 3 mth						0.72- 11.0		
4- 12 mth						0.73- 8.35		
1- 6 yr						0.70- 5.97		
7- 11 yr						0.60- 4.84		
12- 20 yr						0.51- 4.30		
Adults				0.27-4.2	0.27-4.2			
Thyroxin binding globulin	serum	RED with Clot Separator	mg/dl	1.6 - 2.5	1.6 - 2.5	1.6 - 2.5	2 weeks	
Thyroxine-AK	serum	RED with Clot Separator		negative	negative	negative	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Tiagabin	serum	RED with Clot Separator	ng/ml	see report	see report	see report	2 weeks	
Titane	serum	RED with Clot Separator	ug/l	< 7.7	< 7.7	< 7.7	2 weeks	
Topiramate (LCMS)	serum	RED with Clot Separator	mg/l	1 - 10	1 - 10	1 - 10	2 weeks	
Toxoplasmosis avidity	serum	RED with Clot Separator		see report	see report	see report	2 weeks	
Toxoplasmosis igG abs	serum	RED with Clot Separator	Index	see report	see report	see report	36 HRS	
Toxoplasmosis igM abs	serum	RED with Clot Separator	Index	see report	see report	see report	36 HRS	
TPHA i. serum	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
TPO abs (Antimicrosomal abs)	serum	RED with Clot Separator	ng/dl	< 60	< 60		2 weeks	
Tramadole	serum	RED with Clot Separator	ng/ml	60 - 600	60 - 600	60 - 600	2 weeks	
Transferrin	serum	RED with Clot Separator, or dark GREEN top	g/l	220 - 400	220 - 400	220 - 400	36 HRS	
Transglutaminase abs (EIA)	serum	RED with Clot Separator	Index	< 10	< 10	< 10	2 weeks	
Tranylcypromine	serum	RED with Clot Separator	ng/ml	20 - 50 ng/ml	20 - 50 ng/ml	20 - 50 ng/ml	2 weeks	
Treponema pall. igG abs (WB)	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Treponema pall. igM abs (WB)	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Triazolam	serum	RED with Clot Separator	ng/ml	2 - 20	2 - 20	2 - 20	2 weeks	
Trijodthyronine abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Trileptal	serum	RED with Clot Separator	mg/l	< 3.0	< 3.0	< 3.0	2 weeks	
Trimipramin	serum	RED with Clot Separator	ng/ml	therapeutical range: 20 - 200	therapeutical range: 20 - 200	therapeutical range: 20 - 200	2 weeks	
Troponin I	serum	RED with Clot Separator		see report	see report	See report	1 HR	
Trypanosomiasis	serum	RED with Clot Separator		see report	see report	see report	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Trypsin i. serum	serum	RED with Clot Separator	µg/l	10 - 57	10 - 57	up to 70	2 weeks	
Tryptophan	serum	RED with Clot Separator	µmol/l	up to 80	up to 80	up to 70	2 weeks	
TSH-receptor abs	serum	RED with Clot Separator	IU/l	< 1.0 = negative 1.0 - 1.5 = borderline > 1.5 = positive	< 1.0 = negative 1.0 - 1.5 = borderline > 1.5 = positive	< 1.0 = negative 1.0 - 1.5 = borderline > 1.5 = positive	2 weeks	
Tularemia igG abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Tularemia igM abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Tyrosin-Phos-phatase (IA-2A)	serum	RED with Clot Separator	U/ml	< 0.75	< 0.75	< 0.75	2 weeks	
Urea	serum		mmol/l				90 minutes	
Adults				2.8-7.8	2.8-7.8			
Newborns						1.4-4.3		
Infant/Child						1.8-6.4		
Urine	Urine		mmol/day			250-570		
Uric Acid	serum	RED with Clot Separator	µmol/l	220-450	150-350	120-320	90 minutes	
Vancomycin	serum	RED with Clot Separator	µg/ml	lower peak value: 5-10 upper peak value: 30-40 toxic: > 80	lower peak value: 5-10 upper peak value: 30-40 toxic: > 80	lower peak value: 5-10 upper peak value: 30-40 toxic: > 80	90 minutes	
Varicella zoster igG abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Varicella zoster igM abs	serum	RED with Clot Separator	Index	< 0.9	< 0.9	< 0.9	2 weeks	
Verapamil	serum	RED with Clot Separator	ng/ml	50 - 350	50 - 350	50 - 350	2 weeks	
Very long chain fatty acids	serum	RED with Clot Separator		see report	see report	see report	2 weeks	
Vigabatrin	serum	RED with Clot Separator	µg/ml	3 - 25	3 - 25	3 - 25	2 weeks	
Vitamin A (Retinol)	serum	RED with Clot Separator	µg/l	200 - 1200	200 - 1200	200 - 1200	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Vitamin B1 (Thiamine)	EDTA-blood	LAVENDER	µg/l	28 - 85	28 - 85	28 - 85	2 weeks	
Vitamin B12	EDTA-blood	LAVENDER	pg/mL				36 HRS	
<1 yr					F	228-1515		
					M	293-1210		
2- 3 yr					F	416-1210		
					M	264-1215		
4- 6 yr					F	313-1410		
					M	245-1075		
7- 9 yr					F	247-1175		
					M	271-1170		
10- 12 yr					F	196-1020		
					M	183-1090		
13- 18 yr					F	182-820		
					M	214-864		
Adults (Europe)				191-663	191-663			
Adults (USA)				211-946	211-946			
Vitamin B2 (FAD)	ED-TA-blood	RED with Clot Separator	µg/l	137 - 370	137 - 370	137 - 370	2 weeks	
Vitamin B6	serum	RED with Clot Separator	µg/l	5.0 - 30.0	5.0 - 30.0	5.0 - 30.0	2 weeks	
Vitamin C	serum	RED with Clot Separator	mg/l	5 - 15	5 - 15	5 - 15	2 weeks	
Vitamin D3 (1,25 OH)2	serum frozen	LAVENDER	pg/ml	16 - 65	16 - 65	20 - 135	36hrs	
Vitamin D3 (25-OH)	serum	RED with Clot Separator	ng/ml	10 - 70 toxic> 100 target val-ue dialysis patients:> 30	10 - 70 toxic> 100 target val-ue dialysis patients:> 30	10 - 70 toxic> 100 target value dialysis pa-tients:> 30	36hrs	
Vitamin E (Toxopherol)	serum	RED with Clot Separator	mg/l	5 - 18	5 - 18	5 - 18	2 weeks	
Vitamin H (Biotin)	serum	RED with Clot Separator	pg/ml	>200	>200	>200	2 weeks	
Warfarin	serum	RED with Clot Separator	µg/ml	1 - 3	1 - 3	1 - 3	2 weeks	
Weil Felix Test	serum	RED with Clot Separator		negative	negative	negative	24 HRS	
West-Nile-fe-ver (HHT)	serum	RED with Clot Separator		<1:10	<1:10	<1:10	2 weeks	

Test Name	Sample	Container (Top Color)	Units	RR Male	RR Female	Reference Ranges	TAT	(S) STAT availability
Yellow fever abs. (IIF)	serum	RED with Clot Separator		<1:10	<1:10	<1:10	2 weeks	
Yersinia igA abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Yersinia igG abs	serum	RED with Clot Separator		negative	negative	negative	2 weeks	
Zinc	serum	RED with Clot Separator	µg/dl	70 - 130	70 - 130	70 - 130	2 weeks	
Zonisamide	serum	RED with Clot Separator	µg/ml	10 - 40	10 - 40	10 - 40	2	

URINE AND BODY FLUID CHEMISTRY

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	TAT	(S) STAT availability
5-Hydroxyin-dol acetic acid	10 ml of 24 hr urine stabilised with 10 ml conc. HCl	Clean Container	mg/24 h	< 9.0	< 9.0	9.0	2 weeks	
Adenosine deaminase level	3 ml of 24-hr urine without stabiliza-tion - shipping frozen	Tap fluids including pleural in clean container		see report	see report	see report	2 weeks	
Adenosine Deaminase abs	Tap fluids e.g. pleural in EDTA	LAVENDER		1:<100 negative	1:<100 negative	1:<100 neg-ative	1 week	
Adrenalin (Epinephrine) i. urine	10 ml of 24 hr urine - stabilized with 10 ml conc. HCl	Clean Container	µg/24 h	< 20	< 20	< 1 yr = < 2.5 1 - 2 yrs = < 3.5 3 - 5 yrs = < 6.0 4 - 7 yrs. = < 10.0 7 - 10 yrs = < 14	2 weeks	
Albumin i. urine	urine	Clean Container	mg/l	normal range: < 20 microalbu-minuria: 20 - 200 macropro-teinuria: > 200	normal range: < 20 microal-buminuria: 20 - 200 macropro-teinuria: > 200	normal range: < 20 microalbu-minuria: 20 - 200 macropro-teinuria: > 200	90 minutes	
Albumin Creatinine Ratio	urine	Clean Container	mg/mmol creati-nine				90 minutes	

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	TAT	(S) STAT availability
<1 mth						<21		
1- 12 mth						<3.8		
1- 5 yr						<3.3		
6- 10 yr						<2.7		
11- 15 yr						<2.1		
Adults						<2.26		
Aldosterone i. urine	5 ml of 24 hr urine stabilised with 1 ml glacial acetic acid	Clean Container	µg/day	5 - 20	5 - 20	5 - 20	2 weeks	
Alpha-1-Mi-croglobulin i. urine	urine	Clean Container	mg/l	< 12	< 12	< 12	2 weeks	
Alpha-2-Mac-roglobulin i. urine	urine	Clean Container	mg/l	< 7.0	< 7.0	< 7.0	2 weeks	
Amino acid i. urine	10 ml urine, to the pre-vention of artifacts sample frozen	Clean Container		see report	see report	see report	2 weeks	
Amino-laevuline acid	10 ml of 24 hr urine stabilised with 1 ml glacial acetic acid	Clean Container	mg/24 h	up to 5.0 (BAT 15 mg/l)	up to 5.0 (BAT 6 mg/l)	see report	2 weeks	
Amphet-amines i. urine (Qualitative)	urine	Clean Container	ng/ml	< 600	< 600	< 600	2 weeks	
Amphet-amines i. urine (Quantitative)	urine	Clean Container		negative	negative	negative	90 min-utes	
Amylase i. urine	urine	Clean Container	U/l	up to 400	up to 400	up to 400	2 weeks	
Arsenic i. urine	urine	Clean Container	µg/l	< 25.0	< 25.0	< 25.0	2 weeks	
Barbiturates i. urine	urine	Clean Container	ng/ml	< 200	< 200	< 200	2 weeks	
Bence-Jones-protein electroph.	urine	Clean Container		see report	see report	see report	1 week	
Benzodiaze-pine i. urine	urine	Clean Container	ng/ml	< 200	< 200	< 200	2 weeks	

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	TAT	(S) STAT availability
Beta-2-Microgobuline i. urine	urine	Clean Container	µg/l	< 300	< 300	< 300	2 weeks	
Cadmium i. urine	urine	Clean Container	µg/l	< 1.3	< 1.3	< 1.3	2 weeks	
Cannabis/ Cannabinoids i. urine	urine	Clean Container	ng/ml	< 50	< 50	< 50	2 weeks	
Catechol-amines, total	10 ml of 24 hr urine - stabilised with 10 ml conc. HC	Clean Container	µg/24 h	117 - 575	117 - 575	117 - 575	2 weeks	
Chlamydia.ag trachomatis (PCR) i. urine	urine	Clean Container		negative	negative	negative	2 weeks	
Chromium i.urine	urine	Clean Container	µg/l	< 4.0	< 4.0	< 4.0	2 weeks	
Citrate	urine	Clean Container	mg/day	90 - 800	90 - 800	90 - 800	2 weeks	
Cocaine i. urine	urine	Clean Container	ng/ml	< 300	< 300	< 300	2 weeks	
Copper i. urine	urine	Clean Container	µg/day	< 40	< 40	< 40	2 weeks	
Cortisol free i. urine	10 ml of 24-hr. urine	Clean Container	µg/24 h	20 - 120	20 - 120	10 - 80	6 HRS	
Cotinine i.urine	urine	Clean Container	µg/ gCrea	Non smok-er: < 50 g/g Crea smoker: > 100 g/g Crea	Non smoker: < 50 g/g Crea smoker: > 100 g/g Crea	Non smoker: < 50 g/g Crea smoker: > 100 g/g Crea	2 weeks	
C-Peptide i. urine	urine	Clean Container	µg/day	23 - 155	23 - 155	23 - 155	2 weeks	
Creati-nine-conc.i. urine	urine	Clean Container	mg/dl	30 - 125	30 - 125	30 - 125		
1st morning urine				3.46- 22.9 mmol/L	2.47- 19.2 mmol/L		90 minutes	
24 hr urine				9.2- 20.7 mmol/d	6.6- 13.9 mmol/d		6 HRS	
Creatinine i. urine	urine	Clean Container	g/24 h	1.0 - 2.0	1.0 - 2.0	1.0 - 2.0	6 HRS	
Cystine i.urine	10 ml urine, to the pre-vention of artifacts sample frozen	Clean Container	mg/day	up to 40	up to 40	up to 40	2 weeks	

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	TAT	(S) STAT availability
Drugs of Abuse Qualitative Screen								
Dopamine i. urine	urine	Clean Container	µg/24 h	< 450	< 450	< 450	2 weeks	
Electrolytes (24hr urine)		Clean Container					6 HRS	
sodium		Clean Container	mmol/ 24hrs	80-180	80-180			
potassium		Clean Container	mmol/ 24hrs	40-80	40-80			
chloride		Clean Container	mmol/ 24hrs	110-250	110-250			
calcium		Clean Container	mmol/ 24hrs	2.5-6.3	2.5-6.3			
Epinephrine (Adrenalin) i. urine	10 ml of 24 hr urine - stabilized with 10 ml conc. HCl	Clean Container	µg/24 h	< 20	< 20	< 1 yr = < 2.5 1 - 2 yrs = < 3.5 3 - 5 yrs = < 6.0 4 - 7 yrs. = < 10.0 7 - 10 yrs = < 14	2 weeks	
Fluoride i. urine	urine	Clean Container		up to 1 mg/l BAT = 7 mg/g Crea	up to 1 mg/l BAT = 7 mg/g Crea	up to 1 mg/l BAT = 7 mg/g Crea	2 weeks	
Immunoglobulin G i. urine	urine	Clean Container	mg/l	< 10.0	< 10.0	see report	2 weeks	
Jodid i.urine	urine	Clean Container	µg/g Krea	11 - 403	11 - 403	11 - 403	2 weeks	
Lead i. urine	urine	Clean Container	µg/l	< 30 (BAT 50)	< 30 (BAT 50)	< 30 (BAT 50)	2 weeks	
Legionella pneum. ag.	urine	Clean Container		negative	negative	negative	2 weeks	
Magnesium i. urine	urine	Clean Container	mg/24 h	120 - 180	120 - 180	120 - 180	2 weeks	
Manganese i. urine	urine	Clean Container	ug/l	< 1.9	< 1.9	< 1.9	2 weeks	
Mercury i.urine	urine	Clean Container	µg/l	up to 1.4 (BAT 30)	up to 1.4 (BAT 30)	up to 1.4 (BAT 30)	2 weeks	
Metanephrines	10 ml of 24 hr urine - stabilized with 10 ml conc. HCl	Clean Container	µg/24 h	< 340	< 340	< 340	2 weeks	

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	TAT	(S) STAT availability
Methadone i. urine	urine	Clean Container	ng/ml	< 300	< 300	< 300	2 weeks	
Methylmalonic acid	urine	Clean Container	mg/g Krea	up to 10	up to 15	up to 10	2 weeks	
Myoglobin i. urine	urine	Clean Container		negative	negative	negative	2 weeks	
Myoglobin i. urine quant	urine	Clean Container	µg/l	< 50	< 50	< 50	2 weeks	
Norepineph-rine i. urine	10 ml of 24 hr urine - stabilised with 10 ml conc. HCl	Clean Container	µg/24 h	< 105	< 105	< 1 yr.: < 10 1 - 2 yr.: < 17 2 - 4 yr.: < 29 4 - 7 yr.: < 45 7 - 10 yr.: < 65 10 -18 yr.:< 105	2 weeks	
Normeta-nephrines	10 ml of 24 hr urine - stabilized with 10 ml conc. HCl	Clean Container	µg/24 h	< 445	< 445	< 445	2 weeks	
Opiates i. urine	urine	Clean Container	ng/ml	< 300	< 300	< 300	2 weeks	
Orotic Acid i. urine	urine	Clean Container		see report	see report	see report	2 weeks	
Osmolality i.urine (*Cal-culated) *Cal-culated urine Osmolality is inaccurate	urine	Clean Container	mos-mol/ kgH20	50 - 1600	50 - 1600	50 - 1600	6 HRS	
Osmolality i.urine (Mea-sured)	urine	Clean Container	mos-mol/ kgH20	50 - 1600	50 - 1600	50 - 1600	2 weeks	
Oxalic acid	urine	Clean Container	mg/day	< 40	< 40	< 40	2 weeks	
Phenol i. urine	urine	Clean Container	mg/l	normal < 15 biological tolerance value: 200	normal < 15 biological tolerance value: 200	normal < 15 biological tol-erance value: 200	2 weeks	
Porphobilino-gen	10 ml of 24 hrs.urine	Clean Container	mg/d	< 1.6	< 1.6	< 1.6	2 weeks	
Porphyrin (total)	10 ml of 24 hr urine preserved in dark	Clean Container	µg/24 h	20 - 100	20 - 100	20 - 100	2 weeks	

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	TAT	(S) STAT availability
Pregnantriol i. urine	10 ml of 24 hr urine stabilized with 1 ml glacial acetic acid	Clean Container	mg/l	up to 2 mg/day	up to 2 mg/day	< 6 years: up to 0.15 mg/day 6 - 11 years: up to 0.4 mg/day 11 - 14 years: up to 1.5 mg/day	2 weeks	
Protein, total i. urine	urine	Clean Container	mg/l	< 120	< 120	< 120	90 minutes	
Protein, total in 24-hrs-urine	10 ml of 24 hrs.urine	Clean Container	mg/24 h	< 140	< 140	< 140	6 HRS	
Serotonin i. urine	10 ml of 24 hr urine -10 ml of 24 hr urine - stabilized with 10 ml conc. HCl	Clean Container	µg/day	< 200 ug/day	< 200 ug/day	< 200 ug/day	2 weeks	
Thallium urine	urine	Clean Container	µg/l	< 10	< 10	< 10	2 weeks	
Tricyclic anti-depres.	urine	Clean Container		see report	see report	see report	2 weeks	
Urate i. urine	urine	Clean Container	mg/24 h	80 - 800	80 - 800	80 - 800	2 weeks	
Uropropor-phyrin	10 ml of 24 hr urine preserved in dark	Clean Container	µg/24 h	5 - 20	5 - 20	5 - 20	2 weeks	
Vanillinmandelic acid (VMA)	10 ml of 24 hr urine - stabilised with 10 ml conc. HCl	Clean Container	mg/24 h	< 7.0	< 7.0	Infant:< 1.5 mg/d or < 1.8 mg/g Crea Children: < 2.2 mg/d or < 11 mg/g Crea	2 weeks	

CSF CHEMISTRY

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	Frequency	TAT	(S) STAT availability
ACE i. csf	csf		mU/ml	< 2.0	< 2.0	< 2.0	Daily	2 weeks	
Albumin i. csf	csf+ serum		g/l	0.066-0.44	0.066-0.44		Daily	90 minutes	
Borrelia burgd. IgG i.csf	csf and serum			negative	negative	negative	Daily	2 weeks	
Borrelia burgd.IgM i.csf	csf and serum			negative	negative	negative	Daily	2 weeks	
C-Reactive Protein i. puncture	puncture specimen		mg/l	up to 6.0	up to 6.0	up to 6.0	Daily	2 weeks	
Crypt. neofor-mans antigen i. csf	csf			see report	see report	see report	Daily	90 minutes	
CytomegaloV PCR i. csf	csf			negative	negative	negative	Daily	2 weeks	
CytomegaloV igG abs i. csf	csf+ serum			negative	negative	negative	Daily	2 weeks	
CytomegaloV igM abs i. csf	csf+ serum			negative	negative	negative	Daily	2 weeks	
EBV igG abs i. csf	csf+ serum			negative	negative	negative	Daily	2 weeks	
EBV igM abs i. csf	csf+ serum			negative	negative	negative	Daily	2 weeks	
Glucose	csf+ serum		mmol/l				Daily	90 minutes	S
1-16 yrs						1.8-4.6			
>16 yrs				2.2-4.6	2.2-4.6				
Herpes sim-plex type 1/2 DNA i. csf	csf			negative	negative	negative	Daily	2 weeks	
Herpes sim-plex type 1/2 igG abs i. csf	csf+ serum			negative	negative	negative	Daily	2 weeks	
Herpes sim-plex type 1/2 igM abs i. csf	csf+ serum			negative	negative	negative	Daily	2 weeks	
Herpes simplex virus PCR i. csf	csf			negative	negative	negative	Daily	2 weeks	
Human Her-pes Virus 6 DNA i. csf	csf			see report	see report	see report	Daily	2 weeks	
Immunoglob-ulin G i. csf	csf+ serum		mg/dl	1.4 - 3.8	1.4 - 3.8	1.4 - 3.8	Daily	2 weeks	

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	Frequency	TAT	(S) STAT availability
Immunoglobulin M i. csf	csf+ serum		mg/dl	up to 0.1	up to 0.1	up to 0.1	Daily	2 weeks	
JC-Virus DNA i. csf	csf			see report	see report	see report	Daily	2 weeks	
Lactate i. csf	csf		mg/dl	10 - 20	10 - 20	10 - 20	Daily	24hrs	
3-10 days			mmol/l			1.1-4.4			
10-30 days						1.1-2.8			
1-12 months						1.1-2.8			
1-15 yrs						1.1-2.8			
>15 yrs				1.1-2.4	1.1-2.4				
Measles igG abs (POD-B) i. csf	csf+ serum		IU/l	negative	negative	negative	Daily	2 weeks	
Measles igM abs i. csf	csf+ serum			negative	negative	negative	Daily	2 weeks	
Measles virus PCR i. csf	csf			see report	see report	see report	Daily	2 weeks	
Mumps virus igG abs i. csf	csf+ serum			negative	negative	negative	Daily	2 weeks	
Mumps virus igM abs i. csf	csf+ serum			negative	negative	negative	Daily	2 weeks	
Myelin associated glycoprotein abs i. csf	csf		Elisa-LU/ml	< 100	< 100	< 100	Daily	2 weeks	
Oligoclonal bands	csf+ serum			negative	negative	negative	Daily	2 weeks	
Protein	csf+ serum		g/l	0.2-0.5	0.2-0.5		Daily	2 HRS	S
TBC PCR i. csf	csf			negative	negative	negative	Daily	36HRS	
TBC PCR i. puncture specimen	puncture specimen			negative	negative	negative	Daily	2 weeks	
Toxoplasmosis igG abs i. csf	csf+ serum			negative	negative	negative	Daily	2 weeks	
Toxoplasmosis igM abs i. csf	csf+ serum			negative	negative	negative	Daily	2 weeks	
TPHA i. csf	csf			negative	negative	negative	Daily	2 weeks	
Varicella zoster PCR i.csf	csf			negative	negative	negative	Daily	2 weeks	
Varicella zoster virus igG abs i. csf	csf+ serum			negative	negative	negative	Daily	2 weeks	

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	Frequency	TAT	(S) STAT availability
Varicella zoster virus igM abs i. csf	csf+serum			negative	negative	negative	Daily	2 weeks	
West nile fever PCR	csf			see report	see report	see report	Daily	2 weeks	
All CSF should be collected in plain bottles and NOT in those with anticoagulants unless otherwise specified.									
All CSF should be accompanied by a simultaneously drawn serum sample unless otherwise specified.									
THERAPEUTIC DRUG MONITORING									
Test Name	**Sample**	**Container**	**Units**	**RR Male**	**RR Female**	**RR Paeds**	**Frequency**	**TAT**	**Collection**
Acetaminophen	serum	RED with Clot Separator	µg/ml	10.0-20.0	10.0-20.0		Daily	24 HRS	
Alprazolam	serum	RED with Clot Separator	µg/l	5 - 50	5 - 50	5 - 50	Daily	2 weeks	
Amikacin	serum	RED with Clot Separator	µg/ml				Daily	90 minutes	
Peak				20-25					Max: 0.5-1 h after end of a 30min infusion (1h after IM dose)
Trough				5.0-10.0					Min: immediately before next dose
Amitriptyline	serum	RED with Clot Seperator	ng/ml	50 - 250	50 - 250	50 - 250	Daily	2 weeks	
Acetylsalicylic acid	serum	RED with Clot Seperator		see report	see report	see report	Daily	2 weeks	1-3 h after oral dose
Atenolol-Level	serum	RED with Clot Seperator	ug/ml	0.05 - 1.0	0.05 - 1.0	0.05 - 1.0	Daily	2 weeks	
Atomoxetine (LCMS)	serum	RED with Clot Seperator	ng/ml	see report	see report	see report	Daily	2 weeks	
Benzodiazepine	serum/ urine	RED with Clot Seperator OR Urine Container		see report	see report	see report	Daily	24 HRS	

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	Frequency	TAT	Collection
Benzhexol (Trihexyphe-nidyl) HPLC	serum	RED with Clot Seperator		see report	see report	see report	Daily	2 weeks	
Buspirone	serum	RED with Clot Seperator	ng/ml	1 - 10	1 - 10	1 - 10	Daily	2 weeks	
Carbamaze-pine	serum		µg/ml	8.0-12.0	8.0-12.0		Daily	2 HRS	Max: 6-8 hafter final dose Min: imme-diately before next dose
Clobazam	serum	RED with Clot Seperator	ng/ml	100 - 400	100 - 400	100 - 400	Daily	2 weeks	
Clomip-ramine	serum	RED with Clot Seperator	ng/ml	20 - 140	20 - 140	20 - 140	Daily	2 weeks	
Clonazepam	serum	RED with Clot Seperator	ng/ml	ther. range: 15- 60	ther. range: 15- 60	ther. range: 15- 60	Daily	2 weeks	
Clozapine	serum	RED with Clot Seperator	ng/ml	ther. range 50-700	ther. range 50-700	ther. range 50-700	Daily	2 weeks	
Cocaine i. serum	serum	RED with Clot Seperator	ng/ml	see report	see report	see report			
Cyclospo-rine A	EDTA	LAVEN-DER	ng/ml				Daily	90 min-utes	
Therapeutic Pre dose (C0) levels									
12hrs after dose				100-400					12hrs after dose
24hrs after dose				100-200					24hrs after dose
Toxic pre dose levels (C0)				> 400					
C2 - month post trans-plantation									
1 and 2				1700-2100					2 h post dose
3				1200-1500					2 h post dose

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	Frequency	TAT	Collection
4 and 5				1000-1200					2 h post dose
6 and >				800-1000					2 h post dose
Digoxin	serum	RED with Clot Separator	ng/ml	0.8-2.0	0.8-2.0		Daily	90 minutes	8-24 h after initial dose
Drugs of Abuse Screen (Qualitative):	Urine	Clean Container		Negative	Negative	Negative	Daily	90 minutes	
Ethosuximide	serum	RED with Clot Separator	µg/ml	40 - 100	40 - 100	40 - 100	Daily	2 weeks	Interval between doses
Gentamicin	serum	RED with Clot Separator	µg/ml				Daily	90 minutes	
Peak				6.0-10.0	6.0-10.0				Max: 30min post IV infusion
Trough				0.5-2.0	0.5-2.0				Min: immediately before next dose
Lidocaine	serum	RED with Clot Separator	µg/ml	1.5-5.0	1.5-5.0	1.5-5.0	Daily	2 weeks	During infusion
Lithium	serum	RED with Clot Separator	mmol/l	0.3-1.2	0.3-1.2		Daily	2 HRS	12 h after final dose
Methotrexate	serum	RED with Clot Separator	µmol/l	High dosage therapy: after 24 hrs. < 10 after 48 hrs. < 1.0 after 72 hrs. < 0.1 <Low dosage therapy (10-20 mg once the week): after 2 h 0.2 - 0.9 after 4 h 0.1 - 0.4 after 24 h ca. 0.1	High dosage therapy: after 24 hrs. < 10 after 48 hrs. < 1.0 after 72 hrs. < 0.1 <Low dosage therapy (10-20 mg once the week): after 2 h 0.2 -0.9 after 4 h 0.1 - 0.4 after 24 h ca. 0.1	High dosage therapy: afert 24 hrs. < 10 after 48 hrs. < 1.0 after 72 hrs. < 0.1 <Low dosage therapy (10-20 mg once the week): after 2 h 0.2 - 0.9 after 4 h 0.1 - 0.4 after 24 h ca. 0.1	Daily	2 weeks	Sampling Time Depends on dose, duration of infusion and clinical status of the patient.

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	Frequency	TAT	Collection
Mycopheno-late-Mofetil	serum		µg/ml	see report	see report	see report	Daily	2 weeks	
N-Acetytyl-procain-am-ide (NAPA)	serum						Daily	2 weeks	
Phenobarbital	serum		µg/ml	10.0-40.0			Daily	90 minutes	Interval between doses
Phenytoin (Total)	serum		µg/ml				Daily	90 minutes	Interval between doses
Premature Infants						6.0-14.0			
Adults				10.0-20.0	10.0-20.0				
Primidone	serum		µg/ml	5.0 - 15.0	5.0 - 15.0	5.0 - 15.0	Daily	2 weeks	
Promethazine	serum		ng/ml	50 - 400	50 - 400	50 - 400	Daily	2 weeks	
Propranolol	serum	RED with Clot Separator	ng/ml	50 - 300	50 - 300	50 - 300	Daily	2 weeks	
Procainamide	serum	RED with Clot Sep-arator					Daily	2 weeks	Max: 1-5 after final dose Min: imme-diately before next dose
Quinidine	serum	RED with Clot Separator	µg/ml	1.0-5.0	1.0-5.0	1.0-5.0	Daily	2 weeks	Max: 1-3 h (slow release products 8h) after final dose Min: im-mediately before next dose
Salicylate	serum	RED with Clot Separator	mg/l	therap. range: 20 - 200	therap. range: 20 - 200	therap. range: 20 - 200	Daily	2 weeks	
Tacrolimus (Prograf/FK 506)	EDTA-blood	LAVEN-DER	ng/ml	5 - 20	5 - 20	see report	Daily	2 weeks	

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	Frequency	TAT	Collection
Theophylline	serum	RED with Clot Separator	µg/ml				Daily	90 minutes	Collect during continuous infusion Max: 1h (slow release products 4h) after final dose
Premature Infants						6.0-11.0			
Adults				8.0-20.0	8.0-20.0				
Topiramate (LCMS)	serum	RED with Clot Separator	mg/l	1 - 10	1 - 10	1 - 10	Daily	2 weeks	
Valproic acid	serum	RED with Clot Separator	µg/ml				Daily	90 minutes	
Plasma Value				50-100	50-100				Max: 1-4h (up to 8 h) after final dose
Free				5-10% of plasma value	5-10% of plasma value				Min: immediately before next dose
Vancomycin	serum	RED with Clot Separator	µg/ml				Daily	90 minutes	
Peak				20-40	20-40				Max: 1 hr post IV infusion
Trough				5.0-10.0	5.0-10.0				Min: immediately before next dose
Warfarin	serum	RED with Clot Separator	µg/ml	1 - 3	1 - 3	1 - 3	Daily	2 weeks	

HAEMATOLOGY									
Test Name	**Sample**	**Container**	**Units**	**RR Male**	**RR Female**	**RR Paeds**	**Frequency**	**TAT**	**(S) STAT availability**
Antithrombin III	1ml citrat-plasma frozen	SKY BLUE	%	80-120	80-120	80-120	Daily	2 weeks	
APC resistance	1ml citrat-plasma frozen	SKY BLUE	Ratio	< 2.4 results < 2.2 indicates an abnormal risk of thrombosis	< 2.4 results < 2.2 indicates an abnormal risk of thrombosis	< 2.4 results < 2.2 indicates an abnormal risk of thrombosis	Daily	2 weeks	
APC resistance with anticoagulant	1ml citrat-plasma frozen	SKY BLUE	Ratio	> 0.8	> 0.8	> 0.8	Daily	2 weeks	
Activated Partial Thromboplastin Time (APTT)	citrated blood 1:10	SKY BLUE	seconds	24.9-36.8	24.9-36.8		Daily	4 HRS	S
Blood Grouping (ABO Rhesus)	EDTA-blood	LAVENDER		See Report				90 minutes	S
Carboxy hemoglobin	EDTA-blood	LAVENDER	%	smoker: 1.5 - 5.0 % Hb non smoker: < 1.5 %	smoker: 1.5 - 5.0 % Hb non smoker: < 1.5 %	smoker: 1.5 - 5.0 % Hb non smoker: < 1.5 %	Daily	2 weeks	
D-Dimer	citrated blood 1:10	SKY BLUE	FEU/ml	<0.5			Daily	4 HRS	S
Erythrocytes			*106/ml			4.3- 6.3	Daily	90 minutes	S
1 d						4.0- 6.8			
2- 6 d						3.7- 6.1			
14- 23 d						3.2- 5.6			
24- 37 d						3.1- 5.1			
40- 50 d						2.8- 4.8			
2- 2.5 mth						3.1- 4.7			
3- 3.5 mth						3.2- 5.2			
5- 7 mth						3.6- 5.2			
8 mth-3 yr						3.7- 5.7			
5 yr						3.8- 5.8			
10 yr						4.1- 5.1			
Adults				4.5- 5.9	4.5- 5.9				

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	Frequency	TAT	(S) STAT availability
Erythrocyte sedimentation rate (ESR)	EDTA-blood	LAVENDER	mm/1 hr				90 minutes	90 minutes	
Adults <50 years				<15	<25				
Adults >50 years				<20	<30				
G6-PDH (Glucose-6-PDH)	EDTA-blood	LAVENDER	mU/Mrd. Ery	221 - 570	221 - 570	221 - 570	Daily	1 week	
Hematocrit (HCT, PCV)	EDTA-blood	LAVENDER	%				Daily	90 minutes	S
1 d						44-72			
2- 6 d						50-82			
14- 23 d						42-62			
24- 37 d						31-59			
40- 50 d						30-54			
2- 2.5 mth						30-46			
3-3.5 mth						31-43			
5- 7 mth						32-44			
8 mth - 3 yr						35-43			
5 yr						31-43			
10 yr						33-45			
Adults				40-52	35-47				
Hemoglobin (Hb)	EDTA-blood	LAVENDER	g/dl				Daily	90 minutes	S
1 d						15.2-23.6			
2- 6 d						15.0-24.6			
14- 23 d						12.7-18.7			
24- 37 d						10.3-17.9			
40- 50 d						9.0-16.6			
2- 2.5 mth						9.2-13.6			
3- 3.5 mth						9.6-12.8			
5- 7 mth						10.1-12.9			
8- 10 mth						10.5-12.9			
11- 13.5 mth						10.7-13.1			
1.5- 3 yr						10.8-12.8			
5 yr						10.7-14.7			

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	Frequency	TAT	(S) STAT availability
10 yr						10.8-15.6			
Adults				14.0-17.5	12.3-15.3				
>70 yrs				12.1-17.6	11.7-16.2				
>75 yrs				11.8-17.5	11.6-16.1				
>81 yrs				11.6-16.3	10.9-15.5				
Leukocyte Count	EDTA-blood	LAVEN-DER	*103/ ml				Daily	90 minutes	S
12 h						13.0- 38.0			
1 d						9.4- 34.0			
1 w						5.0- 21.0			
2 w						5.0- 20.0			
4 w						5.0- 19.5			
2 mth						5.5- 18.0			
4- 12 mth						6.0- 17.5			
2 yr						6.0- 17.0			
4 yr						5.5- 15.5			
6 yr						5.0- 14.5			
8- 12 yr						4.5- 13.5			
14- 16 yr						4.5- 13.0			
18 yr				4.5- 12.5	4.5- 12.5				
20 yr				4.5- 11.5	4.5- 11.5				
Adults				4.4- 11.3	4.4- 11.3				
MCH	EDTA-blood	LAVEN-DER	pg/ cell				Daily	90 minutes	S
1 d						33- 41			
2- 6 d						29- 45			
14- 37 d						26- 38			
40- 50 d						25- 37			
2- 2.5 mth						24- 36			
3- 3.5 mth						23- 36			
5- 10 mth						21- 33			
11 mth - 5 yr						23- 31			
10 yr						22- 34			
Adults				28- 33	28- 33				
MCHC		LAVEN-DER					Daily	90min-utes	S
1 d						31- 35			

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	Frequency	TAT	(S) STAT availability
2- 6 d						24- 36			
14- 23 d						26- 34			
24- 37 d						25- 37			
40 d - 7 mth						26- 34			
8- 13.5 mth						28- 32			
1.5- 3 yr						26- 34			
5- 10 yr						32- 36			
Adults				33- 36	33- 36				
MCV	ED-TA-blood	LAVEN-DER	fl				Daily	90min-utes	S
1 d						98- 122			
2- 6 d						94- 150			
14- 23 d						84- 128			
24- 37 d						82- 126			
2- 2.5 mth						81- 121			
3- 3.5 mth						77- 113			
5- 7 mth						73- 109			
8- 10 mth						74- 106			
11- 13.5 mth						74- 102			
1.5- 3 yr						73- 101			
5 yr						72- 88			
10 yr						69- 93			
Adults				80- 96	80- 96				
Factor II	citrated blood 1:10	SKY BLUE	%	> 70	> 70	> 70		2 weeks	
Factor II - Genotype	EDTA-blood	LAVEN-DER		see report	see report	see report		2 weeks	
Factor IX	citrated blood 1:10	SKY BLUE	%	70 - 130	70 - 130	70 - 130		2 weeks	
Factor V	citrated blood 1:10	SKY BLUE	%	> 70	> 70	> 70		2 weeks	
Factor V Leiden	EDTA-blood	LAVEN-DER						2 weeks	
Factor VII	citrated blood 1:10	SKY BLUE	%	> 70	> 70	> 70		2 weeks	

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	Frequency	TAT	(S) STAT availability
Factor VIII (Wille-brand)	citrated blood 1:10	SKY BLUE	%	(bloodgroup specific nr:) A, B, AB: > 60 % 0 < 50 %	(bloodgroup specific nr:) A, B, AB: > 60 % 0 < 50 %	(bloodgroup specific nr:) A, B, AB: > 60 % 0 < 50 %		2 weeks	
Factor VIII activity	citrated blood 1:10	SKY BLUE	%	70 - 120	70 - 120	70 - 130		2 weeks	
Factor X	citrated blood 1:10	SKY BLUE	%	70 - 130	70 - 130	70 - 130		2 weeks	
Factor XI	citrated blood 1:10	SKY BLUE	%	70 - 130	70 - 130	70 - 130		2 weeks	
Factor XII	citrated blood 1:10	SKY BLUE	%	> 70	> 70	> 70		2 weeks	
Factor XIII	citrated blood 1:10	SKY BLUE	%	70 - 130	70 - 130	70 - 130		2 weeks	
Flow Cytometry: For acute leukemias, chronic leuke-mias and selcted lympho-mas.	Contact the sec-tion								
Ferritin	serum	RED WITH CLOT SERPER-ATOR	ng/ml	22 - 322	10 - 291		Daily	24 HRS	
Fibrinogen	citrated blood 1:10	SKY BLUE	mg/dl	180 - 350	180 - 350	180 - 350		2 weeks	
Flow Cytometry: For acute leukemias, chronic leuke-mias and selcted lympho-mas.	Whole Blood or Bone Marrow Sample	LAVEN-DER		see report	see report	see report		1 week	
Haptoglo-bin	serum	RED WITH CLOT SERPER-ATOR	g/l	0.30 - 2.00	0.30 - 2.00	0.30 - 2.00		2 weeks	

Test Name	Sample	Container	Units	RR Male	RR Female	RR Paeds	Frequency	TAT	(S) STAT availability
HbA2 (Hb-elec-trophore-sis)	EDTA-blood	LAVEN-DER		see report	see report	see report		2 weeks	
Hb-elec-trophoresis	EDTA-blood	LAVEN-DER		see report	see report	see report		1 Week	
HbF (Hb-elec-trophore-sis)	EDTA-blood	LAVEN-DER		see report	see report	see report		2 weeks	
Lupus-an-ticoagu-lans-diag-nostic	6 ml citrated plasma								
frozen	SKY BLUE		see report	see report	see report		2 weeks		
Malaria antigen	ED-TA-blood	LAVEN-DER		negative	negative	negative	Daily	90min-utes	S
Protein C	citrated plasma								
frozen	SKY BLUE	%	69 - 134	70 - 140	69 - 134		2 weeks		
Protein S - activity	citrated plasma								
frozen	SKY BLUE	%	75 - 130	60 - 140			2 weeks		
Prothrom-bin Time (PT)	citrated blood 1:10	SKY BLUE	sec-onds	11.5-15.0	11.5-15.0			90min-utes	S
			activ-ity	120-70 %	120-70 %				
			ratio	0.9-1.15	0.9-1.15				
Thrombin Time (TT)	citrated blood 1:10	SKY BLUE	sec-onds	12.0-16.0	12.0-16.0		Daily	90min-utes	S
			ratio	0.85-1.15	0.85-1.15				

PAEDIATRIC HAEMATOLOGY											
Normal red-cell values in utero or on the first day of life in preterm infants											
	Gesta-tional age (weeks)	RBC	Hb	Hct	MCV	MCH	MCHC	RDW	Reticu-locyte		
		(×1012 /l)	(g/dl)	(%)	(fl)	(pg)	(g/l)	(%)	(%)		
Fetal	18–21	2.85 ± 0.36	11.69 ± 1.27	37.3 ± 4.32	131.11 ± 10.97						
	22–25	3.09 ± 0.34	12.2 ± 1.6	38.59 ± 3.94	125.1 ± 7.84						
	26–29	3.46 ± 0.41	12.91 ± 1.38	40.88 ± 4.4	118.5 ± 7.96						
	30	3.82 ± 0.64	13.64 ± 2.21	43.55 ± 7.2	114.38 ± 9.34						
Post-natal	23–25		14.5 ± 1.6	43.5 ± 4.2	115.6 ± 5.6	38.6 ± 2.2	33.4 ± 0.9	15.9 ± 1.4			
	26–28		15.1 ± 1.6	45 ± 4.5	114 ± 7.6	38.3 ± 2.9	33.6 ± 0.6	16.5 ± 1.9			
Central	29–31		16.2 ± 1.7	48 ± 5	110.4 ± 6.6	37.3 ± 2.5	33.7 ± 0.7	16.4 ± 1.5			
Post-natal	24–25	4.65 ± 0.43	19.4 ± 1.5	63 ± 4	135 ± 0.2				6 ± 0.5		
	26–27	4.73 ± 0.45	19 ± 2.5	62 ± 8	132 ± 14.4				9.6 ± 3.2		
Capil-lary	28–29	4.62 ± 0.75	19.3 ± 1.8	60 ± 7	131 ± 13.5				7.5 ± 2.5		
	30–31	4.79 ± 0.74	19.1 ± 2.2	60 ± 8	127 ± 12.7				5.8 ± 2		
	32–33	5.00 ± 0.76	18.5 ± 2	60 ± 8	123 ± 15.7				5 ± 1.9		
	34–35	5.09 ± 0.5	19.6 ± 2.1	61 ± 7	122 ± 10				3.9 ± 1.6		
	36–37	5.27 ± 0.68	19.2 ± 1.7	64 ± 7	121 ± 12.5				4.2 ± 1.8		
	Term	5.14 ± 0.7	19.3 ± 2.2	61 ± 7.4	119 ± 9.4				3.2 ± 1.4		

Red blood cell values in the first year of life											
	RBC (×1012 /l)	Hb (g/dl)	Hct (%)	MCV (fl)	MCH (pg)	MCHC (g/l)	Reticu-locyte Count (absou-lute or %)				
Term infants											
Day											
1	5.14 ± 0.7	19.3 ± 2.2	61 ± 7.4	119 ± 9.4		31.6 ± 1.9	3.2 ± 1.4				
2	5.15 ± 0.8	19 ± 1.9	60 ± 6.4	115 ± 7.0		31.6 ± 1.4	3.2 ± 1.3				
3	5.11 ± 0.7	18.8 ± 2	62 ± 9.3	116 ± 5.3		31.1 ± 2.8	2.8 ± 1.7				
4	5. ± 0.6	18.6 ± 2.1	57 ± 8.1	114 ± 7.5		32.6 ± 1.5	1.8 ± 1.1				
5	4.97 ± 0.4	17.6 ± 1.1	57 ± 7.3	114 ± 8.9		30.9 ± 2.2	1.2 ± 0.2				
6	5.00 ± 0.7	17.4 ± 2.2	54 ± 7.2	113 ± 10		32.2 ± 1.6	0.6 ± 0.2				
7	4.86 ± 0.6	17.9 ± 2.5	56 ± 9.4	118 ± 11.2		32 ± 1.6	0.5 ± 0.4				
Week											
1–2	4.80 ± 0.8	17.3 ± 2.3	54 ± 8.3	112 ± 19		32.1 ± 2.9	0.5 ± 0.3				
2–3	4.2 ± 0.6	15.6 ± 2.6	46 ± 7.3	111 ± 8.2		33.9 ± 1.9	0.8 ± 0.6				
3–4	4.00 ± 0.6	14.2 ± 2.1	43 ± 5.4	105 ± 7.5		33.5 ± 1.6	0.6 ± 0.3				
4–5	3.60 ± 0.4	12.7 ± 1.6	36 ± 4.8	101 ± 8.1		34.9 ± 1.6	0.9 ± 0.8				
5–6	3.55 ± 0.2	11.9 ± 1.5	36 ± 6.2	102 ± 10.2		34.1 ± 2.9	1 ± 0.7				
6–7	3.4 ± 0.4	12 ± 1.5	36 ± 4.8	105 ± 12		33.8 ± 2.3	1.2 ± 0.7				
7–8	3.4 ± 0.4	11.1 ± 1.1	33 ± 3.7	100 ± 13		33.7 ± 2.6	1.5 ± 0.7				
8–9	3.4 ± 0.5	10.7 ± 0.9	31 ± 2.5	93 ± 12		34.1 ± 2.2	1.8 ± 1				
9–10	3.6 ± 0.3	11.2 ± 0.9	32 ± 2.7	91 ± 9.3		34.3 ± 2.9	1.2 ± 0.6				
10–11	3.7 ± 0.4	11.4 ± 0.9	34 ± 2.1	91 ± 7.7		33.2 ± 2.4	1.2 ± 0.7				
11–12	3.7 ± 0.3	11.3 ± 0.9	33 ± 3.3	88 ± 7.9		34.8 ± 2.2	0.7 ± 0.3				

Month											
4	4.35 ± 0.38	11.6 ± 0.8	34.1 ± 2.2	78.2 ± 4.6	26.9 ± 1.5		44 000 ± 20 000d &#2;l-1				
5	4.51 ± 0.27	11.9 ± 0.9	35.6 ± 2.2	77.8 ± 2.8	26.5 ± 1.3		34 000 ± 15 000 &#2;l-1				
6	4.58 ± 0.32	11.5 ± 0.8	33.8 ± 2.2	74.7 ± 2.9	24.9 ± 1.4		34 000 ± 15 000 &#2;l-1				
7–8	4.89 ± 0.31	12.8 ± 0.6	37.9 ± 3.8	77.5 ± 3.6	26.2 ± 2.0		36 000 ± 14 000 &#2;l-1				
9–10	4.85 ± 0.22	12.5 ± 1	37.2 ± 3.7	76.8 ± 5.2	25.9 ± 1.7						
1Year	5.00 ± 0.23	12.3 ± 1	37.3 ± 4.1	74.6 ± 5	24.8 ± 2.7						
Preterm infants											
Day											
1	4.71 ± 0.75	18.2 ± 2.7		115 ± 5	38.9 ± 1.7	33.5 ± 1.2					
3	4.4 ± 0.83	16.2 ± 2.9		112 ± 4	39.0 ± 3.4	33.8 ± 1					
7	4.45 ± 0.83	16.3 ± 2.9		110 ± 5	37.3 ± 1.8	33.9 ± 1.3					
14	4.10 ± 0.69	14.5 ± 2.4		106 ± 5	36.3 ± 1.9	33.9 ± 1					
21	3.71 ± 0.59	12.9 ± 2		102 ± 5	35.3 ± 2.2	34.2 ± 1.1					
28	3.17 ± 0.6	10.9 ± 1.9		100 ± 5	35.1 ± 1.9	34.4 ± 1.0					
35	2.97 ± 0.45	10 ± 1.4		98 ± 5	34.4 ± 1.5	34.5 ± 0.7					
42	2.94 ± 0.49	9.5 ± 1.5		97 ± 5	32.2 ± 1.7	33.7 ± 0.9					
49	3.21 ± 0.59	10.1 ± 1.7		95 ± 5	32.1 ± 1.6	33.5 ± 1					

Normal blood count values from birth to 18 years

Age	Hemo-globin	RBC	Hema-tocrit	MCV	WBC	Neutro-phils	Lym-pho-cytes	Mono-cytes	Eosino-phils	Baso-phils	Plate-lets
	(g/dl)	(×1012/l)		(fl)	(×109/l)	(×109/l)	(×109/l)	(×109/l)	(×109/l)	(×109/l)	(×109/l)
Birth (term)	14.9–23.7	3.7–6.5	0.47–0.75	100–125	10–26	2.7–14.4	2.0–7.3	0–1.9	0–0.85	0–0.1	150–450
2 weeks	13.4–19.8	3.9–5.9	0.41–0.65	88–110	6–21	1.5–5.4	2.8–9.1	0.1–1.7	0–0.85	0–0.1	170–500
2 months	9.4–130	3.1–4.3	0.28–0.42	84–98	5–15	0.7–4.8	33–10.3	0.4–1.2	0.05–0.9	0.02–0.13	210–650
6 months	10.0–13.0	3.8–4.9	0.3–0.38	73–84	6–17	1–6	3.3–11.5	0.2–1.3	0.1–1.1	0.02–02	210–560
1 year	10.1–13.0	3.9–5.1	0.3–0.38	70– 82	6–16	1–8	3.4–10.5	0.2–0.9	0.05–0.9	0.02–0.13	200–550
2–6 years	11.0–13.8	3.9–5.0	0.32–0.4	72– 87	6–17	1.5–8.5	1.8–8.4	0.15–1.3	0.05–1.1	0.02–0.12	210–490
6–12 years	11.1–14.7	3.9–5.2	0.32–0.43	76–90	4.5–14.5	1.5–8.0	1.5–5.0	0.15–1.3	0.05–1.0	0.02–0.12	170–450
12–18 years											
Female	12.1–15.1	4.1–5.1	0.35–0.44	77–94							
					4.5–13	1.5–6	1.5–4.5	0.15–1.3	0.05–0.8	0.02–0.12	180–430
Male	12.1–16.6	4.2–5.6	0.35–0.49	77–92							

MICROBIOLOGY			
Test	**Specimen**	**Container**	**Turn Around Time**
BLOOD CULTURES			
Culture and sensitivity	Blood Adults Blood Pediatric	8-10 mL of blood in Bact-Alert-p aerobic bottle and anaerobic bottles. Neonate (< 1 month) 1 to 2 mL Infant (1 month to 2 years) 3 to 5 mL Children (2 to 12 years) 5 to 10 mL (5 mL per bottle) Adolescent (> 12 years) 20 mL (10 mL per bottle)	Once a positive blood culture bottle is identified, gram stain is done and results are phoned immediately. Culture and sensitivity results in 48 - 72 Hrs after gram staining.
EAR NOSE THROAT			
Culture and sensitivity	Ear swab	Swab in Stuart transport medium	Gram stain: same day; Prelim: 24 hours; Final: 48-72h
Fungus culture	Ear swab	Swab in Stuart transport medium	Gram stain: 24 hours; Final: 4 weeks
Fungus culture	Mouth swab	Swab in Stuart transport medium	Gram stain: 24 hours; Final: 4 weeks
Culture and sensitivity , MRSA	Nasopharyngeal or nasal swab	Swab in Stuart transport medium	Gram stain: same day; Prelim: 24 hours; Final: 48-72h
Culture and sensitivity	Sinus or antral lavage	Swab in Stuart transport medium	Gram stain: same day; Prelim: 24 hours; Final: 48-72h
Fungus culture	Sinus or antral lavage	Swab in Stuart transport medium	Gram stain: 24 hours; Final: 4 weeks
Culture and sensitivity	Throat swab	Swab in Stuart transport medium	Prelim: 24 hours; Final: 48-72h
EYE			
Culture and sensitivity	Conjunctival swab and lids	Swab in Stuart transport medium	Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours
Culture and sensitivity	Corneal scraping	- 3 smear(s); inoculate culture media	Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours
Fungus culture	Corneal scraping	- 3 smear(s); inoculate culture media	Fungal stain: 24 hours; Final: 4 weeks
FAECES (STOOL)			
Culture and sensitivity	Faeces (Stool)	Enteric Pathogen Transport (EPT) Media or clean stool container	48 - 72 HRS
Viral Serology (Rotavirus and Adenovirus)	Faeces (Stool)	Clean stool container	90 minutes
Occult Blood	Faeces (Stool)	Clean stool container	90 minutes
Salmonella typhi antigen test	Faeces (Stool)	Clean stool container	90 minutes
Helicobacter pylori antigen test	Faeces (Stool)	Clean stool container	90 minutes
Culture and sensitivity (C&S), VRE, MRSA	Rectal swab	Clean stool container	48 - 72 HRS

GENITAL TRACT			
Culture and sensitivity	Cervical or urethral swab	Swab in Stuart transport medium	Specimens examined for N. gonorrhoeae only. Cultures for Mycoplasma or Urea plasma not available.
Culture and sensitivity	Upper genital tract swab or aspirate	Swab in Stuart transport medium	Wet preparation Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours
Group B streptococci (GBS)	Vaginal or vulval swab	Swab in Stuart transport medium	Wet preparation Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours
Trichomonas, Candida or bacterial vaginosis	Vaginal or vulval swab	Swab in Stuart transport medium	Wet preparation Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours
Chlamydia trachomatis antigen detection	High Vaginal Swab (female)	Swab in transport vial provided by kit manufacturer; specific kit for females	24 HRS
Chlamydia trachomatis antigen detection	Urethral Swab (male)	Swab in transport vial provided by kit manufacturer; specific kit for males	24 HRS
LOWER RESPIRATORY TRACT			
Culture and sensitivity	Bronchial brushings	Bijou bottle with 1 mL Ringer's lactate	Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours
Culture and sensitivity	Bronchoalveolar lavage (BAL)	Sterile container, 15 mL	Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours
Fungal culture	Bronchoalveolar lavage (BAL)	Sterile container, 1 mL	Gram stain: 24 hours; Final: 4 weeks
Mycobacteria (AFB) ZN staining and or Culture	Bronchoalveolar lavage (BAL)	Sterile container, 2 mL	Prelim: 48 hrs, Final: 4 - 8 weeks
Fungal culture	Bronchoscopy aspirates, washings or lung needle aspirates	Sterile container, 15 mL	Fungal stain: 24 hours; Final: 4 weeks
Culture and sensitivity	Bronchoscopy aspirates, washings or lung needle aspirates	Sterile container, 1 mL	Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours
Mycobacteria (AFB) ZN staining and or Culture	Bronchoscopy aspirates, washings or lung needle aspirates	Sterile container, 2 mL	Prelim: 1 week; Final: 4 - 8 weeks
Culture and sensitivity	Sputum (including endotracheal tube)	Sterile container or endotracheal tube	Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours
Fungal culture	Sputum (including endotracheal tube)	Sterile container or endotracheal tube	Fungal stain: 24 hours; Final: 4 weeks
Mycobacteria (AFB) ZN and or Culture	Sputum (including endotracheal tube)	Sterile container or endotracheal tube	Prelim: 1 week; Final: 4 - 8 weeks
SPINAL FLUID AND STERILE BODY FLUIDS			
Culture and sensitivity (C&S)	Spinal fluid (CSF) or other sterile body fluids	Sterile container	Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours

Fungal culture	Spinal fluid (CSF) or other sterile body fluids	Sterile container	Fungal stain: 24 hours; Final: 4 weeks
Mycobacteria (AFB) ZN staining and or Culture	Spinal fluid (CSF) or other sterile body fluids	Sterile container	Prelim: 1 week; Final: 4 - 8 weeks
Cryptococcal Antigen Test	Spinal Fluid (CSF)	Sterile container	4 HRS
Indian Ink Stain for Cryptococcus neoformans	Spinal Fluid (CSF)	Sterile container	4 HRS
URINE			
Mycobacteria (AFB)	1st Morning Urine	Sterile container	Prelim: weekly; Final: 4 - 8 weeks
Ova and parasites (Schistosomiasis)	Mid-day urine	Sterile container	90minutes
Culture and sensitivity (C&S)	Mid-stream urine (MSU)	Sterile container	48 - 72 HRS
Fungus culture	Mid-stream urine (MSU)	Sterile container	48 - 72 HRS
Pregnancy Detection Test	Urine	Sterile container	90 minutes
Microscopy &Urinalysis (dip stick)	Urine	Sterile container	90 minutes
WOUNDS AND TISSUES			
Mycobacteria (AFB)	Aspirate	Sterile container	Prelim: 48 hours; weekly; Final: 4 - 8 weeks
Culture and sensitivity (C&S)	Deep wound and aspirate	Swab in Stuart transport medium	Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours
Fungal culture	Deep wound and aspirate	Swab in Stuart transport medium	Fungal stain: 24 hours; Final: 4 weeks
Culture and sensitivity (C&S)	Superficial wound swab and drainage	Swab in Stuart transport medium	Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours
Culture and sensitivity (C&S)	Tissue and biopsy	Sterile container with sterile saline	Gram stain: same day; Prelim: 24 hours; Final: 48 - 72 hours
Fungal culture	Tissue and biopsy	Sterile container with sterile saline	Fungal stain: 24 hours; Final: 4 weeks
Mycobacteria (AFB)	Tissue and biopsy	Sterile container with sterile saline	Prelim: 48 hours; Weekly; Final: 4 - 8 weeks
Note: All ICUroutine request are handled as STAT test.			

APPENDIX 2 CRITICAL LABORATORY VALUES

CHEMISTRY		
Test	**Alert values**	**Critical values**
Amylase (VITROS)	Greater than 318 U/L	Greater than 1060 U/L
Calcium		Greater than 3.25 mmol/L
Less than 1.65 mmol/L		
Creatinine		Greater than 650 umol/L
Ethyl alcohol		Greater than 33 mmol/L
Glucose	Adults/Adolescence (12-18 years): Greater than 20.0 mmol/L	Children (<12 years): Less than 2.0 mmol/L Adults/Adolescence (12-18 years): Greater than 30.0 mmol/L
Iron		Greater than 55 umol/L (if age < 10 years)
Lipase	Greater than 150U/L	Greater than 500U/L
Magnesium		Less than 1.0 mmol/L
Pottassium	Less than 2.8 mmol/L Greater than 6.2 mmol/L (non-haemolysed)	Less than 2.5 mmol/L Greater than 6.6 mmol/L (non-haemolysed)
Sodium		Less than 120 mmol/L Greater than 160 mmol/L
Total Bilirubin		0-2 days old – Greater than 260 umol/L 3 days old – Greater than 310 umol/L >3 days old – Greater than 340 umol/L
Urea		Greater than 35.0 mmol/L
Urinalysis glucose + ketone combination (blood glucose takes precedence for adults and adolescence only)		Adults/Adolescence (12-18 years): not known to be diabetic and children (<12 years) : Glucose >55 mmol/L and Ketone >4 mmol/L
Lead		Greater than3.00 umol/L
Chloride		Low: <76 mmol/L High: > 125 mmol/L
C-=Reactive protein high sensitivity (hsCRP)		>150mg/ml
Calcium ionized		<0.8 or > 1.54 mmol/L
Osmolality, plasma or serum		<250 or >335 mOsm/kg water
Lactate serum and plasma		>2.9 mmol/L

THERAPEUTIC DRUG LEVELS		
DRUG TEST	**ALERT VALUES**	**CRITCAL VALUES**
Benzodiazepines		Greater than 7.0 umol/L
Carbamazepine		Greater than 63 umol/L
Despramine & imipramine		Graeter than 1.8 umol/L
Digoxin		Greater than 3.5 nmol/L (if time of last dose is > 6hours)
Gentamicin		Greater than 2.0 mg/L (> 8 hours post dose)
Lithium	Greater than 2.0 mmol/L	Greater than 2.5 mmol/L
Phenobarbital		Graeter than 250 mmol/L
Phenytoin		Graeter than 130 umol/L
Primidone	Greater than 70 umol/L (reflex to Phenobarbital if primidone>70)	Greater than 110 umol/L (reflex to Phenobarbital if primidone>70)
Qinidine		Greater than 30.0 umol/L
Rapamycin	Greater than 25.0	Grreater than 30.0 umol/L
Sirolimus	Ug/L	
Salicylate	Greater than 2.2 mmol/L	Greater than 3.0 mmol/L
Theophylline	Greater than 110 mmol/L	Greater than 220 ummol/L
Tobramycin		Greater than 2.0 mg/L (> 8 hours post dose)
Valproic acid	Greater than 1000 umol/L	Greater than 1400 umol/L

HEMATOLOGY		
Test	**Alert values**	**Critical values**
Total leucocytes count		Greater than 25 x 10/l less 2.5 x 10/l
Absolute lymphocyte count		Greater than 250 x 109/L
Absolute Neutrophil count	Less than 1.0 x 109/L Greater than 50.0 x 109/L	Less than 0.5 x 109/L
Greater than 100 x 109/L		
Hemoglobin	Less than 8.0 mg/dl	
Greater than 20.0mg/dl	Less than 6.0 mg/dl	
Platelet count	Less than 100 x 109/L	Less than 20 x 109/L
Prothrombin Time /INR	Greater than 4.5	Greater than 6.0
Thromboplastin time, _Partial (PTT)	Greater than 80 seconds	Greater than 100 seconds
BLOOD FILM REVIEW CRITERIA		All positive malaria smears. Presence of intracellular bacteria in WBC. Presence of atypical leukocytes like blasts.

MICROBIOLOGY

1. ALL Positive blood cultures and sterile site culture positives and antigen detections.
2. ALL Positive central nervous system (CSF, brain, etc.) cultures or antigen detections including Positive cultures from tiss ues (liver, bone biopsies)
3. All reports diagnostic of TB – Zn smear, TB culture or out sourced molecular tests.
4. Beta hemolytic Streptococci Group A from Abscess/Would/Skin cultures, sterile sites.
5. Neisseria gonorrhea from eyes
6. Pseudomonas aeruginosa from eyes.
7. The following pathogens from lower respiratory cultures
   - Moderate to heavy Streptococcus pneumonia
   - Moderate to heavy Pseudomonas aeruginosa
   - Moderate to heavy Staphylococcus aureas
8. Usually resistant organisms including:
   - Vancomycin resistant Enterococci (VRE) - rare in our environment
   - Methicillin resistant Staphylococcus aureas (MRSA) not very common for Kenya
   - Penicillin resistant Streptococcus pneumonia in sterile sites
   - Multiple resistant organisms for hospitalized patients usually for IPC nurse
9. Cryptococcus neoformans from any site
10. Stenotrophomonas maltophilia ICU or any inpatient culture
11. Stool pathogens including Salmonella, Shigella, Campylobacter spp, Clostridium difficile, E- Coli 0157:H7, Vibrio spp, Aeromonas spp, Rota and Adeno Viruses, Parasites: Trophozoites of Giardia and Entamoeba histolytica, Cyclospora spp., Cryptosporidium spp., parasitic ova, worm, progloit-tid, and larvae.
12. Vitreous fluids from eyes (positive and negative reports from slide preparation).
13. Any other unusual isolates from reference laboratories (immediately as the report is received).
14. All notifiable infectious diseases are alert reports including Cholera, Influenza or other viral or bacterial infections – confirm from MOH site the list of these conditions and modalities of notifi-cation.

APPENDIX 3 REFERRAL LABORATORIES

DEPARTMENT	NATURE OF SERVICE CONTRACTED	PROVIDER
LABORATORY	HISTOLOGY	AGA KHAN UNIVERSITY HOSPITAL
	BIOCHEMISTRY	
	MOLECULAR	
	MICROBIOLOGY	
	HEMATOLOGICAL	
LABORATORY	BIOCHEMISTRY	LANCET LABORATORY
	HEMATOLOGICAL	
	SEROLOGY	
LABORATORY	SEROLOGY	PATHCARE
LABORATORY	MOLECULAR	METROPOLIS

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